

B-fields In STar-forming Region Observations (BISTRO): Technical Foundations for Quantitative 450/850 μm JCMT SCUBA-2/POL-2 Polarimetry

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ABSTRACT

The 450 μm polarization capability of JCMT SCUBA-2/POL-2 offers unique leverage for probing wavelength-dependent dust-alignment physics when combined with the standard 850 μm data, but the shorter-wavelength products are technically vulnerable to three major classes of systematic uncertainty that directly affect any quantitative comparison with the standard 850 μm band: a non-Gaussian compound 450 μm point-spread function, stronger atmospheric transmission fluctuations, and calibration inconsistencies in the extended-emission surface-brightness regime relevant to molecular-cloud polarimetry. Reliable dual-band scientific interpretation therefore requires a dedicated end-to-end validation of the 450 μm products before comparison with their 850 μm counterparts. This paper establishes three technical foundations that together render the JCMT 450/850 μm dual-band polarization data products fit for quantitative scientific use. The methodology is organized as three interlocking architectural pillars: a photometric foundation, a statistical foundation, and a map-domain comparability foundation, each closed by an explicit empirical or quantitative validation step. Pillar 1 (photometric legitimacy): We demonstrate that the use of $\text{FCF}_{\text{arcsec}}$ and mJy arcsec^{-2} units is physically required for extended molecular-cloud emission, whereas adoption of FCF_{peak} introduces a deterministic wavelength-dependent bias that shifts inferred dust emissivity indices by $\Delta\beta = -1.06$. Pillar 2 (statistical legitimacy): We show that the empirical Stokes I noise across the survey is well reproduced by a two-component model with additive (atmospheric) and multiplicative (transmission-fluctuation) components, validated by jackknife resampling across 174 POL-2 fields, and that the Stokes Q, U noise separates cleanly into an atmosphere-dominated ($p \lesssim 5\%$) and a residual-systematic-dominated ($p \gtrsim 10\%$) regime with an empirical isotropy check ($\sigma_Q \simeq \sigma_U$, $\text{Cov}(Q, U) \approx 0$). Pillar 3 (map-domain comparability): We construct a beam- and spatial-frequency-harmonized 450 μm science product matched to the 850 μm angular response, and quantitatively validate its preservation of local polarization-angle morphology through a three-product internal comparison across the dual-band statistical sample. The Pillar 3 capstone validation shows that replacing the native two-component compound 450 μm beam with an analytically deconvolved–reconvolved $12''.6$ Gaussian beam systematically reduces the source-median absolute polarization-angle residual to $|\widetilde{\Delta\chi}| \sim 4^\circ\text{--}6^\circ$ across the 24-source dual-band statistical sample, with all sources reaching post-harmonization $|\widetilde{\Delta\chi}| < 10^\circ$ at $I/\sigma_I > 30$ (two-sample Kolmogorov–Smirnov $D = 0.31$, $p < 10^{-10}$ on the $12''$ polarization-catalog sampling; $D = 0.45$, $p < 10^{-55}$ on the independent $4''$ Stokes-map sampling for the L1688 representative example). Combined with the photometric and statistical pillars above, this establishes practical

survey-wide systematic floors of $\sigma(P_{450}/P_{850}) \lesssim 0.02$ and $\sigma(\gamma - \alpha) \lesssim 0.04$, with the dominant FCF and beam-mismatch terms canceling to first order in these paired ratio diagnostics. These results demonstrate that the final JCMT 450 μm polarization products are technically robust for quantitative dual-band analysis and that the principal observables used in the companion science papers— P_{450}/P_{850} , $\gamma - \alpha$, and the inter-band polarization-angle morphology—carry explicit systematic floors smaller than the astrophysical signals they are used to interpret. The framework therefore provides the calibrated, noise-characterized, and morphology-preserving instrumental basis required for the companion scientific papers in this series, and is applicable to current and future JCMT 450 μm polarimetric observations as well as archival re-analyses of existing POL-2 data.

Keywords: Submillimeter astronomy (1647) — Polarimetry (1278) — Astronomical instrumentation (799) — James Clerk Maxwell Telescope (2109) — SCUBA-2 (2136)

1. INTRODUCTION

1.1. *Physical motivation: why 450 and 850 μm noise characteristics differ fundamentally*

The James Clerk Maxwell Telescope’s SCUBA-2 camera observes simultaneously in two submillimeter continuum bands centered at 450 and 850 μm . Despite the shared instrument, the two bands present radically different noise environments for polarimetric analysis. At 850 μm , atmospheric transmission is high ($\gtrsim 70\%$ in typical Weather Band 1–2 conditions at Maunakea) and varies slowly on the observing timescales relevant to POL-2’s daisy scanning. At 450 μm , atmospheric transmission is markedly lower ($\lesssim 30\%$ even in the best Weather Band 1 conditions) and its variations—driven by shifts in column water vapor on minute-to-minute timescales—are no longer well modeled as a small perturbation. Compounding this, the beam solid angle scales as λ^2 , so the sky-noise contribution per beam at 450 μm is statistically independent from that at 850 μm despite the identical scan pattern. The empirical signature of these effects, established in the present work from 174 POL-2 fields in the JCMT archive, is a median ratio $\sigma_{450}/\sigma_{850} \simeq 38$ (interquartile range 21.6–60.2) with a power-law scaling $\sigma_{450} \propto \sigma_{850}^{2.59 \pm 0.26}$ —*superlinear* in the 850 μm noise, reflecting the non-proportional impact of atmospheric transmission degradation on the shorter-wavelength band (§3.2). This scaling, together with the implication that ~ 30 –60 times more surface-brightness noise must be propagated through any 450 μm polarization analysis, is the physical reason why published POL-2 data-reduction guidance—validated almost exclusively at 850 μm —does not transfer to the 450 μm band without additional characterization and correction steps.

1.2. *Community state: the 850 μm success and the 450 μm methodology gap*

Since the commissioning of POL-2 (Friberg et al. 2016, 2018), the 850 μm band has matured into the community standard for submillimeter emission polarimetry. The BISTRO survey (Ward-Thompson et al. 2017) and its sibling programmes have delivered 850 μm polarization maps of tens of star-forming regions (e.g., Pattle et al. 2017; Kwon et al. 2018; Soam et al. 2018; Pattle et al. 2021, 2023; Choi et al. 2024; Wu et al. 2024; Karoly et al. 2025; Yang et al. 2025; Hwang et al. 2025), and the published `pol2map.py` workflow within Starlink SMURF (Chapin et al. 2013) is now widely adopted. Validation and calibration of this workflow—the instrumental polarization model (Friberg et al. 2018), the modified asymptotic debiasing estimator (Plaszczynski et al. 2014; Montier et al. 2015), the extended-source flux-conversion factors (Mairs et al. 2021)—has focused on the 850 μm band.

The 450 μm band has lagged. Although Hwang et al. (2021) presented the first 450 μm POL-2 polarimetric mapping of OMC 1, a systematic methodological reference for 450 μm data reduction specific to the dual-wavelength polarimetric context has not been published. The additional challenges identified by the present work—the $\sigma_{450}/\sigma_{850}$ scaling above, a wavelength-dependent additive Stokes- Q/U zero-level offset invisible in single-wavelength analyses, and the propagation of residual systematics through dual-band spectral-index diagnostics—are all consequences of the dual-wavelength analysis requirement, not of the 450 μm band in isolation. Absent a systematic methodology, 450 μm polarimetric products either cannot be compared to 850 μm products on equal footing, or must be re-derived ad hoc for each analysis. The need for a standalone technical reference is therefore methodological rather than procedural: without a documented treatment of the 450 μm offset correction, beam harmonization, and noise propagation, dual-band 450/850 μm polarization measurements cannot be placed on a common quantitative footing. The present paper provides that reference framework and establishes the technical basis for the companion science analyses.

1.3. Contribution of this paper

Before any wavelength-dependent polarization signature can be interpreted in terms of dust alignment physics, it is necessary to determine whether the observed differences between 450 and 850 μm polarization observables exceed the combined uncertainties introduced by flux calibration, noise propagation, additive Stokes- Q/U zero-level offset correction, and beam harmonization. The primary goal of this paper is therefore to establish a quantitatively validated technical framework for dual-band JCMT/POL-2 polarimetric comparison—one in which inter-band differences in the BISTRO 450/850 μm polarization survey can be interpreted as astrophysical signal rather than as residual instrumental, calibration, or noise systematics.

We present the complete data-reduction, noise-modeling, and systematic-calibration methodology developed for the BISTRO 450 μm dual-wavelength survey. The methodology rests on three interlocking pillars: photometric legitimacy, statistical legitimacy, and map-domain comparability. Together they establish the end-to-end technical credibility of the JCMT 450 μm dual-band polarization products used in the companion scientific papers.

Pillar 1 – Photometric legitimacy. An empirical verification of the Planck-corrected spectral-index framework $\beta = \alpha - C(T_d)$ (§3.1) resolves a systematic bias of $\Delta\beta = -1.06$ that would otherwise be introduced by naive application of peak-brightness FCFs to extended-emission surface-brightness analysis. This pillar establishes that the inferred fluxes and spectral slopes are physically meaningful for the molecular-cloud emission relevant to dual-band polarimetry.

Pillar 2 – Statistical legitimacy. A hybrid additive-noise estimator for Stokes I (§3.2) combined with a two-component noise model for Stokes Q and U (§3.3) establishes that the 450 μm polarization noise separates cleanly into an atmosphere-dominated regime at low polarization fraction ($p \lesssim 5\%$) and a residual-systematic-dominated regime at high polarization fraction ($p \gtrsim 10\%$). The Stokes Q, U component of this model is validated via an empirical isotropy check ($\sigma_Q \simeq \sigma_U$, $\text{Cov}(Q, U) \approx 0$), and the Stokes I component via jackknife resampling across 174 POL-2 fields. The superlinear $\sigma_{450}/\sigma_{850}$ scaling (§1.1) emerges naturally from this framework as a diagnostic that can be transferred to any future 450 μm POL-2 polarimetric reduction. This pillar establishes that the retained pixels are statistically trustworthy for downstream dual-band analyses.

Pillar 3 – Map-domain comparability legitimacy. Pillar 3 has two stages. In the *construction* stage (§3.4) we identify and correct a wavelength-dependent additive offset in the 450 μm Stokes- Q and Stokes- U maps, where required, at the $\sim 1 \text{ mJy arcsec}^{-2}$ level. This offset is largely invisible in single-wavelength analyses but is inconsistent with the physical requirement that regions with $Q_{850} = U_{850} = 0$ should also yield $Q_{450} = U_{450} = 0$ under optically thin conditions. No corresponding additive offset correction is applied to Stokes- I . We trace the origin of the Q/U offset to a coupling between noise characteristics and low-frequency filtering in the iterative `pol2map` reduction, design a three-step PCA-based correction procedure that reduces the residual to $\lesssim 0.07 \text{ mJy arcsec}^{-2}$, and independently validate the corrected 450 μm polarization maps against SOFIA/HAWC+ 154/214 μm polarimetry of Ophiuchus A. In the *capstone validation* stage (§3.5) we demonstrate, through a three-product internal comparison applied at three successively broader scales (single-source $12''$ catalog, independent $4''$ map, and 24-source dual-band statistical sample), that the beam-harmonization procedure produces a statistically significant and survey-wide contraction of the inter-product polarization-angle residual distribution: the source-median absolute polarization-angle difference after harmonization is $|\widetilde{\Delta\chi}| \sim 4^\circ\text{--}6^\circ$, with *all* sources reaching $|\widetilde{\Delta\chi}| < 10^\circ$ at $I/\sigma_I > 30$, well below the angle-tolerance scale used in companion [Paper S1](#)’s dual-band angle-consistency analysis.

The systematic-uncertainty budget (§4.1) propagates the contributions of all three pillars through to the primary derived quantities P_{450}/P_{850} and $\gamma - \alpha$, showing that many systematics cancel in these ratio forms and that the residual offset-correction floor is the only contribution that survives non-trivially. As a supporting derivation, a comparison of Ricean expectation formulations (Appendix A) establishes when the Laguerre-polynomial approximation remains valid and when the exact Bessel-function formulation is required for the intermediate- S/N regime that dominates 450 μm data.

The scientific findings enabled by this methodology partition into a unified F1–F6 program across the two science papers in the series: the existence-layer findings F1–F3—three empirical constraints on dust grain alignment physics—are reported in the companion ApJ paper, [Paper S1](#), and the universality-layer findings F4–F6—population-level universality, regime-dependent wavelength response, and disorder-mediated wavelength divergence on the BISTRO POLSRC sample—are reported in the companion ApJ paper, [Paper S2](#); both are in preparation. The present paper is intended as a standalone methodology reference suitable both for future JCMT 450 μm polarimetric observations and for archival re-analyses of existing POL-2 data. Readers interested primarily in the physical interpretation are referred

to Paper S1; readers interested in the universality and environment-dependence tests, per-source data products, and the dual-wavelength magnetic-field orientation atlas are referred to Paper S2.

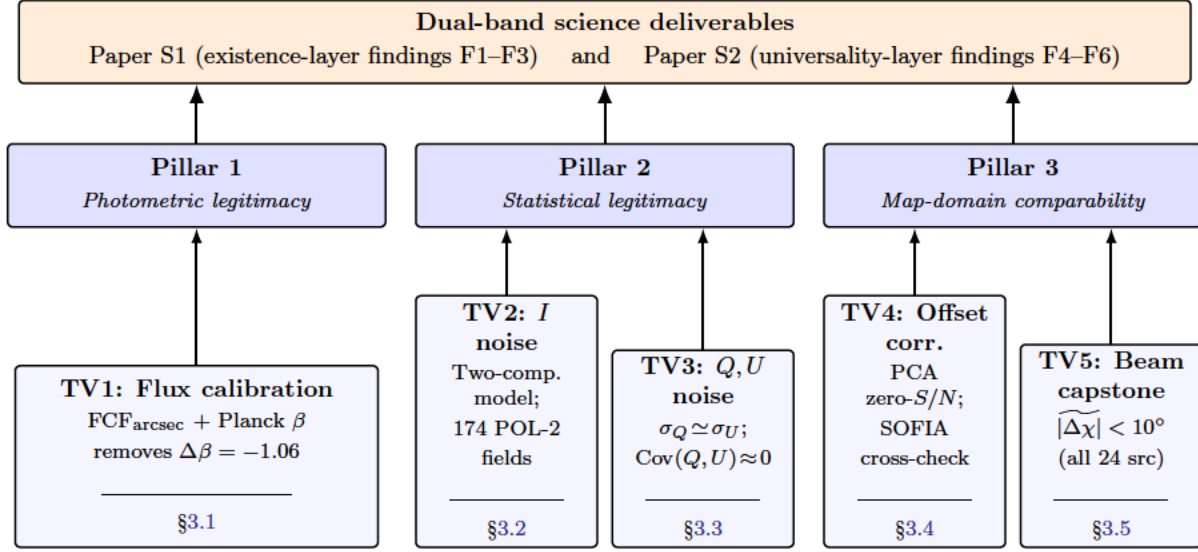


Figure 1. Five-validation ladder $\times$ three-pillar technical-validation architecture of Paper T. The five technical validations of §3 (TV1–TV5, bottom row) provide the per-subsection evidence and group naturally into three architectural pillars (middle row) that organize the per-section narrative. *Pillar 1 (Photometric legitimacy)*: hosts **TV1: Flux calibration** (§3.1), where the use of $\text{FCF}_{\text{arcsec}}$ and mJy arcsec^{-2} units, combined with the Planck-corrected spectral-index framework $\beta = \alpha - C(T_d)$, removes a deterministic $\Delta\beta = -1.06$ bias that naive FCF_{peak} application to extended emission would otherwise introduce, and ensures that all dust spectral indices (α , β , γ) entering the companion analyses are physically meaningful for molecular-cloud emission. *Pillar 2 (Statistical legitimacy)*: hosts **TV2: Stokes- I noise model** (§3.2; two-component additive model validated against the 174-field POL-2 archive) and **TV3: Stokes- Q, U noise model** (§3.3; two-component model verified by the empirical $\sigma_Q \simeq \sigma_U$, $\text{Cov}(Q, U) \approx 0$ isotropy check), together delivering self-consistent statistical-error budgets for every retained polarization measurement. *Pillar 3 (Map-domain comparability)*: hosts **TV4: 450 μm additive Stokes- Q/U zero-level offset correction** (§3.4; PCA-based offset removal to a residual of $\lesssim 0.07 \text{ mJy arcsec}^{-2}$ and beam/spatial-frequency harmonization to a common $12''.6$ Gaussian over $12''.6$ – $300''$, independently validated against SOFIA/HAWC+ polarimetry of Ophiuchus A) and **TV5: Beam-harmonization capstone validation** (§3.5; survey-wide $|\widetilde{\Delta\chi}| \sim 4^\circ$ – 6° floor with all 24 dual-band statistical-sample sources reaching $|\widetilde{\Delta\chi}| < 10^\circ$ at $I/\sigma_I > 30$, two-sample K–S $D = 0.31$, $p < 10^{-10}$ in the $12''$ catalog domain; $D = 0.45$, $p < 10^{-55}$ in the independent $4''$ map domain for the L1688 representative example). The systematic-uncertainty budget (§4.1) propagates TV1–TV5 jointly through to the dual-band derived quantities P_{450}/P_{850} and $\gamma - \alpha$, showing that many systematics cancel in these ratio forms and that the residual offset-correction floor is the only contribution that survives non-trivially. Together, the TV1–TV5 ladder and its three-pillar grouping define the end-to-end technical-credibility framework that underpins the existence-layer findings F1–F3 of Paper S1 and the universality-layer findings F4–F6 of Paper S2. More specifically, TV1 underpins the spectral-index basis (α , γ , β) of F2–F5; TV2–TV3 underpin the polarization-detection significance and angle-precision floors of F1 and F3 in Paper S1 and of F4 and F6 in Paper S2; and TV4–TV5 underpin their matched-resolution F1, F3, and F6 comparisons throughout the dual-band analyses.

1.4. Organization of this paper

The remainder of this paper is organized as follows. Section 2 describes the JCMT SCUBA-2/POL-2 observations, the seven sample counts that recur throughout the dual-wavelength paper series under the source-based and beam-based partition of Table 2 (with one-word short names CENSUS, POLSRC, SPIDEX, DUALBAND for the four principal samples), and the basic reduction parameters. Section 3.1 presents Pillar 1: the flux-calibration framework, including the choice of extended-source FCF, and its empirical verification through Planck-corrected spectral indices; the dual-band derived quantities α , γ , and P_{450}/P_{850} are defined in §3.1.2. Sections 3.2 and 3.3 develop Pillar 2 of the methodology in two parts: the additive-noise characterization of Stokes I (§3.2) and the two-component noise model of Stokes Q and U (§3.3). Section 3.4 presents the construction stage of Pillar 3: the discovery, investigation, and PCA-based correction of the 450 μm additive Stokes- Q/U zero-level offset, including the SOFIA/HAWC+ independent validation.

Section 3.5 provides the capstone validation of Pillar 3, demonstrating—through a three-scale comparison spanning a single-source 12'' catalog example, an independent 4'' map robustness check, and a 24-source dual-band statistical-sample ensemble test—that the beam-harmonization procedure preserves the local polarization-angle morphology to a measured residual floor of $|\widetilde{\Delta\chi}| \sim 4^\circ\text{--}6^\circ$ with all sources reaching post-harmonization $|\widetilde{\Delta\chi}| < 10^\circ$ at $I/\sigma_I > 30$. Section 4 consolidates the three-pillar systematic-uncertainty budget for the dual-band derived quantities P_{450}/P_{850} and $\gamma - \alpha$ (§4.1). Section 5 summarizes the framework and its applicability to future JCMT 450 μm polarimetric analyses. Appendix A compares the exact Bessel and Laguerre-approximation formulations of the Ricean expectation, a supporting derivation underpinning the noise model of Pillar 2.

For navigation, the present paper provides two complementary architectural aids. Figure 1 summarizes the technical-validation framework on two complementary levels: at the per-subsection level it shows the five-validation ladder TV1–TV5 of §3, and at the per-section level it shows their grouping into three architectural pillars, where Pillar 1 corresponds to TV1, Pillar 2 groups TV2 and TV3, and Pillar 3 groups TV4 and TV5. Table 1 provides the inverse-direction cross-index (technical topic to Pillar mapping), enumerating each major technical topic of the dual-band paper series and indicating whether it resides in the present paper or in a companion paper.

Table 1. Cross-reference index of Paper T’s three-pillar technical architecture, with cross-references to companion-paper material

Topic	Pillar	Location in this paper series	Keywords	Note
<i>Paper T — Pillar 1: Photometric legitimacy</i>				
Flux Calibration & Surface Brightness Units	1	§3.1	FCF _{arcsec} ; mJy arcsec ^{−2} ; β	1
<i>Paper T — Pillar 2: Statistical legitimacy</i>				
Stokes I Noise Characterization	2	§3.2	Two-component model; σ_λ ; k_λ	2
Stokes Q, U Noise Model and Isotropy Check	2	§3.3	Polarization noise; $\sigma_Q \simeq \sigma_U$; (Q, U) isotropy	3
Ricean Expectation Formulations	2 (supp.)	App. A	Bessel function; Laguerre polynomial	4
<i>Paper T — Pillar 3: Map-domain comparability legitimacy</i>				
Offset Recognition	3 (constr.)	§3.4	three-wavelength SOFIA/HAWC+ comparison; 450 isolated	5
Hardware-Origin Investigation	3 (constr.)	§3.4	Pointing; IP model; brightness dependence	6
Software-Origin Offset Correction	3 (constr.)	§3.4	Three-step regression; <code>pca.zero_snr</code> ; KS test	7
Beam-Harmonization Capstone Validation	3 (capstone)	§3.5	K–S test; $ \widetilde{\Delta\chi} $ floor; 24-source survey	8
<i>Paper T — Cross-pillar consolidation</i>				
Systematic Uncertainty Budget	1, 2, 3	§4.1	FCF; residual offset; beam mismatch; error propagation	9
<i>Companion-paper material (informational; cross-referenced from Paper T)</i>				
Distance Determination	—	Paper S2 Appendix E	Gaia; Zucker et al. (2020); Prisinzano et al. (2022); VLBI	10
¹² CO($J=3-2$) Contamination	—	Paper S1	850 μm ; Orion A; HARP	11
Stacked P_{450} Histogram	—	Paper S2	Log-normal; percentile statistics	12
Spectral-Index S/N Bias Diagnostics	—	Paper S2	S/N -dependent bias in α and γ	13
Spectral-Index Distance-Bias Analysis	—	Paper S2	Beam-size effects on $\gamma - \alpha$ and β	14
Bounds of $\mathcal{S}$	—	Paper S2	Random-field limit; noise floor	15
$\mathcal{S}$ Kernel Dependence	—	Paper S2	3×3 vs. 5×5 ; robustness	16
$\vec{B}$ -Field Maps Atlas	—	Paper S2	450/850 μm ; distance-ordered	17

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Table 1 continued on next page

Table 1 (*continued*)

Topic	Pillar	Location in this paper series	Keywords	Note
NOTE— “Pillar” column indicates which of the three architectural pillars of §1.3 the topic supports; “constr.” = construction stage, “capstone” = quantitative capstone validation, “supp.” = supporting derivation. Items moved to the companion papers retain their original technical position relative to the dual-band methodology, but their primary discussion now lives in those papers; the Paper T body cross-references them where needed. (1) Rationale for extended-source calibration; demonstration that point-source FCF introduces a systematic $\Delta\beta = -1.06$ bias in dust spectral indices; underpins the photometric foundation. (2) Two-component additive noise model ($\sigma_\lambda^2 + k_\lambda^2 I^2$) for atmospheric and signal-dependent contributions; jackknife-validated across 174 POL-2 fields. (3) Extension of noise framework to polarization; derivation of the two-component Stokes Q, U model and empirical verification of (Q, U) noise isotropy. (4) Comparison of exact Bessel-function and Laguerre-polynomial approximations to the Ricean expectation; demonstration of Laguerre breakdown at intermediate S/N. (5) Identification of $\sim 20^\circ$–30° angular discrepancies between 450 μm and independent SOFIA/HAWC+ 154 μm polarimetry as the empirical trigger for the construction stage of Pillar 3. (6) Systematic tests ruling out telescope pointing, source brightness, and instrumental polarization as offset origins. (7) Three-step regression enforcing physical constraints; identification of <code>pca.zero_snr</code> as the offset source; SOFIA/HAWC+ validation. (8) Three-product internal comparison (native-beam vs. deconvolve-reconvolve vs. final beam- and spatial-frequency-harmonized) on L1688 Ophiuchus A and on a 24-source dual-band statistical sample, yielding the global polarization-angle uncertainty architecture adopted in Papers S1 and S2. (9) Summary of systematic uncertainty contributions from flux calibration, offset-correction residuals, and beam mismatch; propagation to P_{450}/P_{850} and $\gamma - \alpha$; demonstration that many systematics cancel in these ratio quantities. Items (10)–(17) below correspond to the “Companion-paper material” rows of the table, whose “Location” column identifies the home paper for each topic. (10) Hierarchical distance assignment using Gaia-based catalogs with literature overrides for sources with VLBI maser parallaxes. (11) Assessment of molecular-line contamination in the 850 μm band; median contamination of $\sim 9\%$ for high-S/N pixels. (12) Log-normal behaviour of polarization fractions and justification for percentile statistics. (13) S/N-dependent bias diagnostics for α and γ; threshold-determination methodology. (14) Regression analysis separating beam-size effects from intrinsic physical variations in $\gamma - \alpha$ and β. (15) Derivation of the theoretical upper limit $\sim 52^\circ$–63° for the angle-dispersion function $\mathcal{S}$ in random fields, together with the corresponding observational noise floor. (16) Verification that $\mathcal{S}$ measurements are robust to 3×3 versus 5×5 pixel averaging. (17) Atlas of dual-wavelength magnetic-field orientation maps for all 450 μm-detected sources, ordered by distance.				

2. OBSERVATIONS AND SAMPLE

This section establishes the observational foundation for Pillars 1–3 of Figure 1 (cross-indexed in Table 1): the JCMT SCUBA-2/POL-2 observations, the seven sample counts that recur throughout the paper series under the source-based and beam-based partition of Table 2, and the basic reduction parameters that underpin all subsequent pillar developments.

2.1. JCMT SCUBA-2/POL-2 observations

The SCUBA-2 bolometer camera on the James Clerk Maxwell Telescope (JCMT), together with the POL-2 rotating half-wave plate polarimeter, provides dual-band continuum polarimetry at nominal wavelengths of 450 and 850 μm . Because SCUBA-2 simultaneously acquires data at both 450 and 850 μm in every observation, the 450 μm polarization data analyzed in this paper were obtained in parallel with the 850 μm observations carried out by the BISTRO survey and its precursor programmes. The POL-2 instrument and the dual-band observing mode are described in detail by Friberg et al. (2016) and Friberg et al. (2018); the present work concerns the data-reduction and systematic-calibration methodology specific to the 450 μm band.

The data used in this work were selected from POL-2 observations available in the JCMT archive as of 2022 February. For the 450 μm analyses, we restricted the sample to observations taken under the highest quality Weather Band (WB 1), corresponding to zenith atmospheric opacity $\tau_{225\text{ GHz}}^{\text{zenith}} \leq 0.05$. No opacity-based selection was applied to the 850 μm data, which benefit from relaxed atmospheric-transmission requirements. The full dual-band time-series raw dataset, totaling 54 Tb, was downloaded from the Canadian Astronomy Data Centre (CADC) JCMT Science Archive to the Astronomy Data Center of the National Astronomical Observatory of Japan for reduction.

2.2. Sample inventory

Seven sample counts recur throughout this paper and its companion papers, partitioned into a *source-based* layer (each entry has a named astronomical source or POL-2 field) and a *beam-based* layer (matched-resolution 12'' pixel-pair, the smallest unit of independent dual-band measurement); Table 2 summarizes their counts, short names, and primary uses. The four principal samples carry one-word short names (CENSUS, POLSRC, SPIDX, and DUALBAND) used adjectivally throughout the paper series; three subsidiary subsamples (37, 24, 9) are listed without short names and are referred to by phrasal descriptors (e.g., “the 24-source beam-validation subsample of CENSUS”). The selection criteria that produce each sample are described in the text below and in §2.3.

The sample-selection function is governed by three criteria: (i) availability to the lead author as of 2022 February; (ii) 450 μm observations obtained under Weather Band 1 ($\tau_{225\text{ GHz}}^{\text{zenith}} \leq 0.05$); and (iii) sufficient signal-to-noise ratio for dual-band polarization analysis (peak Stokes I signal-to-noise ratio > 100 at 450 μm). The 174-field noise characterization and the 37-source FCF/Planck-correction verification span the full distance range of the sample, from ~ 100 pc low-mass cores to ~ 8 kpc Galactic Center clouds, so the methodology presented in this paper is exercised across the full range of signal levels and sky-background conditions encountered in dual-band POL-2 polarimetry; the scientific-impact consequences of these selection criteria are discussed in Paper S1 and Paper S2.

2.3. Basic reduction parameters and calibration

We processed the raw time-series data using `pol2map.py` (version 2021 July 23) from SMURF within Starlink (Chapin et al. 2013); we note that a subsequent update was released in February 2022 within Starlink 2023, incorporating exposure-time-weighted FCF normalization (Hwang et al. 2025), but our analysis uses the 2021 July version consistently throughout. This version has been used across recent BISTRO survey publications (e.g., Pattle et al. 2023; Choi et al. 2024; Wu et al. 2024; Karoly et al. 2025; Yang et al. 2025; Hwang et al. 2025, and references therein); the reduction follows the established two-step procedure validated for POL-2 data (e.g., Ward-Thompson et al. 2017; Pattle et al. 2017; Kwon et al. 2018; Soam et al. 2018; Pattle et al. 2021). Unlike 850 μm reduction, the 450 μm run requires the `normalise` parameter to compensate for rapid atmospheric-opacity variations between observations; this step is a prerequisite for the offset-correction analysis of §3.4.

For instrumental polarization (IP) correction we apply the “August 2019” model of Friberg et al. (2018), valid for zenith angles between 10° and 60° . Flux calibration uses epoch-dependent flux conversion factors (FCFs) adopted from Mairs et al. (2021): $\text{FCF}_{450\text{ }\mu\text{m}} = 531 \pm 90$ and 472 ± 80 Jy beam $^{-1}$ pW $^{-1}$ before and after 2018 June 30, respectively, where the quoted uncertainties incorporate a 17% systematic component. $\text{FCF}_{850\text{ }\mu\text{m}} = 517 \pm 36$ and

Table 2. Sample inventory across the BISTRO 450/850 μm dual-wavelength paper series

Sample name	Short name	Count	Primary use
<i>Source-based samples (each entry is a named astronomical source or POL-2 field)</i>			
POL-2 archive fields	CENSUS	174	Paper T §3.2–§3.3 (Pillar 2 noise characterization)
FCF/spectral-index subsample	—	37	Paper T §3.1 (Pillar 1 FCF and Planck-correction verification)
Beam-validation subsample	—	24	Paper T §3.5 (Pillar 3 capstone beam-harmonization validation)
Unique star-forming regions	POLSRC	69	Paper S2 (complete dual-wavelength census; F4–F6)
Statistical science subsample	SPIDX	28	Paper S1 (principal α , γ , R_p per-source statistics; F2–F3)
HGBS-overlap subsample	—	9	Paper S2 §3.6 (F6 N_H , T_d per-pixel correlations)
<i>Beam-based sample (matched-resolution 12'' pixel-pair; the smallest unit of independent dual-band measurement)</i>			
Method-3 HGBS pixel-pair sample	DUALBAND	417	Paper S1 §3.2, §3.3; Paper S2 §3.6 (F1 angle, F2 R_p per-pixel; F6)

NOTE—Nested structure of the source-based layer: 174 POL-2 fields in the JCMT archive (2022 Feb., WB1) reduce to 69 *unique* astronomical regions after mosaicking and quality gating; of those 69, 37 have measurable total-intensity and polarized-intensity spectral indices (α , γ); 28 of the 37 form the final science-grade subset used for the principal per-source dual-wavelength statistics in Paper S1; 24 of the 37 form the dual-band statistical sample used for the beam-validation capstone of §3.5; and 9 of the 69 fall within the public *Herschel* Gould Belt Survey (N_H , T_d) sky coverage used for the per-pixel correlation analyses of Paper S2 §3.6. Paper T Pillar 2 uses the full 174-field set because the noise modeling benefits from every independent field sampling; Pillar 1 uses the 37-source subset; Pillar 3 capstone (§3.5) uses the 24-source dual-band statistical sample. The beam-based DUALBAND sample (417 pixel-pair measurements) is constructed at matched 12'' resolution from the 9 HGBS-overlap regions and provides the per-pixel detection units used in Paper S1’s F1 and F2 analyses and in Paper S2’s F6 correlations; “ N_{samp} ” counts in the science papers refer to subsets of these 417 measurements after band-specific detection masks. Dual-wavelength science results are reported either on the 28-source SPIDX subset (Paper S1, principal F2/F3) or on the full 69-region POLSRC census including lower-significance fields (Paper S2, F4–F6).

498 $\pm$ 35 Jy beam $^{-1}$ pW $^{-1}$ for the same epochs, with a 7% systematic component. An attenuation correction accounting for signal loss through the POL-2 optical train is applied to both bands: 1.96 $\pm$ 0.023 at 450 μm and 1.35 $\pm$ 0.008 at 850 μm . Beam sizes are adopted from Mairs et al. (2021): $\theta_{\text{HPBW}}^{450} = 8''.6 \pm 1''.3$ and $\theta_{\text{HPBW}}^{850} = 12''.6 \pm 1''.9$.

For extended-emission surface-brightness analysis, which encompasses all wavelength-dependent quantities derived in Paper T and its companion papers, calibrated maps are converted to surface-brightness units of mJy arcsec $^{-2}$ using the extended-source FCF; the rationale and systematic implications of this choice are detailed in §3.1. For dual-band overlap analyses, Stokes parameters are regridded onto a common 12'' pixel grid, which is Nyquist-sampled for the 850 μm beam and adequate-but-undersampled at 450 μm . For 450- μm -only native-resolution analyses, a 4'' pixel grid is used; this undersamples the 8''.6 beam by a factor of 2.15 but empirically suppresses reduction artefacts in all Stokes parameters.

Polarized-intensity debiasing and polarization-fraction computation use the modified asymptotic estimator (MAS) of Plaszczyński et al. (2014) and Montier et al. (2015); the implementation details relevant to the 450 μm band are given in §3.4 (modified-asymptotic subsection) and a comparison with Bessel-function-based formulations is provided in Appendix A.

3. RESULTS AND ANALYSIS

This section presents the five technical validations (TV1–TV5) that establish the methodological basis of the BISTRO dual-band polarimetric survey. TV1 (§3.1) establishes the photometric and surface-brightness framework required for inter-band comparison. TV2 and TV3 (§§3.2–3.3) develop the empirical and analytic noise models for Stokes I , Q , and U . TV4 (§3.4) identifies and corrects the 450 μm additive Stokes- Q/U zero-level offset, while TV5 (§3.5) quantifies the residual beam-harmonization angle floor. Together these five validations define the survey-wide systematic precision floor used throughout Papers S1 and S2.

3.1. Technical Validation 1: Flux Calibration, Surface Brightness Units, and Dust Spectral Indices

Accurate comparison between the 450 and 850 μm polarization observables requires that both maps be placed on a physically consistent surface-brightness scale before any spectral-index or polarization-ratio analysis is attempted.

We therefore re-examine the JCMT flux-conversion formalism and demonstrate that the commonly adopted point-source calibration convention is not appropriate for the extended molecular-cloud emission that dominates the present survey. This subsection establishes the surface-brightness framework adopted throughout the present paper and the companion science papers, and quantifies its consequences for the derived dust spectral indices (α , β , γ) used in F2–F5—the Pillar 1 entry of Figure 1 and Table 1.

3.1.1. Motivation and overview

In this work, we express surface brightness in units of mJy arcsec^{-2} , rather than the commonly used mJy beam^{-1} . This choice is not merely a matter of convention, but a physical requirement for extended emission. As we demonstrate below, the use of point-source calibration for spatially extended molecular cloud emission introduces a wavelength-dependent bias that propagates directly into dust spectral indices.

This issue becomes apparent only in multiwavelength analyses. Because most previous POL-2 studies have focused on single-wavelength quantities, such systematic effects have remained largely unrecognized.

3.1.2. Dual-band derived quantities: α , γ , and P_{450}/P_{850}

Two derived quantities recur in the Pillar 1 calibration discussion below, in the Pillar 3 construction stage (§3.4), and in the consolidated three-pillar uncertainty propagation (§4.1): the dust spectral indices of total and polarized intensity, and their algebraic relation to the polarization-fraction ratio. We record the defining equations here so that Paper T is self-contained on these quantities; the physics-side discussion of what they measure (grain-emissivity spectral index, size-dependent alignment, etc.) is given in companion Paper S1.

The spectral indices of total and polarized intensity are defined as

$$\alpha \equiv \frac{\log(I_{450}/I_{850})}{\log(\nu_{450}/\nu_{850})} \quad \text{and} \quad \gamma \equiv \frac{\log(P_{450}/P_{850})}{\log(\nu_{450}/\nu_{850})}, \quad (1)$$

where $\nu_{450} = 666 \text{ GHz}$ and $\nu_{850} = 353 \text{ GHz}$ are the central SCUBA-2/POL-2 frequencies, so that $\log(\nu_{450}/\nu_{850}) \approx 0.276$. The polarization-fraction ratio P_{450}/P_{850} follows from Equation 1 directly:

$$\frac{P_{450}}{P_{850}} = \left(\frac{\nu_{450}}{\nu_{850}} \right)^{\gamma - \alpha}. \quad (2)$$

Equation 2 is an algebraic identity—not a physical model—and will be used in §4.1 to propagate FCF and offset-correction systematics from intensities through to P_{450}/P_{850} and $\gamma - \alpha$. Companion Paper S2 (§3.3) additionally defines three distinct statistical estimators of P_{450}/P_{850} (intensity-weighted, correlation-based, and median-of-per-pixel-ratios) that differ in how they weight individual pixels; Paper T’s uncertainty-budget formulae below apply most directly to the intensity-weighted and median estimators, the latter being the default for the dual-band ratio adopted in companion Paper S1.

3.1.3. Point-source versus extended-source calibration

Two FCF flavors and the extended-emission requirement—The JCMT SCUBA-2/POL-2 calibration framework provides two distinct flux conversion factors (Mairs et al. 2021): (i) the peak FCF (FCF_{peak}), optimized for point-source photometry and expressed in $\text{Jy beam}^{-1} \text{ pW}^{-1}$, and (ii) the arcsecond-based FCF ($\text{FCF}_{\text{arcsec}}$), expressed in $\text{mJy arcsec}^{-2} \text{ pW}^{-1}$ and appropriate for spatially extended emission.

The standard `pol2map` pipeline with `JY=yes` applies FCF_{peak} automatically, producing maps in mJy beam^{-1} . This procedure is correct for calibration sources such as Uranus and CRL 2688, but it is not physically appropriate for molecular clouds, which typically extend over many beam areas. Applying FCF_{peak} to extended emission effectively folds the beam solid angle into the intensity scale, obscuring the true surface brightness.

For this reason, we retain detector outputs in pW during map making (`JY=no`) and apply $\text{FCF}_{\text{arcsec}}$ manually during post-processing. For observations obtained after 2018 June 30, we adopt the following arcsecond-based FCF values (Mairs et al. 2021; Friberg et al. 2023):

$$\text{FCF}_{\text{arcsec},450}^{\text{POL2}} = 7.59 \text{ mJy pW}^{-1} \text{ arcsec}^{-2}, \quad (3)$$

$$\text{FCF}_{\text{arcsec},850}^{\text{POL2}} = 2.79 \text{ mJy pW}^{-1} \text{ arcsec}^{-2}, \quad (4)$$

which include the appropriate POL-2 attenuation corrections.

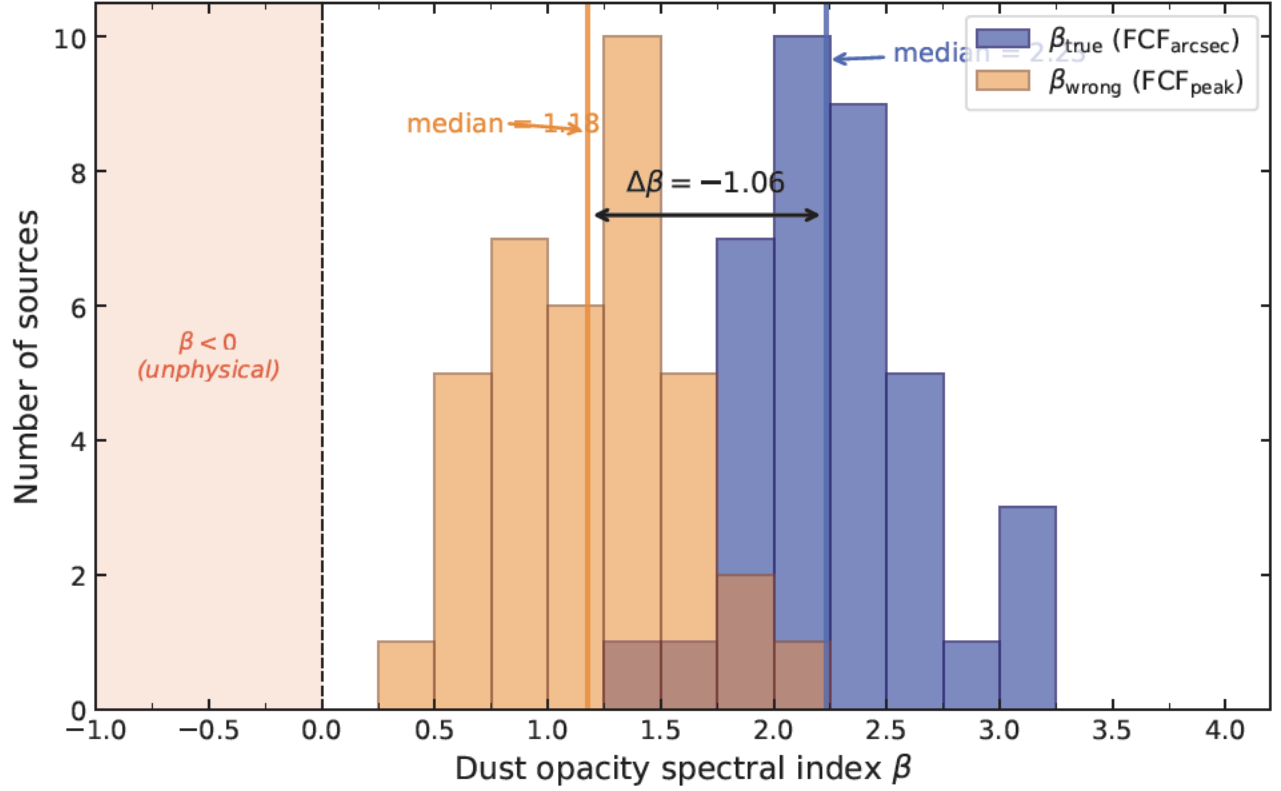


Figure 2. Distribution of the dust opacity spectral index β for the 37 sources in the [Paper S2 §3.4 spectral-index table](#), computed using the full Planck function with per-source dust temperatures ([Paper S2 §3.4](#)). Blue histogram: β_{true} values derived using the extended-source calibration ($\text{FCF}_{\text{arcsec}}$), with a median of $\beta = 2.23$. Orange histogram: β_{wrong} values that would result from incorrect application of the point-source calibration (FCF_{peak}), systematically shifted by $\Delta\beta = -1.06$ (Eq. 8). This figure demonstrates that incorrect use of FCF_{peak} for extended emission introduces a deterministic offset of $\Delta\beta = -1.06$, shifting the entire β distribution toward lower values. The shaded $\beta < 0$ region is shown only as a physical reference; in the present Planck-corrected analysis, no sources fall into this regime. The main result is instead the systematic displacement of the distribution by $\Delta\beta = -1.06$, which is independent of the adopted T_d . The figure does not show source-by-source β uncertainties. Uncertainty in the adopted T_d changes β through the Planck factor $C(T_d)$ and therefore broadens each histogram, but does not alter the calibration-induced displacement between the two histograms. For the representative temperature ranges adopted in companion [Paper S2 §3.4.1](#) ($T_d = 15 \pm 5$ K for LMSF/cold IRDCs and 25 ± 10 K for HMSF/CMZ sources), this contribution is typically $\sigma_{\beta,T} \lesssim 0.3\text{--}0.4$, smaller than the deterministic FCF-induced shift of $\Delta\beta = -1.06$.

A key distinction between the two calibration approaches lies in their wavelength dependence. The ratio between FCF_{peak} and $\text{FCF}_{\text{arcsec}}$ at each wavelength is approximately equal to the effective beam solid angle:

$$R_{450} \equiv \frac{\text{FCF}_{\text{peak},450}}{\text{FCF}_{\text{arcsec},450}} \approx 122 \text{ arcsec}^2, \quad (5)$$

$$R_{850} \equiv \frac{\text{FCF}_{\text{peak},850}}{\text{FCF}_{\text{arcsec},850}} \approx 241 \text{ arcsec}^2. \quad (6)$$

Because the beam solid angle scales approximately as λ^2 , the ratio $R_{450}/R_{850} \approx 0.51$ differs substantially from unity. This wavelength dependence is fully accounted for when using $\text{FCF}_{\text{arcsec}}$, which yields a true surface brightness, but it is *not* accounted for when using FCF_{peak} . As a result, any spectral analysis performed on extended emission calibrated with FCF_{peak} is systematically biased.

If FCF_{peak} is incorrectly applied to extended emission, the intensity ratio between 450 and 850 μm is biased by a factor R_{450}/R_{850} . This propagates directly into the total-intensity spectral index $\alpha \equiv \log(I_{450}/I_{850})/\log(\nu_{450}/\nu_{850})$:

$$\alpha_{\text{wrong}} = \alpha_{\text{true}} + \frac{\log(R_{450}/R_{850})}{\log(\nu_{450}/\nu_{850})} \approx \alpha_{\text{true}} - 1.06. \quad (7)$$

Because the Planck-corrected dust opacity index $\beta = \alpha - C(T_d)$ depends on α only through the temperature-dependent Planck shape factor $C(T_d) \equiv \log[B_{\nu_{450}}(T_d)/B_{\nu_{850}}(T_d)]/\log(\nu_{450}/\nu_{850})$ (cf. companion Paper S2 Eq. 9), which is unaffected by the calibration error, the bias propagates identically to β :

$$\beta_{\text{wrong}} = \beta_{\text{true}} - 1.06. \quad (8)$$

Empirical Verification—We verify the predicted bias using the 37 sources listed in the Paper S2 § 3.4 spectral-index table. The measured distributions yield median $\beta_{\text{true}} = 2.23$ and median $\beta_{\text{wrong}} = 1.18$, in excellent agreement with the expected offset $\Delta\beta = -1.06$ (Figure 2). With the full Planck correction adopted here, this shift no longer drives the sample into the formally unphysical $\beta < 0$ regime, but it still produces systematically low β values that would bias the physical interpretation of the dust properties: and one source (3%) falls below $\beta_{\text{wrong}} < 0.5$.

Physical Interpretation—The correct distribution is centered at values broadly consistent with dense molecular-cloud dust, including coagulated grains with thin ice mantles (Ossenkopf & Henning 1994; Juvela et al. 2015; Köhler et al. 2015), although individual high- β tails should be interpreted with caution because $\beta \gtrsim 3$ at uncertainties (see also Paper S2 § 3.4 for further discussion of these high- β sources). In contrast, the biased distribution shifts toward values characteristic of protoplanetary disks ($\beta \lesssim 1$; Testi et al. 2014), highlighting the risk of misinterpreting calibration artifacts as physical grain evolution.

A critical implication of adopting the full Planck correction is that calibration errors no longer produce obviously unphysical values such as $\beta < 0$. Instead, they yield apparently reasonable β distributions that are nonetheless systematically biased. This makes the distinction between FCF_{peak} and $\text{FCF}_{\text{arcsec}}$ essential for reliable interpretation, as the bias may go undetected in the absence of independent calibration cross-checks.

Numerical Consistency—The small (~ 0.01) differences between the median values quoted here and those listed in the Paper S2 § 3.4 spectral-index table arise from rounding the spectral index α to two decimal places prior to computing β in the table. All statistical quantities in this Appendix are computed from full-precision values and are therefore internally consistent.

Summary—We conclude that the use of $\text{FCF}_{\text{arcsec}}$ and mJy arcsec^{−2} units is a physical necessity for extended emission. Incorrect application of FCF_{peak} introduces a deterministic, wavelength-dependent bias that (1) shifts spectral indices by $\Delta\beta = -1.06$, (2) systematically lowers inferred β values, and (3) can lead to physically misleading interpretations even when the results appear reasonable. These findings emphasize that calibration choices must be treated as an integral part of the physical modeling, rather than a purely technical step.

3.1.4. Summary of Technical Validation 1

This section establishes Pillar 1 of the technical credibility chain. All dual-band intensities and dust-emissivity spectral indices entering the companion science analyses (Papers S1 and S2) are calibrated on a physically appropriate surface-brightness footing: the deterministic $\Delta\beta = -1.06$ shift that naive FCF_{peak} application would impose on extended emission is removed by construction, and the residual photometric-calibration uncertainty enters the systematic-uncertainty budget of §4.1 only through the per-band FCF tolerances themselves, which cancel in the dual-band ratio quantities P_{450}/P_{850} and $\gamma - \alpha$. The α , γ , and P_{450}/P_{850} definitions of §3.1.2 are therefore physically meaningful for the molecular-cloud emission targeted by the companion science papers.

3.2. Technical Validation 2: Noise Characteristics and Selection of Polarization Data Based on Stokes I Intensity

3.2.1. Motivation and overview

Having established the photometric calibration framework of §3.1, we next quantify the statistical noise properties that govern which polarization measurements can be used reliably in the dual-band comparison. The purpose of this

subsection and the following one is to construct an observationally validated noise model that links Stokes- I intensity, polarized intensity, and polarization-angle precision, thereby defining the objective selection thresholds adopted in the companion survey papers.

This subsection opens Pillar 2 of the methodology—*statistical legitimacy*—which is presented in two complementary parts (Pillar 2 entry of Figure 1 and Table 1). Here we develop the additive-noise characterization of Stokes I across the full 174-field POL-2 archive; the companion two-component noise model for Stokes Q and U is presented in §3.3. Together the two parts establish that the retained pixels are statistically trustworthy for the downstream dual-band analyses. Figure 3 provides a subsection-level roadmap.

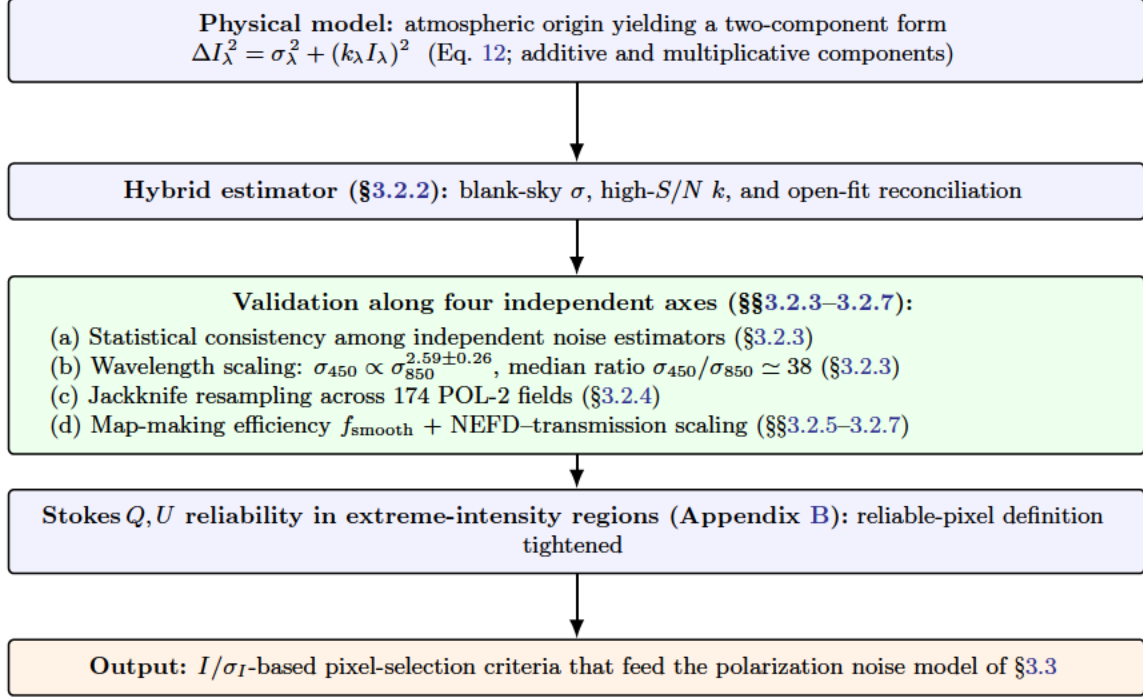


Figure 3. Roadmap for the Stokes I noise characterization (§3.2, Pillar 2a). The atmospheric origin of POL-2 noise leads to a two-component model with additive (σ_λ) and multiplicative ($k_\lambda I_\lambda$) components. A hybrid estimator extracts σ_λ and k_λ from distinct intensity regimes (§3.2.2); the resulting parameters are validated along four independent axes spanning estimator consistency, wavelength scaling, jackknife stability across 174 POL-2 fields, and map-making/atmospheric diagnostics (§§3.2.3–3.2.7). The section closes with an output I/σ_I -based pixel-selection criterion that defines the input to the polarization noise model of §3.3; Stokes Q, U reliability diagnostics for two exceptionally bright 850 μm fields (W51 and DR21) supporting this criterion are presented in Appendix B.

Above a source-dependent threshold, ΔI_λ increases monotonically with I_λ , marking a transition from the atmosphere-dominated noise regime to the signal-dependent noise regime. This behavior originates from small fluctuations in atmospheric transmission along the line of sight. The observed intensity toward a source at a ground-based submillimeter telescope can be expressed as

$$I_{\text{obs}}(t) = I_0 e^{-\tau(t)} + \int_0^s B_\nu(T_{\text{atm}}) e^{-\tau(z,t)} dz, \quad (9)$$

where I_0 is the intrinsic astronomical signal, $\tau(t)$ is the time-variable atmospheric opacity, $B_\nu(T_{\text{atm}})$ is the Planck function at atmospheric temperature, s is the length of the path through the atmosphere, and z is the line-of-sight depth coordinate (with $z = 0$ at the telescope aperture and $z = s$ at the top of the atmosphere). In the optically thin limit ($\tau \ll 1$), the exponential term can be linearized as

$$I_{\text{obs}}(t) \simeq I_0 [1 - \tau(t)] + I_{\text{atm}}(t), \quad (10)$$

where $I_{\text{atm}}(t)$ represents the additive component of atmospheric thermal emission. The dominant residual fluctuations in the POL-2 maps originate not from I_{atm} itself, which is largely removed by common-mode subtraction, but from

imperfect correction of the time-variable transmission term $e^{-\tau(t)}$. After applying an extinction correction using the estimated zenith opacity $\hat{\tau}(t)$, the measured intensity becomes

$$I_{\text{corr}}(t) = I_{\text{obs}}(t) e^{\hat{\tau}(t)} \simeq I_0 [1 + \delta g(t)] + \epsilon(t), \quad (11)$$

where $\delta g(t) \equiv e^{\hat{\tau}(t) - \tau(t)} - 1$ represents residual multiplicative (gain) fluctuations, and $\epsilon(t)$ denotes additive noise from photon statistics, detector readout, and residual sky emission. The corresponding total uncertainty can be written as

$$\Delta I_\lambda^2 = \text{Var}[\epsilon(t)] + \text{Var}[\delta g(t)] I_\lambda^2 = \sigma_\lambda^2 + (k_\lambda I_\lambda)^2, \quad (12)$$

which naturally reproduces the empirical model used throughout this study. The first term, σ_λ , represents the additive noise floor dominated by the atmosphere that yields an approximately constant ΔI_λ at low I_λ , while the second term introduces a linear rise in ΔI_λ with I_λ once the residual gain fluctuation ($k_\lambda I_\lambda$) exceeds σ_λ . Throughout this paper, we use “noise floor” for the asymptotic minimum noise level that remains as integration time and signal averaging increase—distinct from a raw RMS estimate of the data—with the operative source of the floor indicated by an adjective (*additive, instrumental, systematic, empirical, effective, or map-domain*, as appropriate). The transition occurs at

$$I_{\text{trans}} \simeq \frac{\sigma_\lambda}{k_\lambda}, \quad (13)$$

which varies among sources depending on their intrinsic brightness and atmospheric stability. At low intensities, the uncertainty is dominated by additive atmospheric noise (σ_λ), whereas at high intensities it is dominated by multiplicative transmission fluctuations ($k_\lambda I_\lambda$), acting as an effective gain-noise term. The fitted parameters σ_λ and k_λ thus provide quantitative diagnostics of atmospheric stability and calibration precision. The observed consistency of this model across all targets validates its applicability for the noise characterization of POL-2 data.

3.2.2. Implementation of the Hybrid Noise Estimation

Box 2 of the section roadmap (Figure 3) is the hybrid estimator developed in this subsection: it extracts σ_λ and k_λ from distinct intensity regimes through the implementation described below.

The noise estimation procedure implements the two-component model in Eq. (12) through sequential determination of additive and multiplicative noise parameters from distinct intensity regimes, ensuring robust characterization throughout the full dynamic range of POL-2 observations.

First, the additive noise floor σ_λ is determined from the emission-free pixels identified by $S/N < 2$. These low-intensity regions represent the atmosphere-dominated regime where ΔI_λ remains approximately constant. An iterative three-step σ -clipping procedure removes outliers: at each iteration, pixels deviating by more than 3σ from the median ΔI_λ are excluded, and the process repeated until convergence or completion of three iterations. The median of the surviving values ΔI_λ provides a robust estimate σ_{blank} . For fields lacking sufficient emission-free pixels, the 2nd percentile of the full ΔI_λ distribution serves as a conservative lower bound, denoted σ_{medianDI} .

Second, the multiplicative noise coefficient k_λ is extracted from high-intensity pixels where signal-dependent noise dominates. The model in Eq. (12) is fitted to pixels with $S/N > 30$ using Levenberg-Marquardt non-linear least-squares minimization with constraints $\sigma_\lambda > 0$ and $k_\lambda > 0$ to ensure physical validity. The brightest 1% of the intensity distribution is excluded to mitigate potential detector non-linearity or calibration artifacts at extreme intensities. When fewer than 50 high- S/N pixels are available, the selection threshold is lowered adaptively in steps ($S/N > 25$, > 20 , > 15 , > 10) until at least 20 valid pixels are obtained. This adaptive criterion maintains statistical reliability while accommodating faint or compact sources.

Third, an independent verification is performed in the transition regime, where the two noise components contribute comparably. The same functional form is fitted to pixels with signal-to-noise ratios in the range $5 < S/N < 30$ without imposing prior constraints, yielding “open-fit” estimates σ_{open} and k_{open} . This intermediate-intensity fit provides a consistency check between the low- and high-intensity determinations and serves as the primary estimate when blank-sky coverage is insufficient or when the fit residuals deviate from the two-component model.

Fourth, the three independent determinations are reconciled through hierarchical selection logic. The final adopted σ_{final} corresponds to the first physically valid estimate in the priority sequence: σ_{open} , σ_{medianDI} , σ_{blank} . Physical validity requires finite, positive values within empirically established ranges: 0.002–0.1 mJy arcsec^{−2} at 850 μm and 0.05–5 mJy arcsec^{−2} at 450 μm . These bounds, derived from the full BISTRO survey sample, exclude pathological fits while accommodating the natural variation in the observing conditions and the characteristics of the source.

Figures 4 and 5 illustrate typical results of the hybrid noise estimation for two representative bright hybrid-regime sources, Ophiuchus A and Orion A OMC-1, at per-band native (naive-beam) pixel scale ($12''$ for $850\ \mu\text{m}$ and $8''$ for $450\ \mu\text{m}$), with a common ΔI colour scale shared between the two spatial-map panels so that the higher $450\ \mu\text{m}$ noise level is immediately apparent. Together these examples demonstrate that the two-component model captures the qualitative ΔI – I_λ behaviour from blank-sky pixels through high- S/N source pixels; a caveat regarding the conservative- σ design of the advanced fit shown in the scatter panels follows the figures below. All intensity and noise measurements employ consistent flux calibration to enable wavelength-independent comparison. Raw detector units (pW) are converted to surface brightness (mJy arcsec^{-2}) using the post-2018 JCMT calibration scale: $\text{FCF}_{450} = 472 \pm 80$ and $\text{FCF}_{850} = 498 \pm 35\ \text{Jy beam}^{-1}\ \text{pW}^{-1}$ (Mairs et al. 2021), with additional POL-2 attenuation corrections of 1.96 ± 0.023 at $450\ \mu\text{m}$ and 1.35 ± 0.008 at $850\ \mu\text{m}$. For archival data in beam-averaged units, conversion to surface brightness employs solid beam angles $\Omega_{450} = 122$ and $\Omega_{850} = 240\ \text{arcsec}^2$ to maintain internal consistency in heterogeneous preprocessing pipelines.

The noise level adopted at the native pixel scale is scaled to standardized resolutions using $\sigma_{\text{scaled}} = \sigma_{\text{native}} \times \sqrt{A_{\text{target}}/A_{\text{native}}}$, where A denotes the pixel area. This square-root scaling assumes uncorrelated noise, valid in the atmosphere-dominated regime that characterizes the majority of our polarization measurements. The resulting parameters $(\sigma_\lambda, k_\lambda)$ quantitatively characterize the noise behavior of each map, providing the empirical foundation for the subsequent analysis.

Caveat: conservative- σ design of the advanced fit.—The noise-model fit curves shown in panels (a) and (c) of Figures 4 and 5 are intentionally conservative: they include a systematic-uncertainty inflation that places the fit envelope above the empirical blank-sky and final-selected noise floors σ_{blank} and σ_{selected} (horizontal reference lines). As a consequence, the fit curve does not trace the lower envelope of the high- S/N data points in the bright source centres; this offset is by design and is independent of source. Furthermore, panels (a) and (c) reveal that the ΔI_λ distribution at low I_λ is intrinsically heterogeneous: a tight low- ΔI cluster from central source pixels coexists with a sparser high- ΔI cluster from map-edge and low-coverage pixels. The single-pair noise model $\Delta I_\lambda = \sqrt{\sigma_\lambda^2 + (k_\lambda I_\lambda)^2}$ cannot resolve both populations, and the robust fit on the full positive- I catalogue therefore converges to a σ_λ value that reflects the median of this bimodal ΔI distribution rather than either floor in isolation. We retain the resulting $(\sigma_\lambda, k_\lambda)$ as the canonical per-field noise pair for cross-paper consistency, and treat σ_{blank} and σ_{selected} (shown as horizontal reference lines) as the relevant low- I baselines for the blank-sky and high-quality pixel populations, respectively. Sources in other noise regimes—additive-dominated filaments such as Taurus B211/B213, or mosaic-processing-limited fields such as Serpens Main—exhibit the same conservative- σ offset under the uniform noise-regime classification, and the per-field $(\sigma_\lambda, k_\lambda)$ pair propagates without modification into the Stokes- Q, U noise analysis of §3.3 and the survey-wide statistics of §3.2.3. Reliable image-noise estimation for ground-based transition-edge-sensor (TES) bolometer polarimetric maps is intrinsically difficult, and we therefore adopt broad consistency with the official JCMT SCUBA-2/POL-2 sensitivity calculator at the level of the independently inferred map-making/smoothing degradation factors as the practical figure of merit for this method (Appendix C).

3.2.3. Statistical Validation and Wavelength-Dependent Characteristics

Within box 3 of the section roadmap (Figure 3), this subsection delivers validation axes (a) statistical consistency among independent noise estimators and (b) wavelength scaling $\sigma_{450} \propto \sigma_{850}^{2.59 \pm 0.26}$.

The hybrid noise estimation procedure described in §3.2.2 yields multiple independent determinations of the additive noise floor σ_λ for each field. To validate the internal consistency and robustness of these estimates, we perform statistical analysis across the full sample of 174 POL-2 fields.

Consistency of Independent Noise Estimators—Figure 6 presents the statistical distribution and inter-relationships among the four noise estimators: σ_{blank} (from emission-free pixels), σ_{mode} (from the modal value after iterative clipping), median ΔI (from the 2nd percentile), and σ_{final} (the adopted consensus estimate). Panels (a) and (b) show stacked histograms at $850\ \mu\text{m}$ and $450\ \mu\text{m}$ respectively. All estimators converge within narrow ranges: at $850\ \mu\text{m}$ the distributions cluster between ~ 0.005 and $0.1\ \text{mJy arcsec}^{-2}$, while at $450\ \mu\text{m}$ the corresponding range is ~ 0.1 to $3\ \text{mJy arcsec}^{-2}$. The close alignment of the four independent estimates demonstrates that the additive noise floor is a well-defined quantity that can be reliably extracted despite the various observing conditions and source morphologies.

Panels (c) and (d) provide direct one-to-one comparisons between σ_{final} and the three independent estimators. Nearly all points lie along the diagonal unity line, confirming that the hierarchical selection logic successfully converges to a

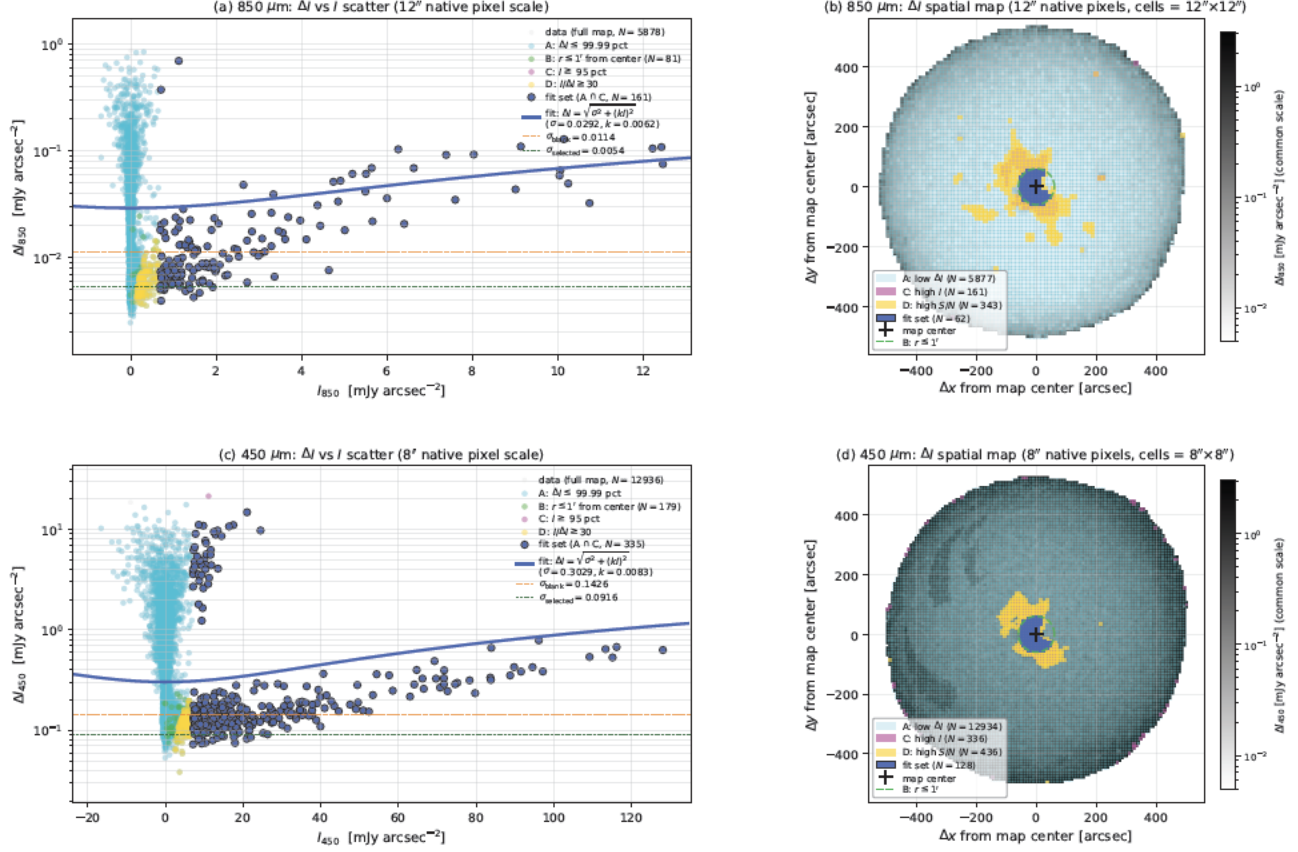
Ophiuchus A: Stokes- I noise behavior at per-band native pixel scale (850 @ 12", 450 @ 8"; naive beam, no beam convolution)

Figure 4. Representative hybrid noise analysis for Ophiuchus A at per-band native pixel scale (naive beam; no beam convolution). (a) Stokes- I noise scatter ΔI_{850} versus I_{850} at the 12" native pixel scale. (b) Stokes- I noise spatial map at 850 μm ; each rendered cell is 12" $\times$ 12" (one 850 μm native pixel). (c) Same as (a) but for 450 μm at the 8" native pixel scale. (d) Same as (b) but for 450 μm at the 8" native pixel scale; each rendered cell is 8" $\times$ 8" (one 450 μm native pixel). The ΔI colour scale of panels (b) and (d) is shared so that the higher 450 μm noise level is immediately apparent. In the scatter panels, gray points show all finite catalogue pixels of the full map (including negative- I noise pixels and the bright-source tail); subsets highlighted by colour correspond to the noise-component classification of §3.2.2: cyan for low- ΔI pixels (Condition A; $\Delta I \leq 99.99$ percentile), line for centre-region pixels (Condition B; $r \leq 1'$ from map centre), magenta for high- I pixels (Condition C; $I \geq 95$ percentile), and gold for high- S/N pixels (Condition D; $I/\Delta I \geq 30$). Royal-blue points show the fit-set visualisation (intersection $A \cap C$; high- I pixels with ΔI within the 99.99 percentile), and the heavy royal-blue curve shows the best-fit two-component noise model $\Delta I_{\lambda} = \sqrt{\sigma_{\lambda}^2 + (k_{\lambda} I_{\lambda})^2}$, fitted via a robust Huber loss on the full positive- I catalogue (Eq. 12); the curve's relation to the bimodal ΔI data structure is discussed in the caveat paragraph following Figure 5. Horizontal dashed and dotted lines mark the empirical blank-sky floor σ_{blank} and the hierarchically-selected final noise σ_{selected} , respectively. In the spatial-map panels (b) and (d), Condition pixels are overlaid as coloured cells matching each band's native pixel size (12" $\times$ 12" at 850 μm , 8" $\times$ 8" at 450 μm) on the grayscale ΔI heatmap; the dashed green circle marks the Condition B boundary ($r \leq 1'$) and the cross marks the map centre. Cross-reference to the polarization-vector atlas: [Paper S2 Figure 27](#).

consistent representation of the underlying additive noise component. The tight correlation extends over more than three decades in noise amplitude at each wavelength, validating both the stability of the fitting procedure and the physical consistency of the two-component noise model.

As an additional validation, we compared the hybrid noise estimates with the noise levels derived directly from the variance maps produced by the standard `pol2map` pipeline. For each field, the median value of $\sqrt{\text{Var}(I)}$ over emission-free regions was computed from the pipeline-generated variance extension. The resulting estimates agree with σ_{final} to within 15% for more than 90% of the fields, confirming that the hybrid estimation procedure recovers

Orion A OMC-1: Stokes- I noise behavior at per-band native pixel scale (850 @ 12", 450 @ 8"; naive beam, no beam convolution)

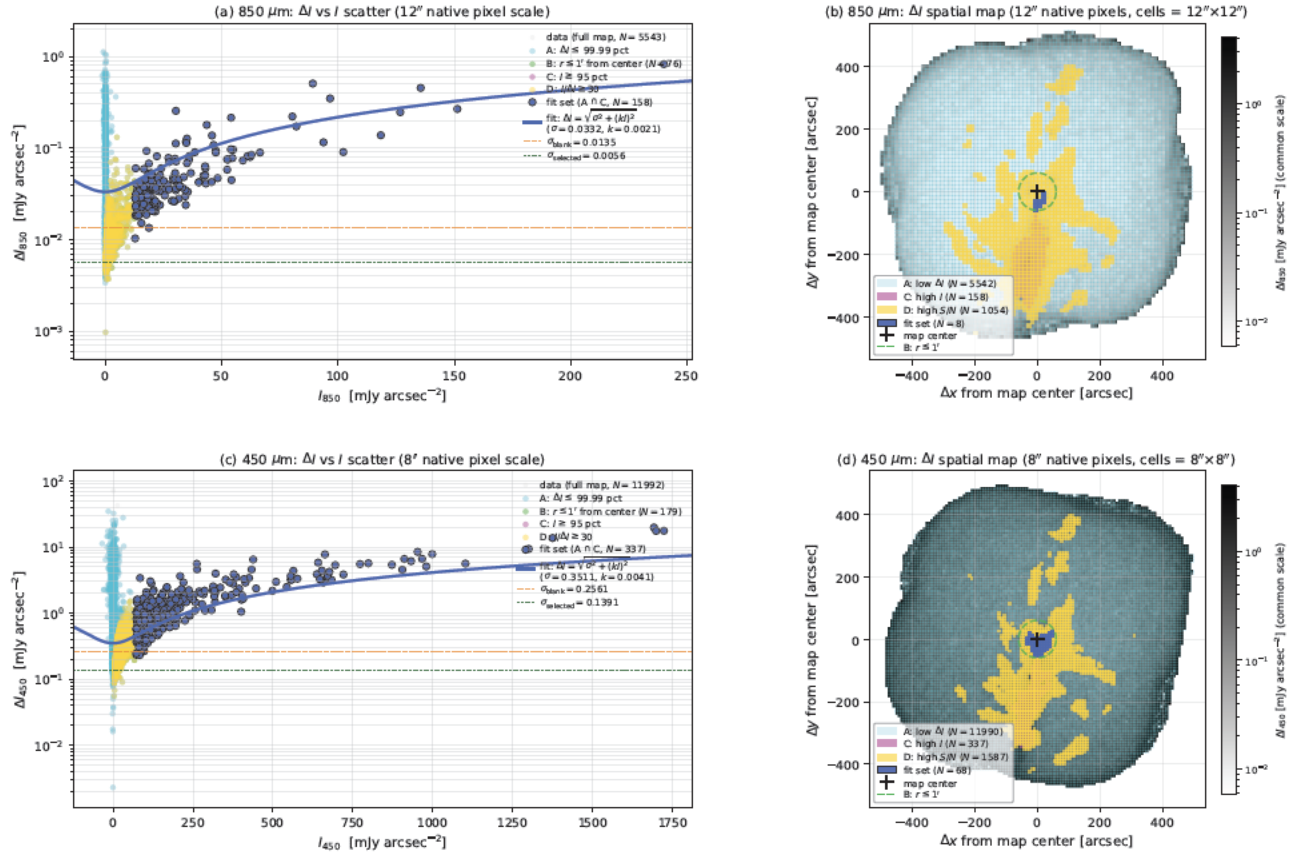


Figure 5. Same as Figure 4 but for the high-mass star-forming Orion A OMC-1 region. The brighter central source produces a substantially larger high- S/N pixel population (Condition D, gold cells in panels b, d) extending over a wider spatial area than in Ophiuchus A, while the order of σ values at both wavelengths remains comparable across the two sources. Cross-reference to the polarization-vector atlas: [Paper S2 Figure 40](#).

noise characteristics consistent with the standard pipeline products while providing the additional diagnostic capability of separating the additive and multiplicative components.

Wavelength Dependence of Additive Noise—Figure 7 examines the correlation between additive noise levels at 450 μm and 850 μm in 174 POL-2 fields. Each point represents one mapped region, with uncertainties estimated from lognormal fits in $\log_{10} \sigma$ space. The orthogonal distance regression (ODR) fit in log-log space yields:

$$\log_{10} \sigma_{450} = (2.59 \pm 0.26) \log_{10} \sigma_{850} + (4.84 \pm 0.53), \quad (14)$$

corresponding to $\sigma_{450} = (6.96 \times 10^4) \sigma_{850}^{2.59}$. The exponent $m = 2.59 \pm 0.26$ significantly exceeds unity, indicating that 450 μm noise increases more rapidly than linear scaling with σ_{850} . This superlinear dependence reflects the combined influence of wavelength-dependent atmospheric transmission fluctuations, detector characteristics, and beam solid angle differences.

The median ratio $\sigma_{450}/\sigma_{850} = 38$ (interquartile range 21.6–60.2) demonstrates that 450 μm maps are typically an order of magnitude noisier in surface brightness than the corresponding 850 μm maps. Fields observed under optimal conditions cluster toward ratios of 20–30, while those obtained during marginal conditions exhibit ratios exceeding 50–60. This broad distribution underscores the strong sensitivity of 450 μm observations to the atmospheric state.

Noise Scaling with Integration Time—The dependence of the final noise floor on the total exposure time is shown in Figure 8. Each point represents the final additive noise estimate adopted σ_{final} for an individual field, plotted

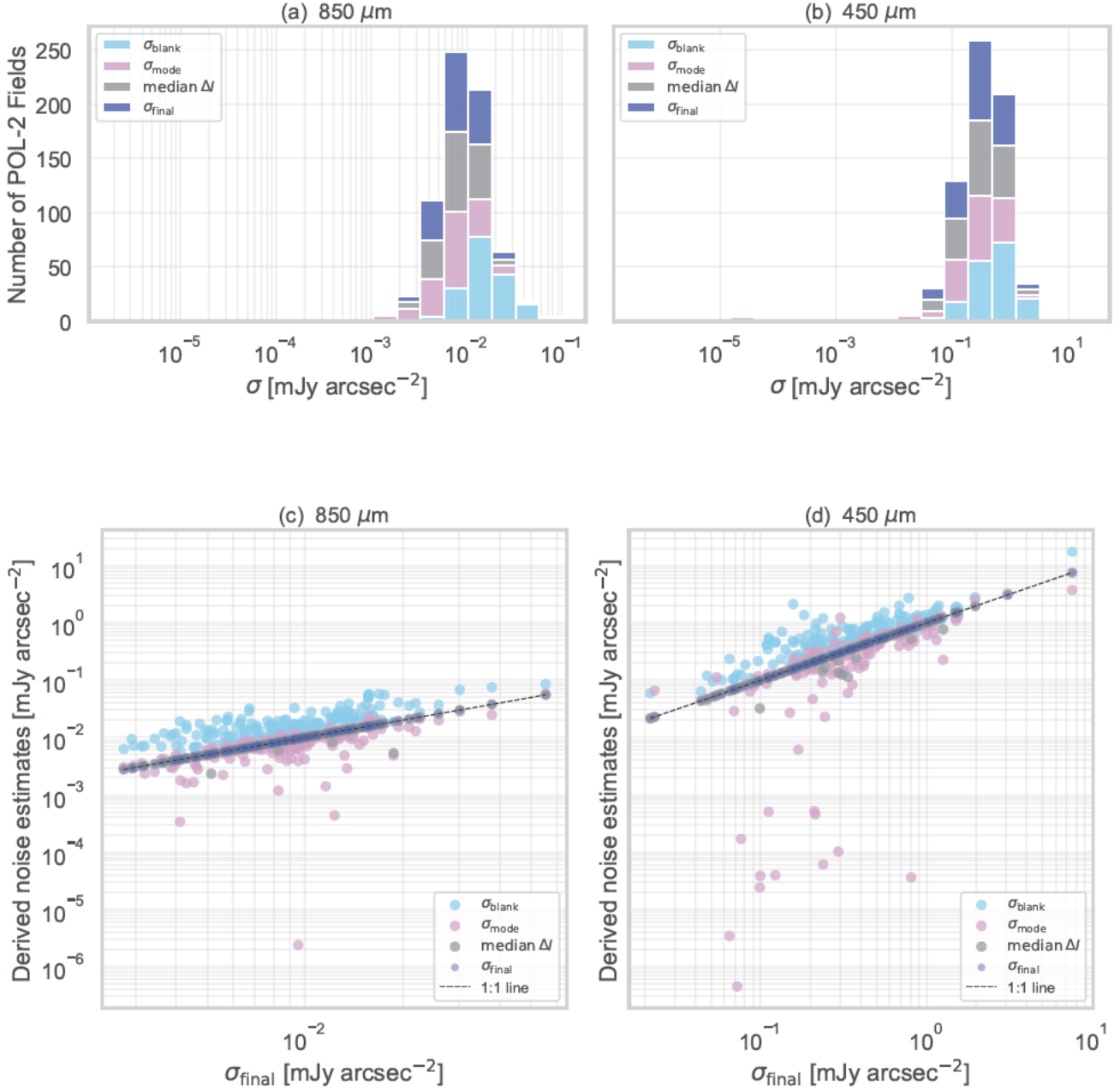


Figure 6. Statistical summary of additive noise estimates derived from 174 POL-2 fields. (a,b) Stacked histograms of additive noise estimators σ_{blank} , σ_{mode} , median ΔI , and the final adopted σ_{final} at 850 μm and 450 μm , respectively. All estimators converge within narrow ranges, indicating a consistent characterization of the additive noise floor across the sample. (c,d) Comparison of derived noise estimates with adopted σ_{final} for the same two bands. The tight alignment of points along the one-to-one line demonstrates that all independent estimators trace the same underlying additive noise component, validating the self-consistency and robustness of the hybrid noise estimation described in §3.2.2.

against the total integration time at the source at the corresponding wavelength. Solid lines indicate the best-fit power-law relations of the form $\Delta I_{\lambda} = a_{\lambda} t^{b_{\lambda}}$, which yield $\Delta I_{450} = (13 \pm 15) t^{-0.34 \pm 0.12}$ mJy arcsec $^{-2}$ and $\Delta I_{850} = (0.08 \pm 0.05) t^{-0.20 \pm 0.06}$ mJy arcsec $^{-2}$. Both slopes are shallower than the $t^{-0.5}$ dependence expected for purely uncorrelated (white) noise, implying that residual atmospheric and instrumental fluctuations do not average down

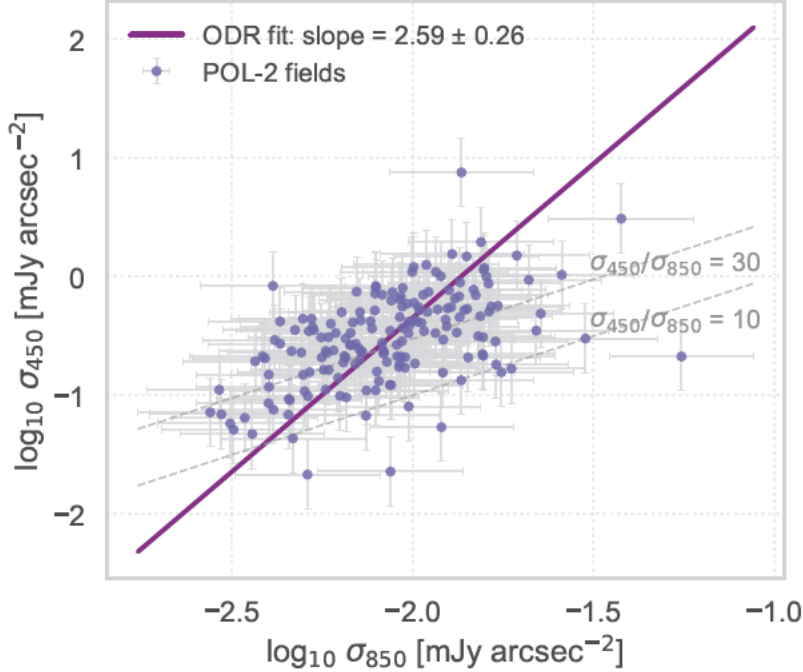


Figure 7. Correlation between the additive noise levels derived at 450 μm and 850 μm across the 174 POL-2 fields. Each point represents one field, with 1σ uncertainties estimated from the lognormal fits in $\log_{10} \sigma$ space. The magenta line shows the orthogonal distance regression (ODR) fit in log-log space, yielding a slope of $m = 2.59 \pm 0.26$ and an intercept of $b = 4.84 \pm 0.53$. This corresponds to the empirical relation $\sigma_{450} = 6.96 \times 10^4 \sigma_{850}^{2.59}$, indicating that the 450 μm noise increases more steeply than a linear scaling with σ_{850} . Dashed gray lines mark fixed $\sigma_{450}/\sigma_{850}$ ratios of 10 and 30, illustrating the typical dynamic range. The median ratio across the sample is $\sigma_{450}/\sigma_{850} = 38$, with an interquartile range of 21.6-60.2. These results reflect the stronger impact of atmospheric transmission fluctuations and detector noise at shorter wavelengths, consistent with the expected behavior of ground-based submillimeter observations.

as efficiently with time. The dashed curves show the ideal white-noise limits normalized to intersect the empirical fits near the median integration time. At both wavelengths, the observed scatter about the best-fit relations spans approximately a factor of 5–10, reflecting heterogeneous observing conditions accumulated over multiple nights and weather grades. The weaker integration-time dependence of the 450 μm noise further supports its stronger sensitivity to correlated atmospheric fluctuations.

The best-fit power-law indices $b_{450} = -0.34 \pm 0.12$ and $b_{850} = -0.20 \pm 0.06$ are significantly shallower than the ideal radiometer prediction $b_{\text{ideal}} = -0.5$ for uncorrelated white noise (shown as dashed lines). This departure demonstrates that achieved map sensitivity is not governed solely by photon statistics but rather by residual correlated atmospheric fluctuations and systematic uncertainties that persist after standard data reduction.

At 450 μm the shallow slope reflects the combination of imperfect atmospheric correction and time-variable transmission fluctuations. Even at 850 μm where atmospheric stability is substantially better, the scaling remains noticeably shallower than photon-noise expectation, indicating that systematic effects dominate the noise budget at integration times exceeding $\sim 10^4$ s. The factor-of-5–10 scatter about the fitted relations arises from field-to-field variations in atmospheric opacity, telescope elevation, and the number of nights contributing to each mosaic.

These shallow exponents imply diminishing returns: increasing the integration time by an order of magnitude reduces the noise by only a factor of ~ 2 at 450 μm and ~ 1.5 at 850 μm , rather than the factor of ~ 3.2 expected for ideal $t^{-0.5}$ scaling. This quantifies the practical limitations imposed by residual atmospheric noise on ground-based submillimeter polarimetry and provides realistic expectations for survey sensitivity forecasts.

An external comparison with the official JCMT SCUBA-2/POL-2 sensitivity calculator is presented in Appendix C. When placed on the same mJy arcsec^{-2} surface-brightness scale, the ITC values provide the expected proposal-level baseline, while the empirical BISTRO science-map noise is higher by factors of $\mathcal{R}_{\text{ITC}} \simeq 7$ at 850 μm and $\simeq 22\text{--}43$ at 450 μm , of the same broad scale as the independently inferred map-making/smoothing degradation factors (Figure 9).

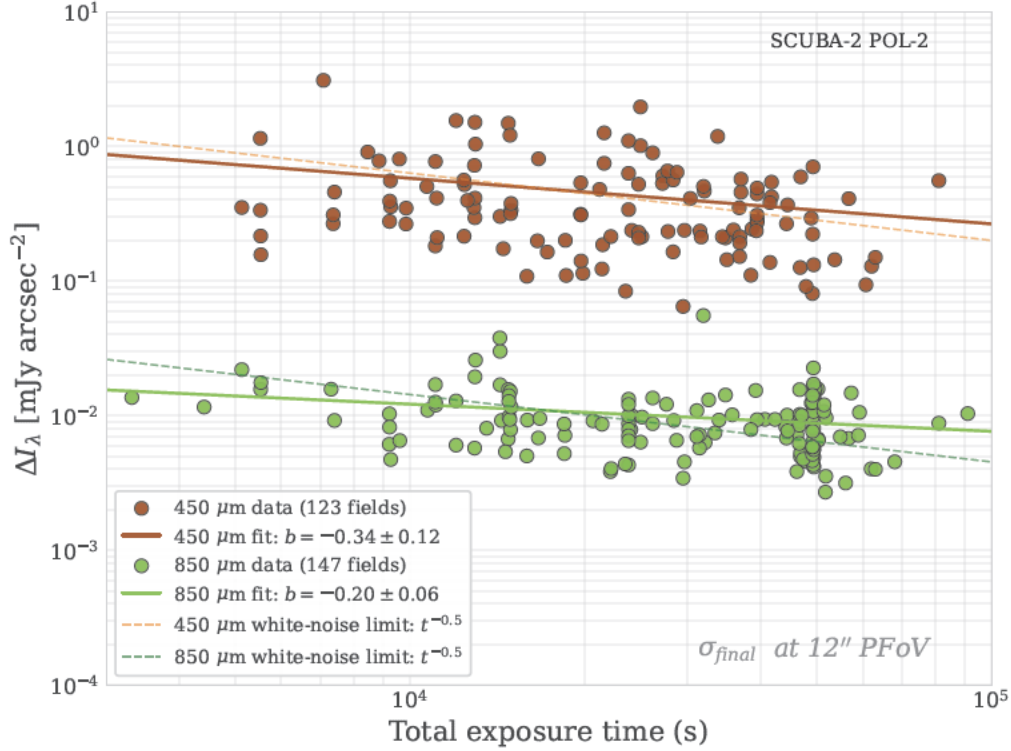


Figure 8. Dependence of the additive noise level ΔI_λ on the total exposure time for the POL-2 observations at 450 μm (sienna) and 850 μm (chartreuse). Each point represents a single mapped field, and solid lines show the best-fit power-law relations $\Delta I_\lambda = at^b$, yielding slopes of $b = -0.34 \pm 0.12$ for 450 μm and $b = -0.20 \pm 0.06$ for 850 μm , based on 123 and 147 fields, respectively. The number of fields analyzed for NEFD characterization exceeds the number of star-forming regions in the main scientific sample because the noise analysis incorporates all POL-2 observations available in the JCMT archive. This dataset includes not only star-forming cloud surveys, but also observations of T Tauri stars, late-type stars, supernova remnants, galaxies, and quasi-stellar objects obtained during the same instrument era. The plotted ΔI_λ values correspond to the final adopted additive noise estimates σ_{final} introduced in Figure 6. Dashed lines indicate the expected white-noise limits ($\propto t^{-0.5}$), normalized to intersect the empirical fits near the median exposure time. Both wavelengths exhibit shallower slopes than the photon-noise prediction, demonstrating that the map sensitivity is not limited by integration statistics but rather by residual atmospheric fluctuations and calibration uncertainties. All values correspond to noise levels evaluated at the 12'' pixel field-of-view (PFoV) scale.

3.2.4. Jackknife Validation of the Additive Noise Model

Within box 3 of the section roadmap (Figure 3), this subsection delivers validation axis (c): jackknife resampling across 174 POL-2 fields.

To assess whether the hybrid noise estimates accurately capture the intrinsic instrumental and atmospheric noise floor, we performed a jackknife analysis based on the individual **Step 1** maps produced by the `pol2map.py` pipeline. Each observing block (MSB), with a typical integration time of ~ 40 minutes, was treated as an independent realization of the noise. The jackknife test evaluates the reproducibility of the measured surface-brightness noise by comparing two independent subsets of the data, thereby providing a direct measure of the per-observation noise level prior to any coaddition (Ward-Thompson et al. 2017).

For each target, all individual observation files were sorted chronologically and divided into two non-overlapping groups (A and B) by alternating assignment: files with even indices were placed in Group A and those with odd indices in Group B. When fewer than four files were available, the sequence was divided into two halves of nearly equal size. Independent coadded catalogs were then constructed for each group:

$$M_A = S + N_A, \quad M_B = S + N_B, \quad (15)$$

where S represents the astronomical signal and N_A, N_B denote independent noise realizations. The difference between the two maps removes the coherent sky emission while preserving the stochastic noise component:

$$D = M_A - M_B = N_A - N_B. \quad (16)$$

Spatial matching between the two datasets was performed using sky coordinates with a positional tolerance of $3''0$, consistent with the astrometric precision of the SCUBA-2 maps (e.g., Chapin et al. 2013; Dempsey et al. 2013). For each matched pair of pixels, the jackknife noise was then estimated as

$$\sigma_{\text{Jack}} = \frac{1}{\sqrt{2}} \text{std}(D), \quad (17)$$

where $\text{std}(D)$ is the standard deviation of the pixel-by-pixel intensity differences. All observations were treated with equal statistical weight, without coverage or S/N weighting, and flux calibration factors, as well as beam-to-surface-brightness conversions, were applied identically to both subsets, following the procedures described in §3.2.2.

The analysis was performed over the entire mapped field for each target, including the outer regions where coverage is lower and noise is higher. This full-field approach provides conservative but representative estimates of the per-observation noise and allows direct comparison between targets, even for extended sources that fill the field of view (e.g., DR21, Galactic Center).

Across the 174 analyzed fields, typical jackknife noise levels range from $\sigma_{\text{Jack}}(I) \sim 10\text{--}100$ mJy arcsec $^{-2}$ per observation at 850 μm and $\sim 100\text{--}1000$ mJy arcsec $^{-2}$ at 450 μm , depending on atmospheric opacity, integration time per file, and source brightness. For example, Ophiuchus A at 850 μm exhibits $\sigma_{\text{Jack}}(I) \simeq 47.6$ mJy arcsec $^{-2}$ per individual MSB. With $N_{\text{obs}} = 20$ independent observations, the expected coadded noise scales as

$$\sigma_{\text{coadd}} \approx \frac{47.6}{\sqrt{20}} \simeq 10.7 \text{ mJy arcsec}^{-2}, \quad (18)$$

consistent with the hybrid noise estimates (Figure 6). The per-observation jackknife noise is typically one to two orders of magnitude larger than the final map noise measured in the coadded, spatially filtered product (e.g., $\sigma_{\text{Jack}} \sim 50$ versus $\sigma_{\text{final}} \sim 0.01$ mJy arcsec $^{-2}$ at 850 μm). This difference arises from the cumulative effects of temporal averaging over $N_{\text{obs}} \sim 10\text{--}50$ observations, spatial filtering that suppresses large-scale atmospheric emission, and common-mode subtraction that removes correlated sky fluctuations. The factor-of- $\sqrt{N_{\text{obs}}}$ improvement expected from ideal averaging is partially offset by correlated atmospheric noise, consistent with the shallow integration-time scaling observed in Figure 8.

The jackknife analysis thus provides an independent validation of the additive-noise model. It confirms that the hybrid estimation procedure correctly characterizes the achievable noise floor after temporal coaddition and spatial filtering. All results are summarized in `jackknife_summary.csv`, which lists, for each field and wavelength, the number of contributing observations, the number of matched pixels, and the derived σ_I , σ_Q , and σ_U values. The following subsection introduces the f_{smooth} metric, which quantifies how efficiently the map-making pipeline converts these intrinsic per-scan noise levels into the final coadded RMS values.

3.2.5. Quantifying Map-Making Efficiency: The f_{smooth} Metric

Within box 3 of the section roadmap (Figure 3), this subsection opens validation axis (d): the f_{smooth} map-making efficiency metric.

To quantify the relative efficiency of the map-making process across the POL-2 imaging, including BISTRO survey, we define the dimensionless “smoothing factor”

$$f_{\text{smooth}} = \frac{\sigma_{\text{jack}}}{\sqrt{N_{\text{MSB}}} \sigma_{\text{blank}}}, \quad (19)$$

which represents the ratio between the noise expected from the per-observation jackknife analysis (§3.2.4) and the residual noise level measured in the final coadded map. In this form, f_{smooth} effectively measures how efficiently the `pol2map` pipeline converts raw, sky-limited time streams into spatially filtered science-quality maps. Values near unity would imply purely statistical noise averaging, whereas larger values indicate additional suppression of correlated noise through common-mode subtraction, high-pass filtering, and beam convolution.

Figure 9 shows the distributions of f_{smooth} for 156 fields (81 at 850 μm , 75 at 450 μm) where the successful jackknife analysis yielded valid per-observation noise estimates. The remaining 18 fields either lacked sufficient independent observations for robust jackknife resampling or exhibited spatial matching ambiguities that precluded a reliable determination of σ_{jack} . The shaded gray band ($5 \leq f_{\text{smooth}} \leq 50$) marks the empirically defined range corresponding

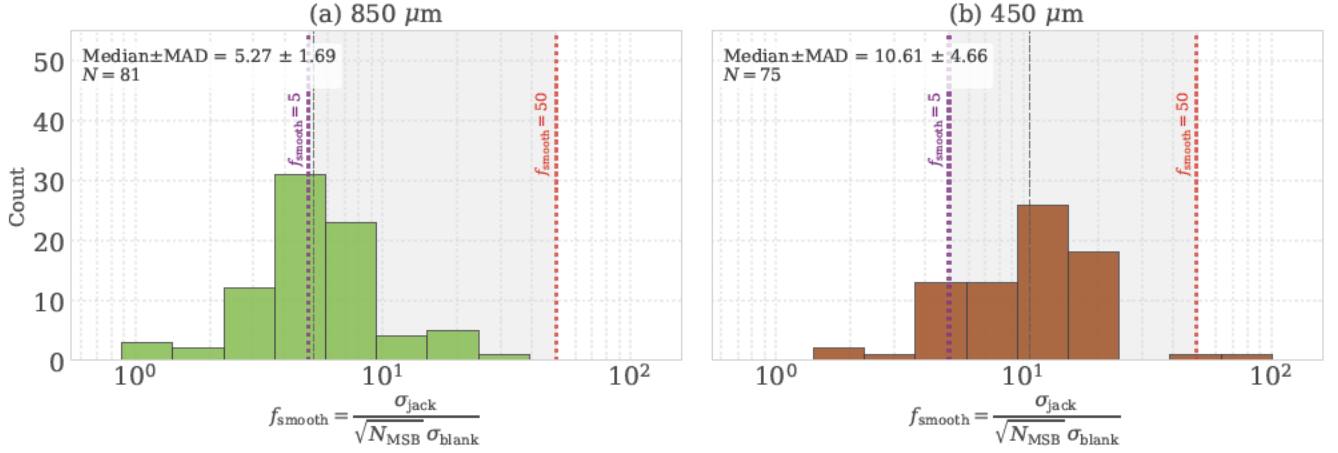


Figure 9. Distributions of the map-making efficiency factor f_{smooth} for (a) 81 fields at $850\ \mu\text{m}$ and (b) 75 fields at $450\ \mu\text{m}$ where successful jackknife analysis yielded valid per-observation noise estimates. The shaded gray region ($5 \leq f_{\text{smooth}} \leq 50$) marks the range considered to represent normal map-making performance, where the final map RMS (σ_{blank}) is consistent with expectations from the jackknife-derived per-observation noise and the number of coadded MSBs. Values below this range ($f_{\text{smooth}} < 5$) indicate overestimated blank-sky RMS levels, while the single outlier above ($f_{\text{smooth}} > 50$; Lynds 43 at $450\ \mu\text{m}$) suggests underestimated noise or incomplete spatial matching in the jackknife analysis. For $850\ \mu\text{m}$, the distribution exhibits median $\pm$ MAD = 5.3 ± 1.7 , with 43% of fields below the nominal range due to the difficulty of masking faint extended emission at longer wavelengths. The $450\ \mu\text{m}$ data show a broader, higher distribution (10.6 ± 4.7), with 81% in the nominal range, indicating less efficient atmospheric-noise suppression at the shorter wavelength. The two distributions are statistically distinct (KS test: $D = 0.56$, $p = 8.6 \times 10^{-12}$).

to “normal” map-making performance, where the final map RMS (σ_{blank}) is consistent with expectations based on the jackknife-derived per-MSB noise and the number of coadded observations. Values below this range ($f_{\text{smooth}} < 5$) imply that the RMS in the blank-sky was overestimated, resulting in overly conservative contour thresholds, while values above ($f_{\text{smooth}} > 50$) indicate underestimated noise or excessive filtering that suppresses true emission. These boundaries were chosen empirically, on the basis of the observed distributions, and have no fundamental physical significance.

At $850\ \mu\text{m}$, the f_{smooth} values exhibit median $\pm$ MAD = 5.3 ± 1.7 , with 57% of the fields in the nominal range and 43% below ($f_{\text{smooth}} < 5$). The large fraction of low- f_{smooth} fields at $850\ \mu\text{m}$ reflects the difficulty of identifying truly emission-free regions at longer wavelengths, where faint extended structures (e.g., cirrus, diffuse molecular envelopes) are more prevalent and challenging to mask, leading to systematically conservative (elevated) σ_{blank} estimates. At $450\ \mu\text{m}$, the distribution is broader and shifted toward higher values (10.6 ± 4.7), with 81% of fields in the nominal range, reflecting the more variable sky background and reduced transmission at shorter wavelengths. A two-sample Kolmogorov-Smirnov test comparing the two distributions yields a statistic of $D = 0.56$ and $p = 8.6 \times 10^{-12}$, confirming that the $450\ \mu\text{m}$ and $850\ \mu\text{m}$ populations are statistically distinct. This difference arises naturally from the higher and more variable sky emission at $450\ \mu\text{m}$, where the SCUBA-2 TES bolometers operate in a more sky-limited regime.

3.2.6. Practical Application of the f_{smooth} Metric

Continuing within box 3 of the section roadmap (Figure 3), this subsection develops the practical interpretation rules for the f_{smooth} metric introduced in §3.2.5 (validation axis (d) of the section roadmap).

For practical visualization and contour definition in the Stokes I maps, we adopt an empirically-based interpretation scheme for f_{smooth} .

Case 1: Overestimated Blank-Sky RMS ($f_{\text{smooth}} < 5$)—For fields with $f_{\text{smooth}} < 5$, the blank-sky RMS is likely overestimated due to residual faint emission in the putatively blank regions. This situation affects 43% of our $850\ \mu\text{m}$ fields, where diffuse extended structures are particularly challenging to mask completely. For these fields, contour thresholds can be relaxed using a scaling factor $\sigma_{\text{effective}} = \sigma_{\text{blank}} \times \sqrt{f_{\text{smooth}}/5}$.

Case 2: Nominal Map-Making Performance ($5 \leq f_{\text{smooth}} \leq 50$)—The fields within this range exhibit map-making performance consistent with expectations from jackknife analysis and represent the majority of our sample (57% at $850\ \mu\text{m}$,

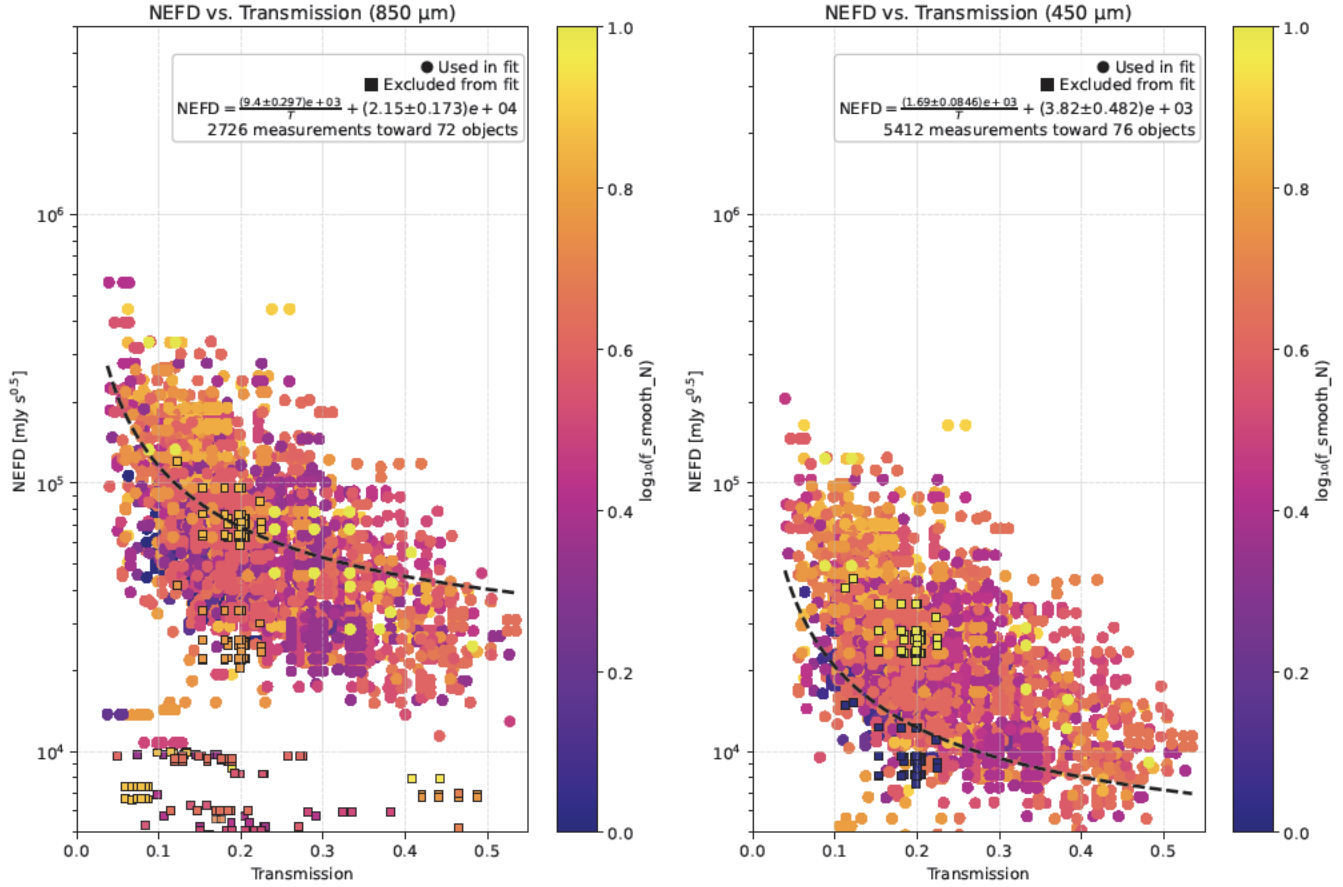


Figure 10. Noise Equivalent Flux Density (NEFD) as a function of atmospheric transmission at 666 GHz for (left) 850 μm and (right) 450 μm POL-2 observations. Each point represents one MSB-level measurement, with filled circles showing data used in the fit and open squares marking excluded outliers (transmission < 0.02 or $\text{NEFD} < 10^4 \text{ mJy s}^{0.5}$ at 850 μm). Points are color-coded by $\log_{10}(f_{\text{smooth}})$ (Section 3.2.5), showing no correlation between map-making efficiency and atmospheric conditions. The dashed black curves indicate the best-fit relation $\text{NEFD} = A/T + B$ (Eq. 20), with parameters from Table 3. The inverse transmission scaling and convergence toward a stable noise floor at high T validate the two-component noise model and confirm that SCUBA-2 performance is photon-noise-limited under good weather conditions. Comparison with SCUBA (Archibald et al. 2002) reveals improved stability at 450 μm (30% reduction in A) and consistent instrument noise floors (B) across detector generations.

81% at 450 μm). For these fields, standard contour levels (3, 5, 10 σ , etc.) based directly on σ_{blank} are appropriate without modification.

Case 3: Underestimated Blank-Sky RMS ($f_{\text{smooth}} > 50$).—Only one field in our sample (Lynds 43 at 450 μm) falls into this category, where the blank-sky RMS appears underestimated or the jackknife analysis encounters spatial matching difficulties. For such fields, more conservative contours are warranted, with $\sigma_{\text{effective}} \approx \sigma_{\text{blank}} \times \sqrt{f_{\text{smooth}}/10}$.

Summary—These scaling relationships are empirical and serve primarily as guidelines for optimizing visualization rather than reflecting fundamental noise properties. The f_{smooth} metric thus provides a quantitative measure of map-making efficiency and a uniform diagnostic of noise calibration consistency across the multi-year POL-2 data set. The statistical properties of these distributions reflect the inherent challenges of submillimeter polarimetric imaging: atmospheric variability at 450 μm produces a broader range of f_{smooth} values, while the difficulty of identifying emission-free regions at 850 μm systematically biases the blank-sky noise estimates upward. The wavelength dependence of the f_{smooth} distributions quantifies these systematic effects and provides a framework for interpreting the noise characteristics in all survey fields.

3.2.7. Validation with Observational Conditions: NEFD–Transmission Scaling

This subsection completes box 3 of the section roadmap (Figure 3) — validation axis (d), the f_{smooth} and NEFD–transmission tests (§§3.2.5–3.2.7) — with the NEFD–transmission scaling test linking per-observation sensitivity to instantaneous sky opacity.

As a final validation of the noise characterization framework developed in the preceding subsections, we examine the relationship between the Noise Equivalent Flux Density (NEFD) measured for individual MSBs and the atmospheric transmission at 666 GHz during those observations. This analysis provides an independent, physically grounded test of the additive and multiplicative noise models by directly linking per-observation sensitivity to the instantaneous sky opacity — a fundamental parameter governing ground-based submillimeter performance.

Physical Motivation and Empirical Model—The NEFD quantifies the instrumental sensitivity required to detect a fluctuation of the flux density of 1σ in one second of integration. For ground-based submillimeter telescopes, NEFD is expected to vary inversely with atmospheric transmission $T = e^{-\tau_\lambda \cdot \text{AM}}$, where τ_λ is the zenith opacity at wavelength λ and AM is the airmass. In the low-opacity regime typical of Mauna Kea ($T \gtrsim 0.2$), the dominant noise sources include: (1) *sky photon noise*, proportional to $\sqrt{I_{\text{sky}}}$, where I_{sky} increases with atmospheric emission (i.e., decreases with T); and (2) *instrument readout noise*, arising from TES bolometer electronics, cryogenic amplifiers, and digitization. These considerations motivate a simple two-parameter model:

$$\text{NEFD}(T) = \frac{A}{T} + B, \quad (20)$$

where A represents the sky-photon-limited component (sensitivity to atmospheric opacity) and B the instrumental noise floor. This form has been found empirically robust across submillimeter facilities (Archibald et al. 2002; Dempsey et al. 2013), offering a physically interpretable parameterization without requiring detailed radiative transfer modeling.

Data and Methodology—The NEFD estimates for each MSB were derived by combining the results of the hybrid noise analysis (§3.2.2) with the observation metadata of JCMT. For each MSB, the noise level σ_{final} in unit of mJy arcsec^{−2} was converted to NEFD in mJy s^{1/2} as

$$\text{NEFD} = \sigma_{\text{final}} \Omega_{\text{beam}} \sqrt{t_{\text{int}}}, \quad (21)$$

where Ω_{beam} is the beam solid angle (83 arcsec² at 450 μm and 157 arcsec² at 850 μm), and t_{int} is the total on-source integration time per MSB (typically 1800 s).

Atmospheric transmission at 666 GHz was computed by scaling using the 225 GHz optical depth (τ_{225}) measured by the CSO radiometer, using the empirical relation (Dempsey et al. 2013):

$$\tau_{666} = 26.0 (\tau_{225} - 0.012), \quad (22)$$

and corrected for airmass via $T = \exp(-\tau_{666} \cdot \text{AM})$. The merged dataset comprises 5412 MSB-level measurements toward 76 fields at 450 μm and 2726 measurements toward 72 fields at 850 μm , spanning $0.05 \lesssim T \lesssim 0.5$. MSBs with extremely poor weather ($T < 0.02$) or, for the 850 μm sample, unphysically low NEFD values ($< 10^4$ mJy s^{1/2}) were excluded from the fitting. Parameter estimation employed Markov Chain Monte Carlo (MCMC) sampling with `emcee` (Foreman-Mackey et al. 2019), using 50 walkers, 2000 iterations, and a 500-step burn-in. This Bayesian approach yields robust posteriors for A and B even under heteroscedastic scatter caused by variable atmospheric conditions.

Results—Figure 10 presents the NEFD– T scaling at both wavelengths, color-coded by $\log_{10}(f_{\text{smooth}})$ to highlight the independence between map-making efficiency and atmospheric transmission. The best-fit parameters obtained from the MCMC sampling are summarized in Table 3.

Several important physical trends emerge. First, the wavelength dependence of the atmospheric term A is evident: $A_{850} \simeq 9.4 \times 10^3$ is approximately five times larger than $A_{450} \simeq 1.7 \times 10^3$. This difference reflects the distinct absorption regimes of water vapor: 850 μm (345 GHz) lies near a steep wing of the 325 GHz water line, where fractional changes in T strongly modulate sky noise, whereas 450 μm (666 GHz) sits near a local transmission minimum where opacity is higher but varies more slowly with τ_{225} .

Second, the instrumental noise-floor values B — $B_{450} \simeq 3.8 \times 10^3$ and $B_{850} \simeq 2.2 \times 10^4$ mJy s^{0.5}—are consistent with the TES bolometer readout noise, roughly scaling with the solid angle of the beam and the integration time. The stability of B for nearly a decade in the dynamic range of the NEFD demonstrates that the dominant performance limit is atmospheric, not instrumental.

Table 3. Best-fit NEFD–Transmission Parameters for Figure 10

Band	A [$\text{mJy s}^{0.5}$]	B [$\text{mJy s}^{0.5}$]	N_{MSB}
450 μm	$(1.69 \pm 0.09) \times 10^3$	$(3.82 \pm 0.48) \times 10^3$	5492
850 μm	$(9.39 \pm 0.30) \times 10^3$	$(2.16 \pm 0.18) \times 10^4$	2726

NOTE—Best-fit parameters of the model $\text{NEFD} = A/T + B$ (Equation 20) obtained by Markov-chain Monte Carlo sampling against per-MSB measurements. N_{MSB} is the number of MSB-level points retained in the fit after the L1689B-exclusion filter; the 850 μm fit additionally imposes a noise-floor cut $\text{NEFD} \geq 10^4 \text{ mJy s}^{0.5}$ to remove outliers near the array detection limit.

Finally, the distribution structure reveals that at low transmission ($T \lesssim 0.2$), NEFD scatter increases due to short-timescale sky fluctuations and turbulence, while at high transmission ($T \gtrsim 0.4$), NEFD converges to a well-defined floor corresponding to the photon-noise-limited regime. The color-coding by f_{smooth} confirms that map-making efficiency is uncorrelated with atmospheric transparency, indicating that pipeline performance is driven primarily by scan strategy and field structure rather than sky conditions.

Comparison with Previous SCUBA Results—Archibald et al. (2002) reported a similar relation for the original SCUBA instrument, with $A_{450} \sim 2.5 \times 10^4$ and $A_{850} \sim 1.1 \times 10^3 \text{ mJy s}^{1/2}$. Relative to those results, the SCUBA-2 parameters exhibit: (1) a $\sim 30\%$ smaller A_{450} , consistent with the higher optical throughput and lower noise equivalent power (NEP) of TES bolometers; (2) an A_{850} larger by roughly an order of magnitude, reflecting the greater sky-loading variability over the larger 850 μm array; and (3) comparable B values (within a factor of 2–3), confirming that the fundamental instrument noise floor has remained stable across two detector generations. The residual RMS of $\lesssim 10 \text{ mJy s}^{0.5}$, compared with $\sim 15\text{--}20 \text{ mJy s}^{1/2}$ for SCUBA, further demonstrates the improved uniformity and opacity correction achieved by SCUBA-2.

Implications for Data Quality Assessment and Future Observations—The validated NEFD – transmission relation offers three key applications for JCMT operations and data calibration. First, it enables real-time sensitivity forecasts from current τ_{225} values, supporting dynamic queue scheduling. Second, it provides a quantitative criterion for anomaly detection: MSBs deviating by more than 3σ from the model (square symbols in Figure 10) flag potential issues such as pointing errors, transient clouds, or incomplete reductions. Third, it allows direct conversion between expected atmospheric transparency and the integration time required to reach a desired S/N ratio for polarization measurements.

The agreement between the per-MSB NEFD scaling and the final coadded map noise (σ_{final} ; Section 3.2.2) confirms that the hybrid noise estimation framework is physically self-consistent, capturing both atmospheric and instrumental components across the full dynamic range of BISTRO data. This final validation closes the methodological loop, demonstrating that the derived noise model is both observationally verified and operationally predictive for future submillimeter polarimetry with SCUBA-2 and successor instruments, such as SCUBA-3 (Li et al. 2024).

Comparison with the SCUBA-2 Integration Time Calculator—A direct quantitative comparison between our empirical noise characterization and the predictions of the JCMT SCUBA-2 Integration Time Calculator (ITC; Dempsey et al. 2013) is not straightforward for several reasons. First, the ITC estimates point-source sensitivity in units of mJy beam^{-1} , whereas our analysis characterizes extended-source noise in mJy arcsec^{-2} . Second, the ITC assumes idealized single-observation conditions, while our measurements reflect the co-addition of multiple MSBs obtained under heterogeneous weather grades spanning several years. Third, POL-2 observations incur additional signal attenuation from the half-wave plate and analyzer optics ($\times 1.96$ at 450 μm and $\times 1.35$ at 850 μm), which is not incorporated in the standard SCUBA-2 ITC designed for total-intensity observations.

Nevertheless, the instrument noise floors B_{450} and B_{850} derived from the NEFD–transmission analysis (Table 3) are qualitatively consistent with the ITC predictions for Grade 1–2 weather conditions after accounting for the POL-2 attenuation factors and beam-to-surface-brightness conversions. The shallower-than- $t^{-0.5}$ integration-time scaling ob-

served in Figure 8 demonstrates that achieved map sensitivity is limited by residual correlated atmospheric fluctuations rather than uncorrelated photon statistics—a regime where the idealized radiometer equation underlying the ITC does not apply. This departure quantifies the practical sensitivity ceiling for ground-based submillimeter polarimetry and should inform realistic exposure-time planning for future observations.

3.2.8. Summary of Technical Validation 2

This section establishes the Stokes-I part of Pillar 2 of the technical credibility chain. The Stokes- I two-component additive-noise model and its hybrid estimator have been validated across the 174-field POL-2 archive along four independent axes (§§3.2.3–3.2.7); the resulting per-pixel additive-noise floor σ_λ defines the input mask to the polarization noise model of §3.3 and ultimately to the dual-band S/N thresholds adopted throughout Papers S1 and S2. The cumulative Pillar 2 closure—combining the present Stokes- I result with the Stokes- Q, U result of §3.3—is given in §3.3.7.

3.3. Technical Validation 3: Noise Model for Stokes Q and U

3.3.1. Motivation and overview

This subsection completes Pillar 2 of the methodology (TV3 in the TV1–TV5 spine) by extending the Stokes I noise characterization of §3.2 to the polarization Stokes parameters: a two-component noise model for Stokes Q and U that separates atmosphere-dominated and residual-systematic-dominated regimes, with empirical isotropy and jackknife cross-validation (Pillar 2 entry of Figure 1 and Table 1).

We proceed in five logical steps that parallel the Stokes- I characterization of §3.2. §3.3.2 derives the physical origin of polarization noise from atmospheric transmission fluctuations. §3.3.3 formulates the two-component additive+multiplicative model for Stokes Q and U . §3.3.4 identifies the observational regimes in which each component dominates and connects the model to the dual-band masking strategy adopted in Papers S1 and S2. §3.3.5 presents the implementation and cross-validation strategy, while §3.3.6 tests the isotropy assumptions ($\sigma_Q \simeq \sigma_U$, $\text{Cov}(Q, U) \approx 0$) empirically across wavelength and beam configurations. The subsection closes with §3.3.7, the cumulative Pillar 2 verdict.

3.3.2. Physical Origin of Polarization Noise

The noise characteristics of the polarized emission measurements inherit the fundamental properties established for Stokes I in §3.2, with additional considerations arising from the physics of polarized radiative transfer through the atmosphere of the Earth. The critical insight is that atmospheric transmission fluctuations affect all Stokes parameters identically because they arise from opacity variations along the entire line of sight—a scalar multiplicative effect that preserves the polarization state. Consider the observed polarized emission through a time-variable atmosphere. Following the formalism of Eq.(9), the Stokes parameters propagate as follows.

$$\begin{bmatrix} I_{\text{obs}}(t) \\ Q_{\text{obs}}(t) \\ U_{\text{obs}}(t) \end{bmatrix} = e^{-\tau(t)} \begin{bmatrix} I_0 \\ Q_0 \\ U_0 \end{bmatrix} + \begin{bmatrix} I_{\text{atm}}(t) \end{bmatrix}, \quad (23)$$

where atmospheric emission contributes only to Stokes I because thermal emission from randomly oriented atmospheric molecules is unpolarized. This fundamental asymmetry—multiplicative attenuation that affects all Stokes parameters versus additive contamination that affects only I —shaped the noise characteristics of polarimetric observations.

3.3.3. Derivation of the Two-Component Model

After extinction correction with estimated opacity $\hat{\tau}(t)$, analogous to Eq.(11), the corrected Stokes parameters become

$$Q_{\text{corr}}(t) = Q_{\text{obs}}(t) e^{\hat{\tau}(t)} \simeq Q_0 [1 + \delta g(t)] + n_Q(t), \quad (24)$$

$$U_{\text{corr}}(t) = U_{\text{obs}}(t) e^{\hat{\tau}(t)} \simeq U_0 [1 + \delta g(t)] + n_U(t), \quad (25)$$

where $\delta g(t) \equiv e^{\hat{\tau}(t) - \tau(t)} - 1$ represents the residual gain fluctuation—critically, this is *identical* for all Stokes parameters because atmospheric opacity is wavelength-dependent but polarization-independent. The additive terms $n_Q(t)$ and $n_U(t)$ arise from the readout noise of the SCUBA-2 detector, photon statistics, and residual instrumental effects after

calibration. Taking the variance of Eqs. 24–25 yields the fundamental two-component model:

$$(\Delta Q_\lambda)^2 = \sigma_{Q,\lambda}^2 + k_g^2 Q_\lambda^2, \quad (\Delta U_\lambda)^2 = \sigma_{U,\lambda}^2 + k_g^2 U_\lambda^2, \quad (26)$$

where $\sigma_{Q,\lambda}$ and $\sigma_{U,\lambda}$ quantify the additive noise floor, and $k_g^2 \equiv \text{Var}[\delta g(t)]$ is the gain variance coefficient shared with Stokes I .

3.3.4. Observational Regimes and Physical Interpretation

The applicability and interpretation of Eq.(26) depend critically on the polarization characteristics of the observed field:

Low Polarization Regime ($p \lesssim 5\%$)—In molecular clouds where typical polarization fractions are a few percent (Crutcher et al. 2004; Planck Collaboration et al. 2020), the polarized intensities scale linearly with the total intensity:

$$Q \approx p I \cos(2\phi), \quad U \approx p I \sin(2\phi), \quad (27)$$

where p and ϕ vary slowly on the map. Substituting into Eq.(26) and recognizing that $\langle p^2 \cos^2(2\phi) \rangle \approx \langle p^2 \sin^2(2\phi) \rangle \approx \langle p \rangle^2/2$ for isotropic field orientations (here $\langle \cdot \rangle$ denotes the ensemble average over the orientation angle ϕ ; the symbol is reused with locally specified meaning elsewhere in the paper), we obtain

$$(\Delta Q_\lambda)^2 \approx \sigma_{Q,\lambda}^2 + (k_g \langle p \rangle I_\lambda)^2 \equiv \sigma_{Q,\lambda}^2 + (k'_{Q,\lambda} I_\lambda)^2, \quad (28)$$

where $k'_{Q,\lambda} = k_g \langle p \rangle$ represents an effective coefficient that links polarization noise to total intensity. This formulation enables direct application of the Stokes I noise characterization procedure while maintaining physical transparency.

High Polarization Regime ($p \gtrsim 10\%$)—Near bright embedded sources or in magnetically dominated regions where polarization can exceed 10% (Hull et al. 2014), the direct coupling between ΔQ and Q (similarly for U) becomes dominant. Here, Eq. (26) must be applied without approximation, requiring a sufficient dynamic range in the polarized signal to constrain k_g independently from the total intensity.

Transition Intensity and Observable Predictions—Analogously to Eq. 13 for Stokes I , the transition from atmosphere-dominated to gain-dominated noise occurs at

$$Q_{\text{trans}} \simeq \frac{\sigma_{Q,\lambda}}{k_g}, \quad U_{\text{trans}} \simeq \frac{\sigma_{U,\lambda}}{k_g}. \quad (29)$$

For the low polarization approximation, this corresponds to total intensity

$$I_{\text{trans}}^{(\text{pol})} \simeq \frac{\sigma_{Q,\lambda}}{k_g \langle p \rangle} \approx \frac{I_{\text{trans}}}{\langle p \rangle}, \quad (30)$$

where I_{trans} is the transition intensity for Stokes I from Eq.(13). This predicts that polarization measurements become gain-limited at intensities ~ 20 – 100 times higher than for total intensity (for $p \sim 1$ – 5%), explaining why most POL-2 observations remain in the atmosphere-dominated regime for polarization even when Stokes I exhibits significant gain noise.

3.3.5. Validation and Implementation Strategy

The two-component model makes several testable predictions:

1. The ratio $\sigma_{Q,\lambda}/\sigma_{U,\lambda} \approx 1$ due to instrumental isotropy;
2. The gain coefficient k_g extracted from Q and U should match that of Stokes I ;
3. The wavelength scaling $\sigma_{Q,450}/\sigma_{Q,850}$ should follow atmospheric opacity trends, typically 6–12 $\times$ based on observing conditions;
4. Pixels with $I < I_{\text{trans}}^{(\text{pol})}$ should exhibit constant ΔQ and ΔU , while those above should show linear scaling.

For practical implementation, we adopt a hierarchical approach: First, determine $\sigma_{Q,\lambda}$ and $\sigma_{U,\lambda}$ from emission-free regions where $S/N(I) < 2$. Second, for fields with $p < 5\%$, fit the model referenced with I [Eq. (28)] to pixels with $S/N(I) > 30$. Third, for high-polarization regions or validation purposes, fit Eq. (26) directly using pixels with $S/N(PI) > 10$, where $PI = \sqrt{Q^2 + U^2}$ is the debiased polarized intensity. This strategy maximizes statistical power while maintaining physical rigor in various astrophysical environments.

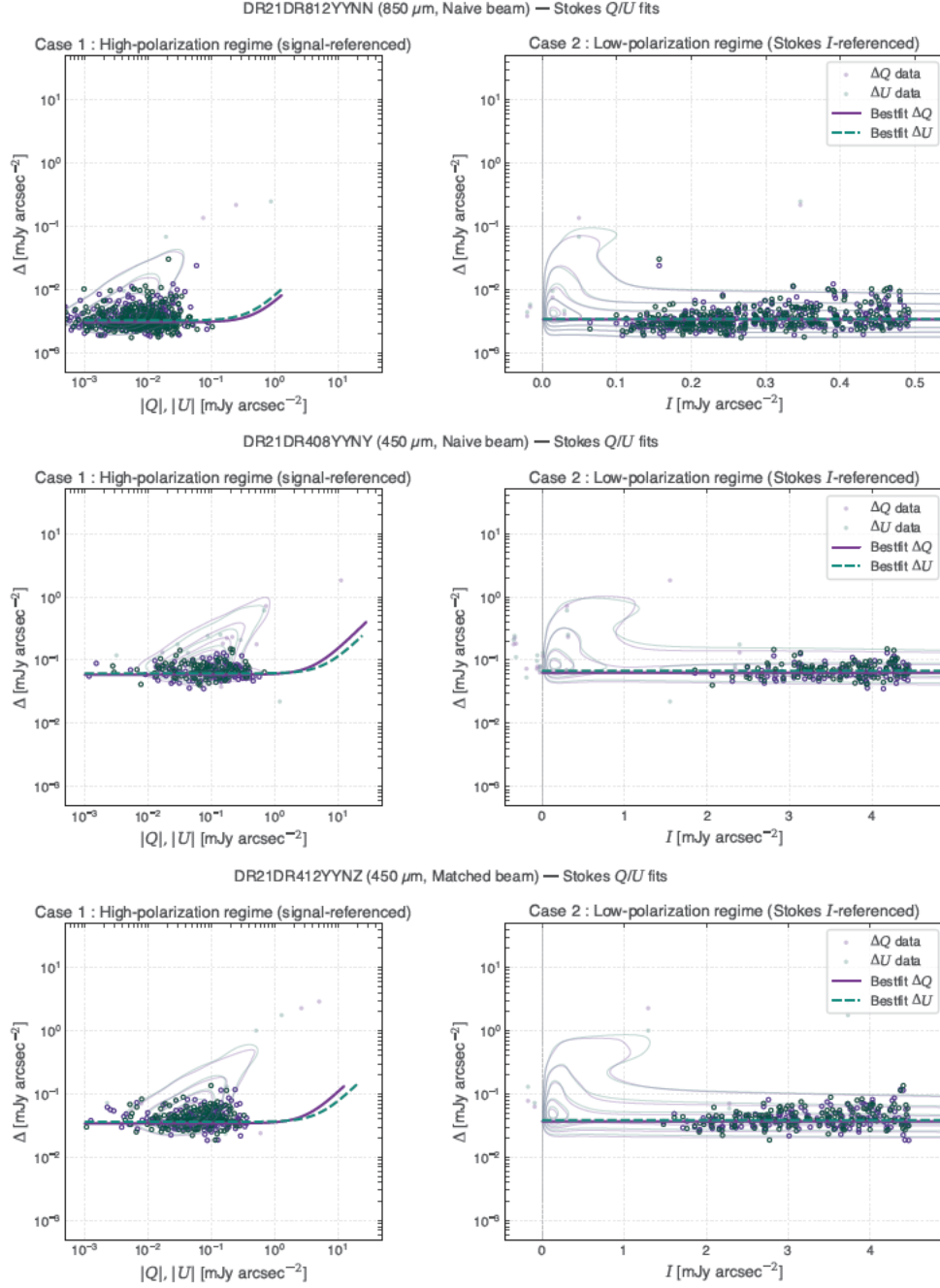


Figure 11. Application of the hybrid noise model to Stokes Q and U measurements in the DR21 field across three observational configurations: (a) 850 μm naive beam, (b) 450 μm naive beam, and (c) 450 μm matched beam (see [Paper S1 § 2 \(dual-band matched-beam preparation\)](#)). Each row contains two panels following the dual-regime analysis framework established for Stokes I (Figure 4). The left panels show Case 1 (high-polarization regime) where uncertainties ΔQ and ΔU are plotted against the respective signal amplitudes $|Q|$ and $|U|$ on logarithmic axes. The right panels display Case 2 (low-polarization regime) with ΔQ and ΔU as functions of Stokes I using a linear horizontal axis. Kernel density contours indicate the data distribution for Stokes Q (crimson) and U (royal blue), with individual measurements shown as semi-transparent points. Open circles mark the subset of data points satisfying the fitting criteria described in §3.3. Solid and dashed curves represent the best-fit two-component noise models for Q and U , respectively, following the functional form $\Delta^2 = \sigma^2 + (k \cdot S)^2$ where S denotes the appropriate signal reference ($|Q|$, $|U|$, or I). The vertical dashed line in the right panels marks $I = 0$. DR21 represents a high-brightness star-forming region where the two-component model successfully characterizes the noise properties across both wavelengths despite the complex emission structure. The consistency of fit parameters between naive and matched beam configurations validates the robustness of the noise characterization methodology. Note that this analysis uses only pixels satisfying the fitting criteria; the anomalous behavior in the central regions of DR21 due to map-making artifacts is discussed separately in Appendix B.

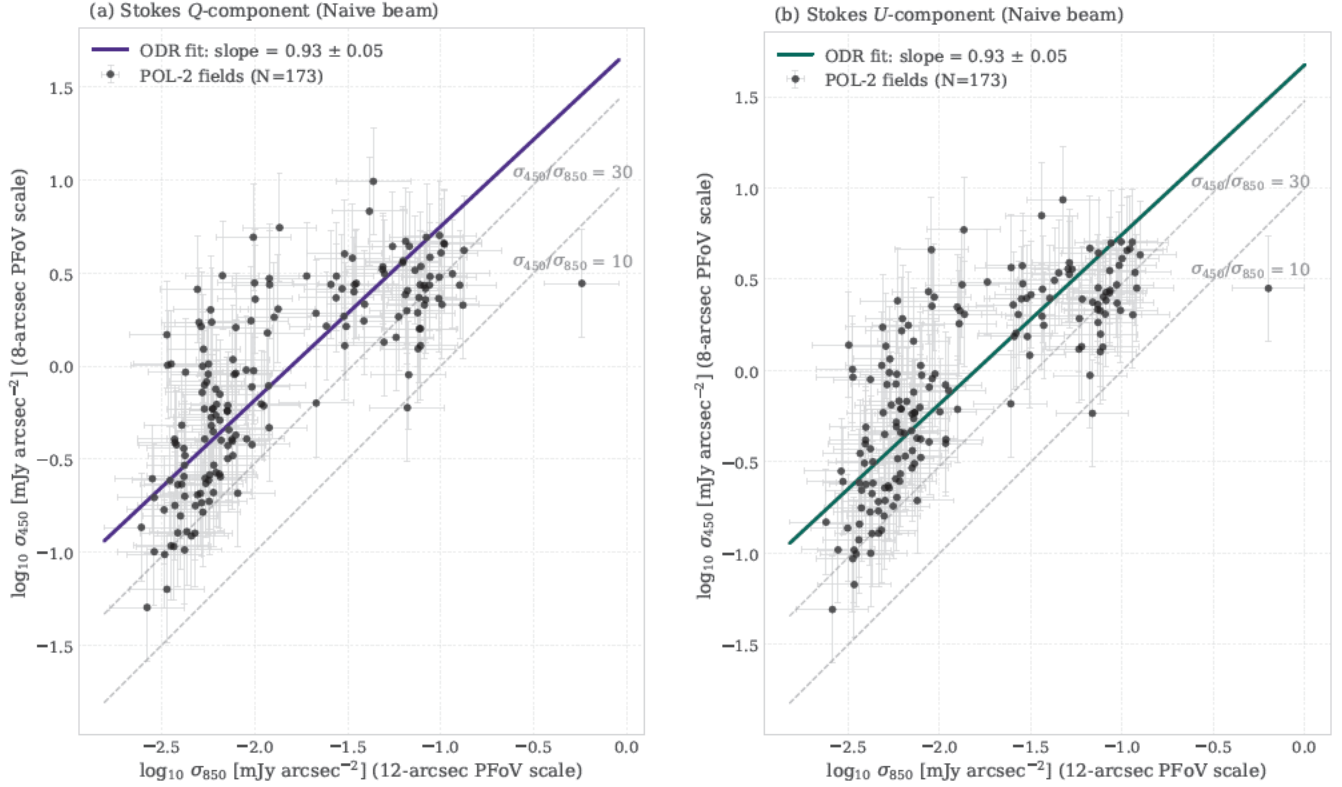


Figure 12. Correlation between the additive noise levels derived at 450 μm and 850 μm for the polarization components across 173 POL-2 fields analyzed with naive-beam configuration. Panel (a) shows the Stokes Q component and panel (b) shows the Stokes U component, with each point representing one field and 1σ uncertainties estimated from the noise model fits in $\log_{10} \sigma$ space. The magenta lines show orthogonal distance regression (ODR) fits in log-log space, yielding slopes of $m = 0.93 \pm 0.05$ for both components. This near-unity slope indicates that the wavelength scaling of additive polarization noise follows $\sigma_{Q,450} \propto \sigma_{Q,850}^{0.93}$ and $\sigma_{U,450} \propto \sigma_{U,850}^{0.93}$, demonstrating that both polarization components exhibit identical wavelength dependence. Dashed gray lines mark fixed $\sigma_{450}/\sigma_{850}$ ratios of 10 and 30 for reference. The median ratios across the sample are $\sigma_{Q,450}/\sigma_{Q,850} = 12.8$ and $\sigma_{U,450}/\sigma_{U,850} = 12.9$, substantially lower than the ratio observed for Stokes I (median 38, Figure 7), reflecting the different noise contributions to polarized versus total intensity measurements. These results provide direct validation of prediction (3) in §3.3, confirming that the additive noise floors for polarization scale with atmospheric opacity following the expected submillimeter observational characteristics.

3.3.6. Empirical Verification of Polarization-Noise Isotropy

Having established the two-component formulation in Eq. (26), we now examine whether the fundamental assumptions $\sigma_Q \simeq \sigma_U$ and $\text{Cov}(Q, U) \approx 0$ are empirically justified. This isotropy validation forms the observational basis for applying the simplified noise-propagation expressions that are routinely used in submillimeter polarimetry (see, e.g., Bastien et al. 2005; Ward-Thompson et al. 2017).

Motivation and Approach—The two assumptions above are not trivial: $\sigma_Q \simeq \sigma_U$ implies that the additive detector and photon noises are statistically isotropic in the (Q, U) plane, while $\text{Cov}(Q, U) \simeq 0$ ensures that the polarization vector $\vec{P} = (Q, U)$ behaves as a Gaussian random vector with circular symmetry. In this case, the joint probability density function,

$$f(q, u) = \frac{1}{2\pi\sigma^2} \exp\left[-\frac{(q - q_0)^2 + (u - u_0)^2}{2\sigma^2}\right], \quad (31)$$

leads directly to the well-known Ricean distribution of the polarized intensity and the von Mises-Fisher form for the polarization angle (see §3.3). When $\sigma_Q \neq \sigma_U$ or $\text{Cov}(Q, U) \neq 0$, Eq. (31) must be replaced by the more general expression

$$f(q, u) = \frac{1}{2\pi\sigma_Q\sigma_U\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left\{ \frac{(q - q_0)^2}{\sigma_Q^2} + \frac{(u - u_0)^2}{\sigma_U^2} - \frac{2\rho(q - q_0)(u - u_0)}{\sigma_Q\sigma_U} \right\}\right], \quad (32)$$

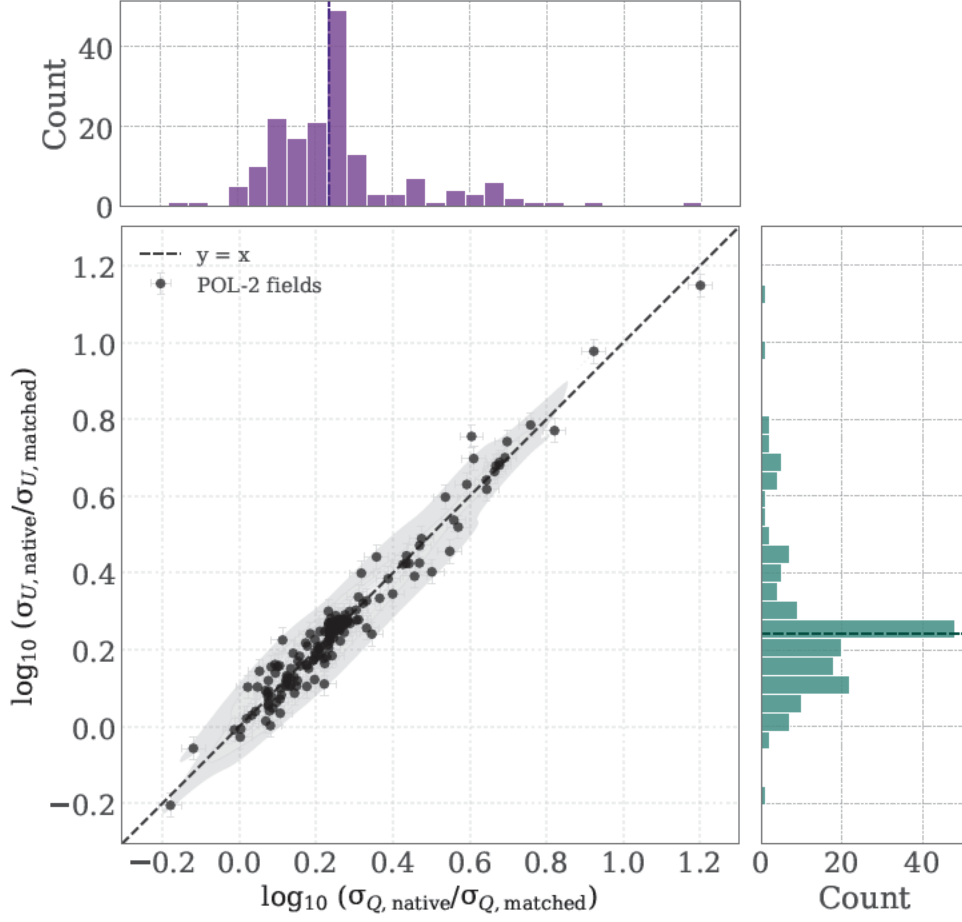


Figure 13. Comparison of additive noise levels between the native-beam and beam-matched reductions of the $450\,\mu\text{m}$ POL-2 maps for 172 fields. The central panel shows the joint distribution of $\log_{10}(\sigma_{Q,\text{native}}/\sigma_{Q,\text{matched}})$ versus $\log_{10}(\sigma_{U,\text{native}}/\sigma_{U,\text{matched}})$, where each point represents one field with error bars indicating the propagated 5% fractional uncertainty in the noise estimates. The black dashed line marks the $y = x$ relation, and purple contours show the kernel density estimate of the distribution. Marginal histograms display the distributions for Stokes Q (top) and Stokes U (right), with orange dashed lines indicating the median values. The median logarithmic ratios are $\log_{10}(\sigma_{Q,\text{native}}/\sigma_{Q,\text{matched}}) = 0.235$ and $\log_{10}(\sigma_{U,\text{native}}/\sigma_{U,\text{matched}}) = 0.243$, corresponding to linear ratios of ~ 1.7 for both components. The tight clustering of points along the diagonal demonstrates that the beam-matching process (Paper S1 § 2 (dual-band matched-beam preparation)) scales the noise levels uniformly for both polarization components, preserving the isotropy of the noise characteristics established in §3.3. This systematic factor of 1.7 reflects the combined effect of beam convolution and the adjustment of pixel scales between the two data products.

where ρ is the correlation coefficient between Q and U . The simplified form in Eq. (31) is therefore valid only if isotropy and independence hold. Testing this condition observationally is thus essential.

Field-by-field Isotropy Tests—We first derived the additive noise parameters $\sigma_{Q,\lambda}$ and $\sigma_{U,\lambda}$ for all POL-2 fields using the two-component model (Eqs. 28–26). For each data set (the naive beam data at 450 and 850 μm , and the matched data at the 450 μm band; Table 4), we evaluated the logarithmic ratio $\log_{10}(\sigma_Q/\sigma_U)$. An ideal isotropic system has a median of zero. To test whether the median differs significantly from zero, we applied the Wilcoxon signed-rank test.

Results—Table 4 summarizes the statistics. The mean absolute values of $\log_{10}(\sigma_Q/\sigma_U)$ are 0.026, 0.031, and 0.025 for the three data sets (the 450 μm native-beam, 450 μm beam-matched, and 850 μm native-beam configurations, respectively), indicating typical deviations of only 6% from perfect isotropy. The corresponding fractions of fields satisfying $|\log_{10}(\sigma_Q/\sigma_U)| < 0.05$ are 89% (156/175), 85% (155/182), and 91% (185/203), respectively, further confirming that nearly all fields exhibit statistically indistinguishable noise levels between Stokes Q and U . The distributions are sharply centered near zero, with median offsets $|\text{median}| < 0.005$ and scatter $\text{MAD} \simeq 0.03$. These correspond to only 7% fractional variations ($10^{0.03} \approx 1.07$), and the p -values ($p > 0.2$) indicate that isotropy cannot be rejected for any

Table 4. Statistical validation of noise isotropy across POL-2 configurations.

Data Set	N_{field}	Median $\log_{10}(\sigma_Q/\sigma_U)$	MAD ^a	Wilcoxon p ^b	Notable outliers ^c
450 μm native compound beam	175	+0.0014	0.031	0.45	NGC 1333
450 μm beam-matched (12''6)	182	−0.0018	0.038	0.90	Rosette N, Serpens Main, Oph C
850 μm native beam	203	+0.0048	0.031	0.29	Oph C, SDC 24.489, IRDC G35.39−0.33

^aMedian absolute deviation of the per-field $\log_{10}(\sigma_Q/\sigma_U)$ distribution.

^bThe p -value is from the two-sided Wilcoxon signed-rank test against the null hypothesis of zero median offset.

^cRepresentative fields with $|\log_{10}(\sigma_Q/\sigma_U)| > 0.1$ (less than 5 % of each sample), sorted by $|\log_{10}|$ descending; cell entries show the top three science outliers. Full per-dataset listings: 450 μm native compound beam (175 fields, 1 science outlier = 0.6 %): NGC 1333 (mosaicked field; [Paper S2 Figure 37](#)). 450 μm beam-matched (182 fields, 3 science outliers = 1.6 %): Rosette North (mosaic component field; [Paper S2 Figure 75](#)), Serpens Main (mosaicked field; [Paper S2 Figure 46](#)), L1688 Ophiuchus C ([Paper S2 Figure 29](#)); the non-BISTRO submm calibrators CTA 102 and IK Tauri and the data-quality-excluded field G16 also exceed the threshold and are excluded for this paper. 850 μm native compound beam (203 fields, 5 science outliers = 2.5 %): L1688 Ophiuchus C ([Paper S2 Figure 29](#)), SDC 24.489 ([Paper S2 Figure 47](#)), IRDC G035.39−00.33 (mosaicked field), B211/213 North (mosaic component field; [Paper S2 Figure 25](#)), IC 1396E ([Paper S2 Figure 67](#)).

configuration. Outliers with $|\log_{10}(\sigma_Q/\sigma_U)| > 0.1$ were identified (e.g., the Rosette Mosaic field and Ophiuchus C), which represents less than 5% of the total sample.

The consistency in these datasets confirms that the instrumental noise in the (Q, U) plane is statistically circular. The DR21 case study in Figure 11 illustrates that the two-component model recovers the same additive noise levels for both Stokes components, even in bright, spatially complex fields.

Wavelength and Beam Effects—The isotropy established above is further supported by the cross-wavelength comparison in Figure 12, which shows that $\sigma_{Q,450}$ and $\sigma_{U,450}$ scale with their 850 μm counterparts as $\sigma_{450} \propto \sigma_{850}^{0.93}$. This slope near unity confirms the prediction (3) in §3.3, demonstrating that the additive polarization noise follows the same opacity dependence as the total-intensity noise. Figure 13 then shows that beam-matching at 450 μm uniformly scales both σ_Q and σ_U by a factor of ≈ 1.7 , without altering their ratio. Together, these tests demonstrate that isotropy and scaling are preserved across wavelength and beam configurations; the corresponding polarization-angle preservation under the same beam-matching procedure is quantified in §3.5.

Theoretical Consistency and Approximate Treatment—The empirical equality $\sigma_Q \simeq \sigma_U$ and the negligible correlation $\rho_{QU} \approx 0$ justify the use of the simplified Gaussian model [Eq. (31)] instead of the general elliptical form [Eq. (32)]. Under these isotropic and uncorrelated conditions, the variance propagation for P simplifies to

$$\sigma_P^2 \simeq \frac{\sigma^2}{I^2}, \quad \sigma_Q^2 \simeq \sigma_U^2 \equiv \sigma^2, \quad \text{Cov}(Q, U) \approx 0, \quad (33)$$

or equivalently, $\sigma_P \simeq \sigma/I$ in terms of the standard deviation. Likewise, the variance of the polarization angle,

$$\sigma_\phi^2 = \frac{1}{(2P^2I^2)^2} [Q^2\sigma_U^2 + U^2\sigma_Q^2 - 2QU \text{Cov}(Q, U)], \quad (34)$$

simplifies to the canonical approximation:

$$\sigma_\phi \simeq \frac{1}{2} \frac{\sigma_P}{P}. \quad (35)$$

These expressions remain valid as long as $|\rho_{QU}| < 0.1$ and $|\log_{10}(\sigma_Q/\sigma_U)| < 0.05$. The isotropy results in Table 4 fall well within this domain, indicating that the approximation $\text{Cov}(Q, U) \approx 0$ is observationally and statistically justified.

Implications—First, the equality $\sigma_Q \simeq \sigma_U$ ensures that the Ricean distribution of polarized intensity is applicable to the POL-2 data, providing the correct form for bias corrections at low S/N ratios. Second, the negligible covariance guarantees that the von Mises-Fisher distribution describes the angular statistics of the polarization vector. Third, the combination of isotropy and independence validates the practical use of Eq. (35) in subsequent sections where angle-dispersion functions and polarization-fraction correlations are analyzed. These results justify the use of simplified error

propagation throughout the POL-2 polarimetric survey, eliminating the need for full covariance matrices in routine data processing. In essence, the empirical tests confirm that the instrumental noise in Stokes Q and U behaves as a circular Gaussian random field and that the simplified statistical framework adopted throughout this work is physically and mathematically sound.

3.3.7. Summary of Technical Validation 3

Together with §3.2, this section establishes Pillar 2 of the technical credibility chain. Every retained polarization measurement entering the companion science analyses (Papers S1 and S2) carries a self-consistent statistical-error budget derived from a single integrated noise model: an additive Stokes I characterization across the 174-field POL-2 archive, extended into the polarization plane by a two-component Stokes Q, U model whose foundational $\sigma_Q \simeq \sigma_U$ and $\text{Cov}(Q, U) \approx 0$ assumptions have been verified empirically and across wavelength/beam configurations. Pixel-level outliers in the polarized intensity products of the companion papers therefore cannot be attributed to under-quantified statistical noise: any apparent dual-band tension above the noise envelope quantified here is, by construction, a candidate astrophysical signal rather than a residual statistical artefact.

3.4. Technical Validation 4: 450 μm Offset Discovery and Correction

3.4.1. Motivation and overview

Before inter-band polarization-angle and intensity comparisons can be interpreted physically, the additive zero levels of the 450 and 850 μm Stokes- Q and Stokes- U maps must be validated. We therefore identify and correct, where required, a systematic 450 μm additive Stokes- Q/U zero-level offset that, although invisible to single-wavelength analyses, would otherwise propagate as spurious inter-band differences into the dual-band polarization observables.

Throughout this section, the term “450 μm offset correction” refers specifically to the additive correction applied to the Stokes- Q and Stokes- U maps where required; no corresponding additive correction is applied to Stokes- I , and the correction is not applied uniformly to all sources.

This subsection opens Pillar 3 of the methodology—*map-domain comparability legitimacy*—which is presented in two stages (Pillar 3 entry of Figure 1 and Table 1). The present subsection is the *construction* stage: discovery and correction of the wavelength-dependent 450 μm additive Stokes- Q/U zero-level offset, together with the dual-band beam- and spatial-frequency harmonization that yields the final 12''6-resolution science product. The *capstone validation* stage of Pillar 3, which quantitatively demonstrates that the constructed product preserves the polarization-angle information required by the dual-band science analyses, is presented in §3.5. Figure 14 provides a subsection-level roadmap.

3.4.2. Recognition of Systematic Inconsistency in the 450 μm Data

Box 1 of the section roadmap (Figure 14) is the problem-recognition stage developed in this subsection: a three-wavelength polarization-angle comparison between POL-2 (450/850 μm) and independent SOFIA/HAWC+ data (154/214 μm) shows that POL-2 850 μm agrees with SOFIA at both far-infrared bands, while POL-2 450 μm does not. This pattern, in which the intermediate wavelength alone is discrepant in an otherwise consistent three-wavelength sequence, is unphysical and exposes the wavelength-dependent 450 μm offset addressed below.

We discovered the inconsistency by comparing POL-2 magnetic-field orientations with independent SOFIA/HAWC+ Band D (154 μm) polarimetry of ρ Ophiuchi A from Santos et al. (2019)—a wavelength intermediate between the two POL-2 bands and at comparable angular resolution ($\theta_{\text{FWHM}} \approx 13''.6$). The 850 μm POL-2 vectors agreed with the 154 μm SOFIA orientations throughout the mapped region, but the 450 μm POL-2 vectors exhibited systematic angular discrepancies of $\sim 20^\circ$ – 30° relative to both reference wavelengths. Because all three wavelengths should trace the same underlying field for optically thin dust emission, this three-wavelength signature isolated a systematic artifact specific to the 450 μm measurements rather than an intrinsic wavelength-dependent astrophysical effect. Quantitative confirmation via Kolmogorov–Smirnov tests, which demonstrate that the uncorrected 450 μm orientations are statistically inconsistent with both reference wavelengths, is given in §3.4.8; this established the need for the dedicated correction framework developed in the remainder of this subsection.

3.4.3. Hardware Origins Excluded by Null Tests

Box 2 of the section roadmap (Figure 14) is the hardware-origin hypothesis. Dedicated null tests rule out three categories of instrumental effects as origins of the 450 μm offset: telescope pointing and attitude parameters, source-brightness dependence, and the elevation-dependent instrumental-polarization (IP) model of Friberg et al. (2018).

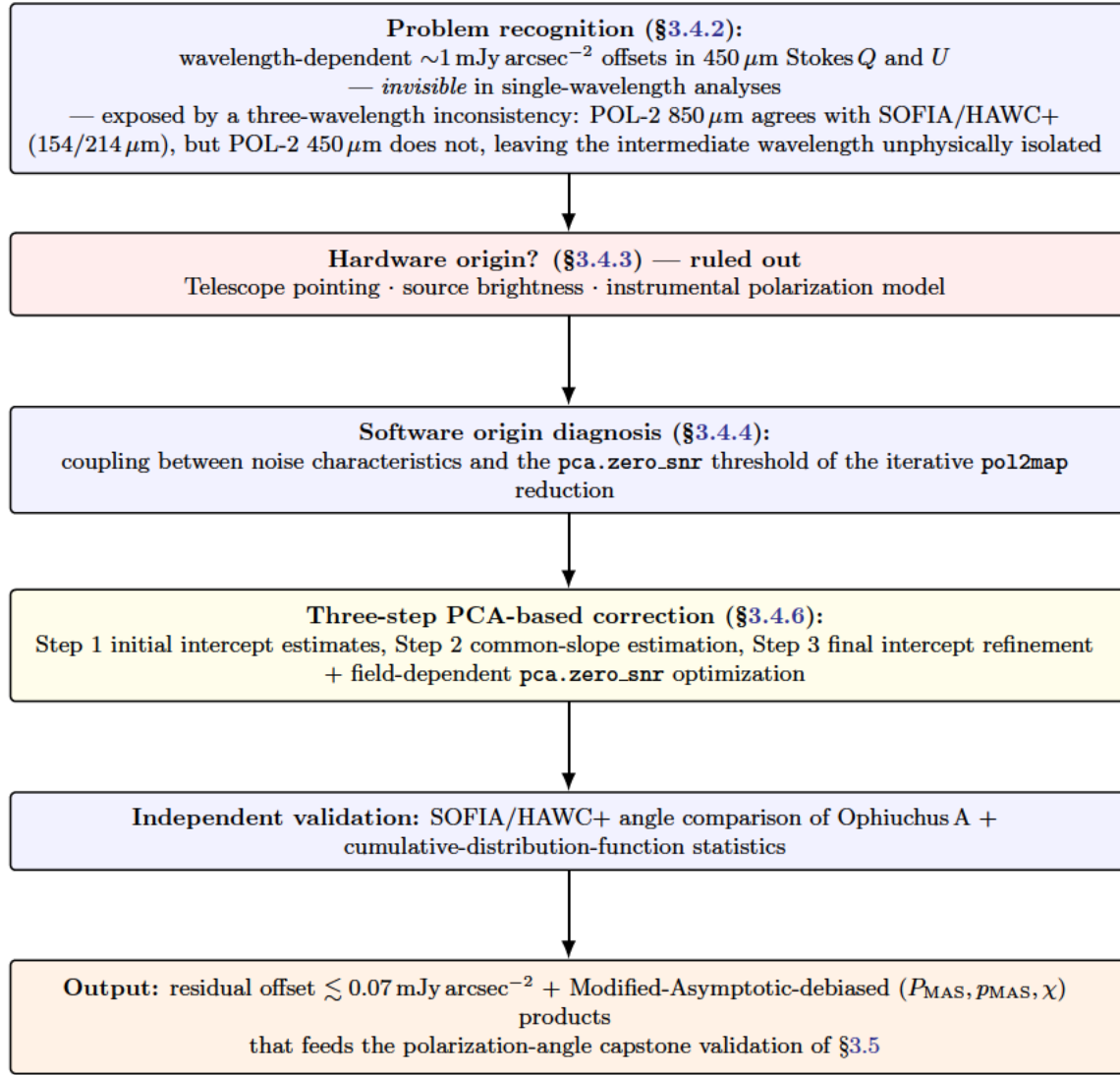


Figure 14. Roadmap for the Pillar 3 construction stage (§3.4). The narrative proceeds top-to-bottom from problem recognition (§3.4.2), through hardware-origin elimination (§3.4.3; details in Appendix D), software-origin diagnosis and the three-step PCA-based correction (§§3.4.4–3.4.6), and SOFIA/HAWC+ independent validation (§3.4.8), to the final Modified-Asymptotic-debiased polarization products that feed the Pillar 3 capstone validation in §3.5.

Across all three tests the residual signature stays an order of magnitude below the observed $\gtrsim 1 \text{ mJy arcsec}^{-2}$ offsets; full diagnostic figures, per-test methodology, and quantitative tolerances are presented in Appendix D. These multiple independent null results redirect the investigation to the data-reduction pipeline. Boxes 3–6 of the section roadmap (Figure 14) cover the resulting software-origin diagnosis: the next three subsections (§§3.4.4–3.4.6) present the observational evidence for the systematic offsets, the physical constraints that require zero intercepts in dual-band Stokes comparisons, and the three-step correction procedure that ensures the final Stokes maps satisfy the requirement that $Q_{850} = U_{850} = 0$ implies $Q_{450} = U_{450} = 0$ under optical thinness (see Paper S1 §2); the SOFIA/HAWC+ independent validation and the resulting Modified-Asymptotic-debiased polarization products fed forward to §3.5 are presented in the subsections that follow.

3.4.4. Observational Evidence for Systematic Intensity Offsets

§3.4.3 demonstrates that significant 450 μm Stokes offsets are detectable at the level of individual Minimum Science Blocks (MSBs), before any coadding of data. Figure 28 shows that these MSB-level offsets do not exhibit dependence on telescope azimuth, elevation, parallactic angle, or tracking geometry, confirming that they cannot originate from hardware- or attitude-related effects (see also §D.4).

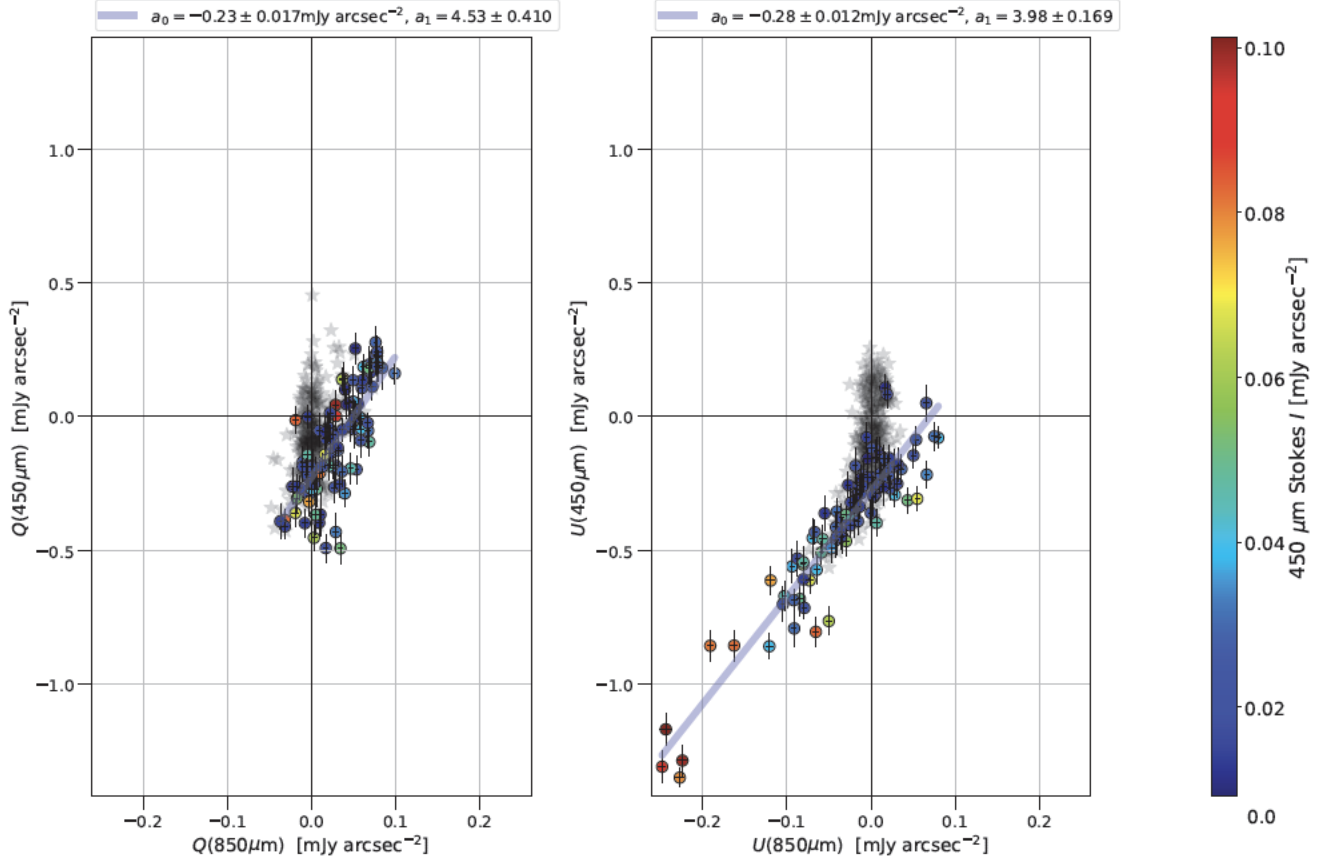


Figure 15. Initial regression analysis for Ophiuchus A (Step 1 of the offset-correction procedure). The panels show the beam-matched relations between Q_{450} and Q_{850} (left) and between U_{450} and U_{850} (right), with $450\ \mu\text{m}$ intensities color-coded. Weighted fits (`curve_fit`) yield significant non-zero intercepts: $a_0^Q = +0.23 \pm 0.02\ \text{mJy arcsec}^{-2}$ and $a_0^U = +0.28 \pm 0.01\ \text{mJy arcsec}^{-2}$ (Eq. 36), indicating raw $450\ \mu\text{m}$ zero-level offsets. These offsets violate the physical constraints enumerated in §3.4.5, motivating the three-step correction procedure (§3.4.6).

After matching the beam size and spatial-frequency range following the established POL-2 methodology (Ward-Thompson et al. 2017; Pattle et al. 2017), systematic offsets become evident in the coadded maps. For every bright field, linear regressions of (Q_{450}, Q_{850}) and (U_{450}, U_{850}) reveal significant, repeatable non-zero intercepts. Figure 15 illustrates this behavior for Ophiuchus A, one of the highest signal-to-noise regions in our sample.

Using weighted least-squares regression (`scipy.curve_fit`), we obtain for Ophiuchus A:

$$a_0^Q = +0.23 \pm 0.02\ \text{mJy arcsec}^{-2}, \quad a_0^U = +0.28 \pm 0.01\ \text{mJy arcsec}^{-2}, \quad (36)$$

with corresponding slopes

$$a_1^Q = 4.53 \pm 0.41, \quad a_1^U = 3.98 \pm 0.17. \quad (37)$$

Both intercepts differ from zero at $\gtrsim 10\sigma$, demonstrating wavelength-dependent zero-level shifts in the $450\ \mu\text{m}$ Stokes parameters relative to $850\ \mu\text{m}$.

These non-zero intercepts violate fundamental physical expectations for optically thin dust emission. In §3.4.5, we enumerate the constraints that require any sky position with $Q_{850} = U_{850} = 0$ to also satisfy $Q_{450} = U_{450} = 0$. No astrophysical mechanism produces wavelength-dependent zero-level polarization shifts between 450 and $850\ \mu\text{m}$ in the optically thin regime (Hildebrand et al. 1999; Vaillancourt et al. 2008). The offsets must therefore arise from the data reduction process itself, motivating the correction framework described in §3.4.6.

3.4.5. Physical Constraints Requiring Zero Intercepts

The detection of non-zero intercepts in the (Q_{450}, Q_{850}) and (U_{450}, U_{850}) regressions (Eqs. 36–37; Figure 15) is incompatible with optically thin submillimeter dust polarization at the two POL-2 bands. Under standard assumptions—

wavelength-independent grain-alignment direction (consistent with RAT theory; [Andersson et al. 2015](#); [Planck Collaboration et al. 2020](#)), optical thinness at both wavelengths ($\tau_{450} \lesssim 0.01$ even at $N_{\text{H}_2} \sim 10^{24} \text{ cm}^{-2}$; [Hildebrand 1983](#); [Ossenkopf & Henning 1994](#)), and Stokes Q , U as projections of a wavelength-common polarization pattern (constraint 6 below; [Hildebrand 1988](#))—the same intrinsic polarization geometry is sampled at both POL-2 bands, with only the proportionality between PI_{450} and PI_{850} free to vary through wavelength-dependent alignment efficiency (the slopes a_1^Q , a_1^U of Eq. 37).

It follows directly that any pixel satisfying $Q_{850} = U_{850} = 0$ must also satisfy $Q_{450} = U_{450} = 0$. The linear regressions of Q_{450} versus Q_{850} and of U_{450} versus U_{850} must therefore pass through the origin, implying $a_0^Q = a_0^U = 0$ as a physical requirement. The observed non-zero intercepts (Eq. 36) consequently signal residual instrumental or processing contamination in the 450 μm Stokes parameters; combined with the null hardware tests of §3.4.3, this motivates the three-step software correction described in §3.4.6. The full enumeration of the six physical assumptions underlying this zero-intercept condition (sources of 850 μm reference polarization, 450 μm contamination channels, optical thinness, line-of-sight emissivity integration, Stokes projection, and wavelength-independent alignment direction)—as well as the supporting citations and the chain of reasoning identifying the dominant violated assumption—is given in Appendix E.

3.4.6. Three-Step Offset Correction Procedure

To isolate and remove the 450 μm Q/U additive offsets identified in §3.4.4, we implement a three-step procedure that incorporates the physical constraint of wavelength-independent polarization angles (constraint 6 in §3.4.5) while improving statistical precision. This approach builds upon established debiasing techniques for polarization measurements ([Plaszczynski et al. 2014](#); [Montier et al. 2015](#)), but addresses the unique systematic effects in dual-band POL-2 observations.

Step 1: Initial Intercept Estimates—We perform weighted least-squares regression of the form:

$$Q_{450} = a_0^Q + a_1^Q Q_{850} + \epsilon_Q, \quad U_{450} = a_0^U + a_1^U U_{850} + \epsilon_U, \quad (38)$$

where weights are $1/\sigma_{Q,450}^2$ and $1/\sigma_{U,450}^2$ following standard weighting schemes ([Pattle et al. 2021](#)). This yields field-specific signed intercepts (a_0^Q , a_0^U) and slopes (a_1^Q , a_1^U), as illustrated in Figure 15 for Ophiuchus A. The intercepts are interpreted as raw 450 μm zero-level offsets that violate the physical constraints enumerated in §3.4.5.

Step 2: Estimation of a Common Slope—Constraint (6) in §3.4.5 requires that the polarization position angle be wavelength-independent. This implies that the proportionality between 450 and 850 μm Stokes parameters must be identical for the Q and U components: $a_1^Q = a_1^U \equiv a_1^{\text{common}}$. After subtracting the Step 1 intercepts:

$$Q_{450}^{\text{corr}} = Q_{450} - a_0^Q, \quad U_{450}^{\text{corr}} = U_{450} - a_0^U, \quad (39)$$

we combine the $(Q_{450}^{\text{corr}}, Q_{850})$ and $(U_{450}^{\text{corr}}, U_{850})$ samples to derive a single common slope a_1^{common} through joint regression (Figure 16). The doubled sample size improves the precision of the slope determination by more than an order of magnitude compared to the individual estimates from Step 1 (compare Eq. 37).

Step 3: Final Intercept Refinement—Holding a_1^{common} fixed, we re-estimate the intercepts using one-parameter fits:

$$Q_{450} = a_0^{Q,\text{final}} + a_1^{\text{common}} Q_{850}, \quad U_{450} = a_0^{U,\text{final}} + a_1^{\text{common}} U_{850}. \quad (40)$$

The reduced degrees of freedom tighten the uncertainties on the intercepts, yielding the final offset corrections ($a_0^{Q,\text{final}}$, $a_0^{U,\text{final}}$) applied to the 450 μm maps. The corrected Stokes parameters used for scientific analysis are then:

$$Q_{450}^{\text{corr}} = Q_{450} - a_0^{Q,\text{final}}, \quad U_{450}^{\text{corr}} = U_{450} - a_0^{U,\text{final}}. \quad (41)$$

These corrected parameters satisfy the physical requirement (derived in §3.4.5) that the regression intercepts be zero, enabling robust dual-band polarization analysis.

Application to Ophiuchus A—Applied to Ophiuchus A—one of the highest- S/N regions in the sample—the three-step procedure converges as follows. Step 1 yields raw intercepts $a_0^Q = +0.23 \pm 0.02 \text{ mJy arcsec}^{-2}$ and $a_0^U = +0.28 \pm 0.01 \text{ mJy arcsec}^{-2}$ with individual slopes $a_1^Q = 4.53 \pm 0.41$ and $a_1^U = 3.98 \pm 0.17$ (Eq. 36, 37; Figure 15); Step 2 yields a precision-improved common slope $a_1^{\text{common}} = 4.120 \pm 0.048$ (Eq. 42; Figure 16); and Step 3 produces final intercept

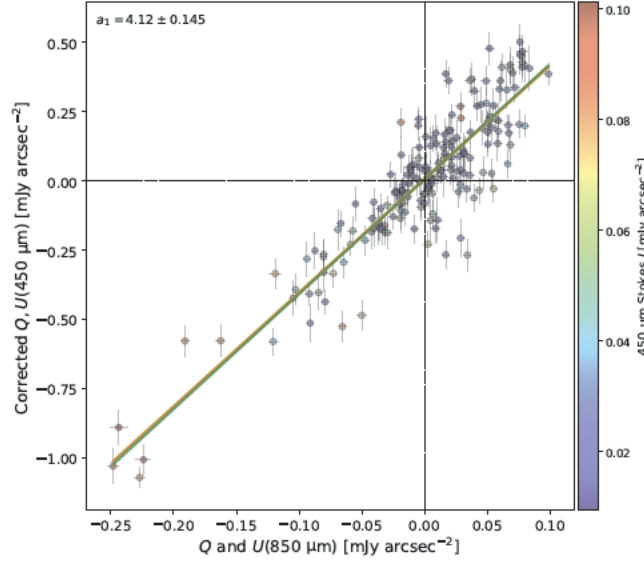


Figure 16. Combined regression of intercept-corrected Stokes parameters for Ophiuchus A (Step 2 of the three-step procedure). After subtracting the Step 1 intercepts (Eq. 39), the $(Q_{450}^{\text{corr}}, Q_{850})$ and $(U_{450}^{\text{corr}}, U_{850})$ samples are merged. The unified regression yields a common slope $a_1^{\text{common}} = 4.120 \pm 0.048$, with precision improved by more than an order of magnitude compared to the individual slopes from Step 1 (Eq. 37: $a_1^Q = 4.53 \pm 0.41$, $a_1^U = 3.98 \pm 0.17$). This precision improvement reflects the doubled sample size from combining Q and U domains, which encode the same underlying polarization pattern (constraint 6 in §3.4.5) with independent noise realizations.

corrections $a_0^{Q,\text{final}} = -0.21 \pm 0.01$ and $a_0^{U,\text{final}} = -0.28 \pm 0.01$ mJy arcsec $^{-2}$ (Eq. 43; Figure 17). When subtracted from the 450 μm Stokes parameters via Eq. (41), these corrections reduce the slope-constrained residual intercepts of the corrected maps to $|a_0^{\text{sc}}| \lesssim 0.02$ mJy arcsec $^{-2}$ (§3.4.8), consistent with zero at the $\lesssim 2\sigma$ level relative to the per-fit statistical uncertainty. This convergence on real data demonstrates that the three-step framework removes the systematic zero-level offset to within ~ 0.02 mJy arcsec $^{-2}$ while preserving the wavelength-independent polarization pattern required by the physical constraints of §3.4.5, so that Q_{450} and U_{450} vanish wherever their 850 μm counterparts do (Eq. 41).

$$a_1^{\text{common}} = 4.120 \pm 0.048, \quad (42)$$

$$a_0^{Q,\text{final}} = -0.21 \pm 0.01 \text{ mJy arcsec}^{-2}, \quad a_0^{U,\text{final}} = -0.28 \pm 0.01 \text{ mJy arcsec}^{-2}. \quad (43)$$

The PCA-threshold optimization procedure—including the empirical origin of the offsets in the FLT/PCA/COM masking and Fourier-domain interpolation, the grid-search formalism (Eqs. F11–F12), the explored parameter grid (Table 9), the mosaicking limitations that motivate the single-field-only application, and the optimal-threshold ranges as a function of field morphology—is documented in Appendix F, with the resulting Category 1–3 atlas given in Figures 29–32. The high-level field categorization that this procedure produces is summarized below.

3.4.7. Offset Significance and Field Categorization

Comparing a_0^{tot} against σ_{eff} (Eqs. F11–F12, defined in Appendix F) across the threshold grid produces a three-way classification of the 119 surveyed fields (Figures 29–32). We first require that the default threshold `pca.zero_snr` = 3 pass the noise-consistency test $|a_0| < \sigma_{\text{eff}}$; fields that fail this test (e.g., Ophiuchus A, OMC 3) are flagged as offset-affected regardless of behavior at larger thresholds. Among the remaining sample, fields whose minimum a_0^{tot} across the grid still exceeds σ_{eff} form Category 1, fields whose minimum falls below σ_{eff} but at thresholds other than three form Category 2, and fields satisfying $|a_0| < \sigma_{\text{eff}}$ throughout the entire tested range form Category 3. Applying these criteria yields 43 Category 1, 51 Category 2, and 25 Category 3 fields. These results are consistent with the MSB-level azimuth–elevation diagnostics of Figure 28: several fields that appear offset-free at the MSB level (e.g., OMC 1, NGC 6334 field 2 as a single field, NGC 7538) develop measurable offsets in their coadded products, indicating nonlinear accumulation of PCA interpolation residuals during coaddition. The mosaicking-step limitations of the public `pol2map` pipeline

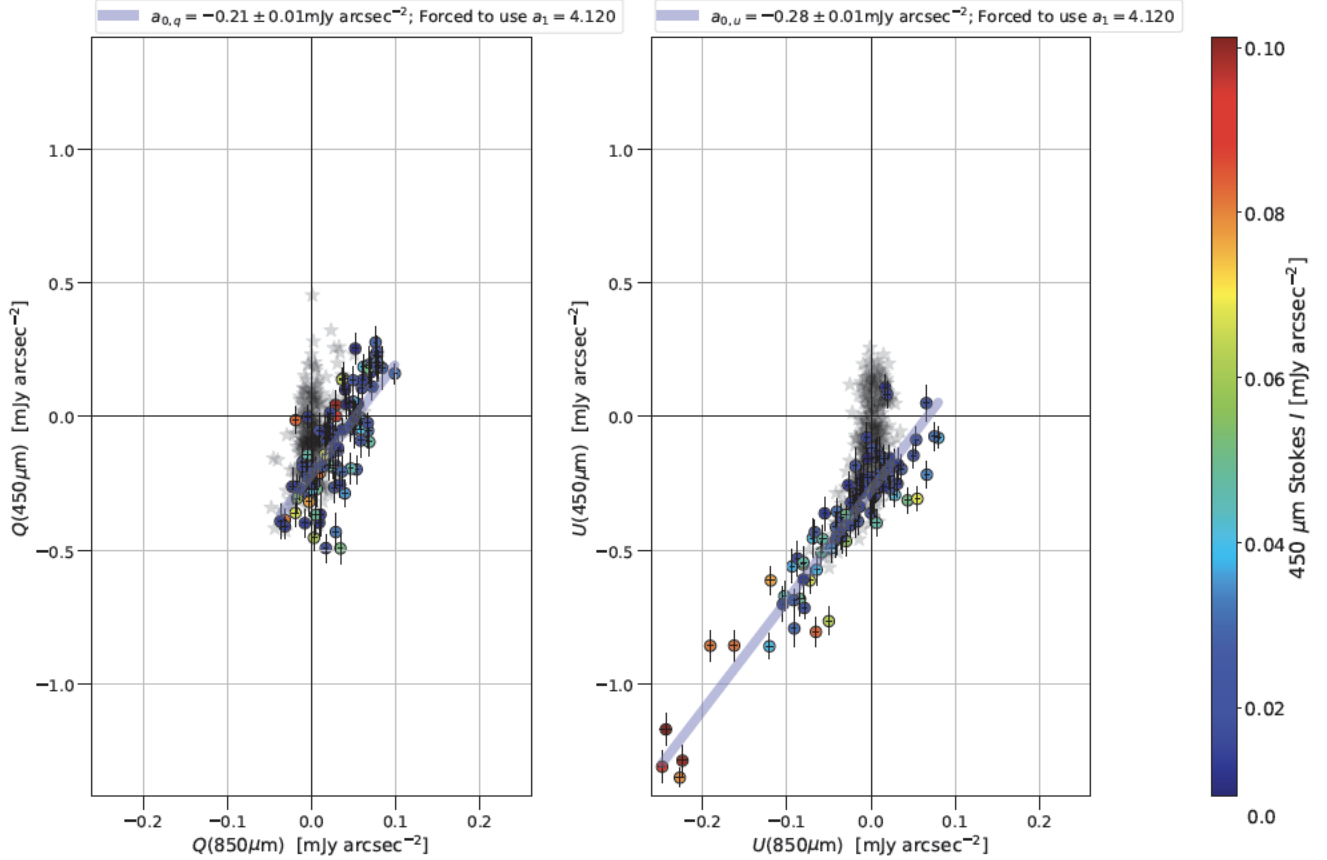


Figure 17. Final intercept refinement for Ophiuchus A (Step 3 of the three-step procedure), using the common slope $a_1^{\text{common}} = 4.120$ derived in Step 2 (Eq. 42; Figure 16). With the slope fixed, regressions reduce to one-parameter fits yielding $a_0^{Q,\text{final}} = -0.21 \pm 0.01 \text{ mJy arcsec}^{-2}$ and $a_0^{U,\text{final}} = -0.28 \pm 0.01 \text{ mJy arcsec}^{-2}$ (Eq. 43). These near-zero residual intercepts confirm that the three-step framework successfully removes the $450 \mu\text{m}$ zero-level offset while preserving the wavelength-independent polarization pattern required by constraint (6) in §3.4.5.

at $450 \mu\text{m}$, which restrict the optimization framework to single-field reductions, are documented in Appendix F; we therefore exclude mosaicked and mosaic-component reductions from the offset diagnostics and apply the corrected three-step framework only to single-field $450 \mu\text{m}$ data, with diagnostic case studies of the failure modes (OMC 1, NGC 1333, NGC 2264) presented in Appendix G.

Data Reduction Workflow—The final science products adopt the standard `pol2map` three-step pipeline with two field-dependent modifications: `flt.filt_edge_largescale=300` (enforcing the common $12''6\text{--}300''$ spatial-scale range together with `smooth450=yes`) and field-specific `pca.zero_snr/pca.zero_snrlo` thresholds from the optimization grid of Table 9; mosaicked and mosaic-component reductions, which do not respond to PCA-threshold modifications (§3.4.7), are excluded from this characterization. The full reproducible `pol2map` command set, together with the debiasing (§3.4.8) and `JY=no` flux-calibration choices, is documented in Appendix H.

3.4.8. Residual $450 \mu\text{m}$ Offsets After Field-Dependent PCA Optimization

The complete offset-removal framework—beam- and spatial-frequency matching, field-dependent PCA-threshold optimization (Table 9), and the three-step regression of §3.4.6—suppresses the dominant wavelength-dependent zero-level biases for all single-field $450 \mu\text{m}$ observations. The corrected Stokes parameters are

$$Q_{450}^{\text{corr}} = Q_{450} - a_0^{Q,\text{final}}, \quad U_{450}^{\text{corr}} = U_{450} - a_0^{U,\text{final}}, \quad (44)$$

where $(a_0^{Q,\text{final}}, a_0^{U,\text{final}})$ are the final intercepts of Eq. (40). The slope-constrained regression itself produces small additional residual intercepts $(a_0^{Q,\text{sc}}, a_0^{U,\text{sc}})$ that we document but do not subtract. Across the 42-source single-field

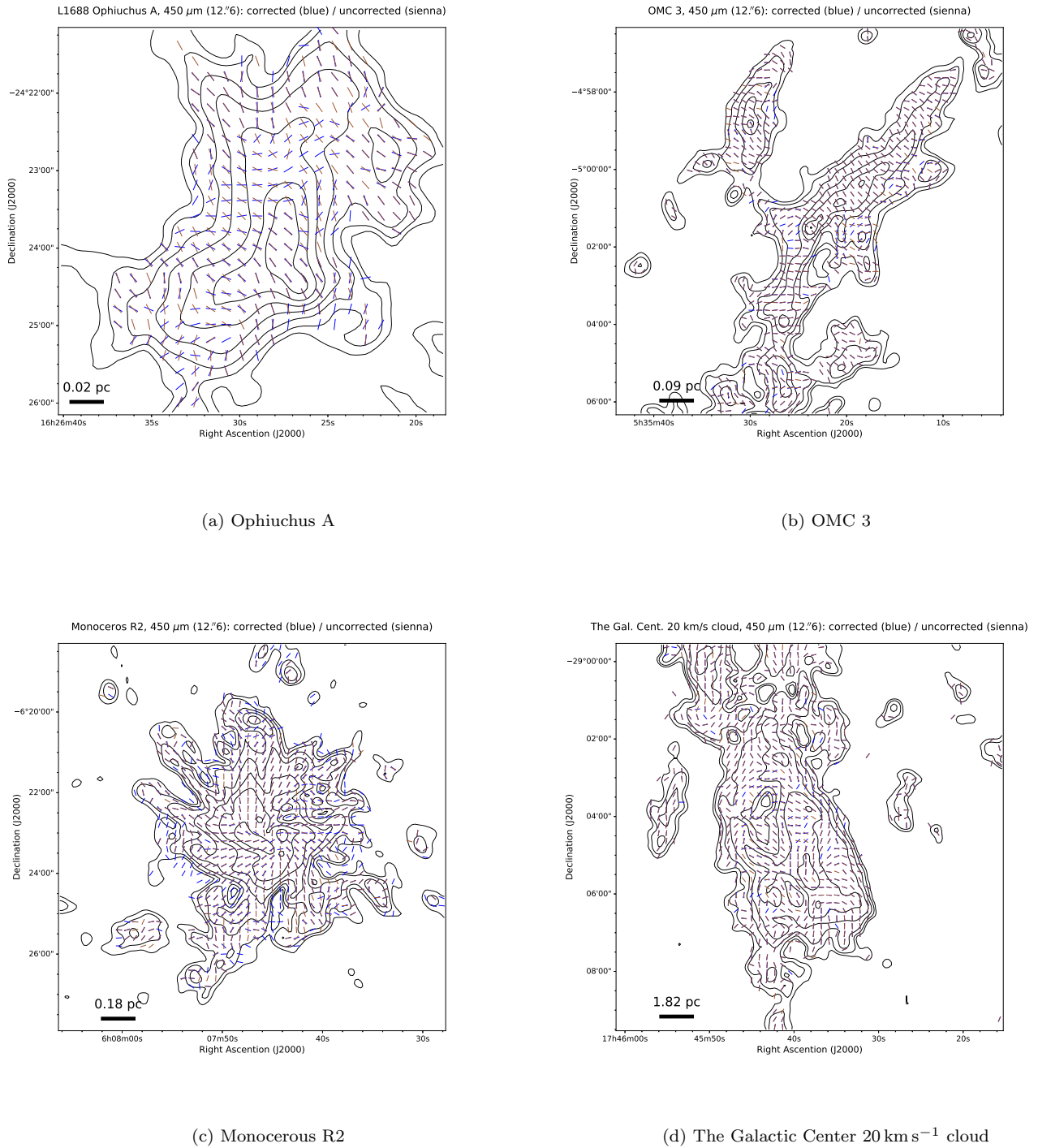


Figure 18. Comparison of 450 μm magnetic field orientations before offset correction, shown in brown, and after the correction procedure, shown in blue, for four representative single-field targets. Magnetic field pseudo-vectors are shown only where $I/\sigma_I > 30$, overlaid on 450 μm Stokes I contours at $10 \times 2^n \sigma_I$ for $n = 0, 1, \dots, 10$. Corrected maps employ optimized `pca.zero_snr` thresholds followed by the three-step offset-removal procedure (§3.4.4). Offset correction primarily affects low-intensity regions while preserving magnetic-field structure in bright emission.

sample, these residuals cluster tightly around zero, with sample means $\langle \cdot \rangle$ and median absolute amplitudes

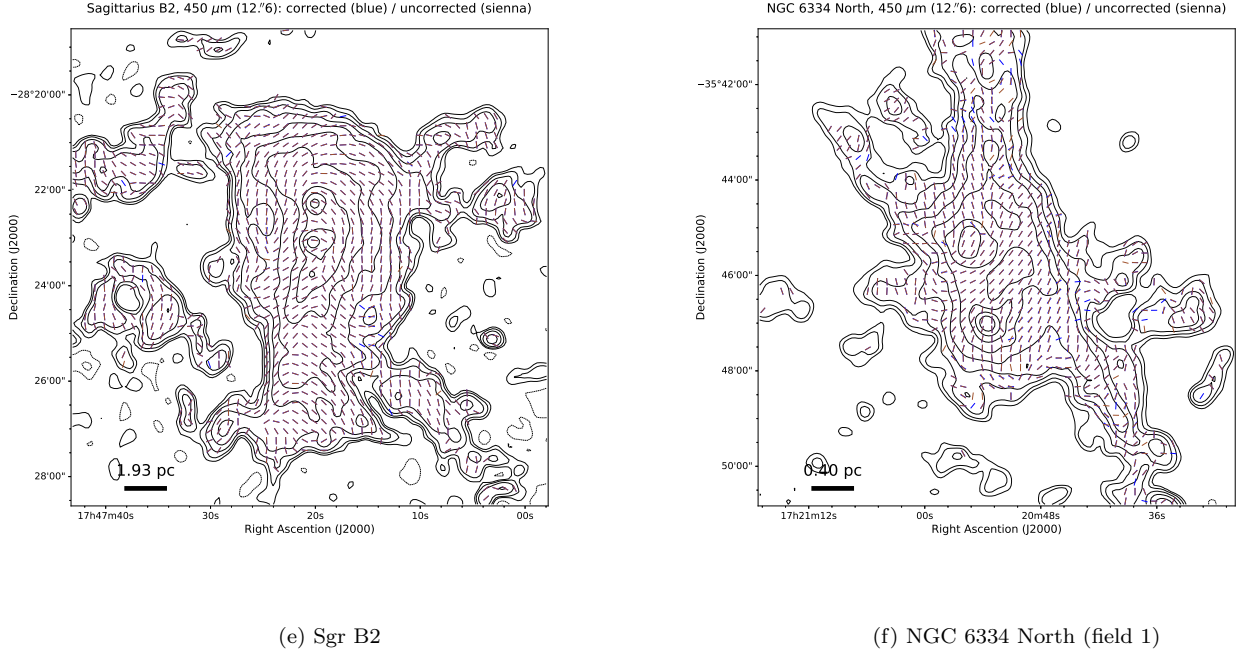


Figure 18. (Continued.) Same as Figure 18, but for two distant, high-dynamic-range regions. Offset correction yields the largest fractional improvements in faint cloud outskirts ($I/\sigma_I \lesssim 100$), while dense cores remain essentially unchanged. These examples illustrate the regime where 450 μm zero-level correction is essential for accurate dual-band polarimetry.

$$\langle a_0^{Q,sc} \rangle = -0.033, \quad \langle a_0^{U,sc} \rangle = -0.071 \quad (\text{mJy arcsec}^{-2}), \quad (45)$$

$$\text{median } |a_0^{Q,sc}| = 0.037, \quad \text{median } |a_0^{U,sc}| = 0.053 \quad (\text{mJy arcsec}^{-2}), \quad (46)$$

$$\langle \Delta a_0^{Q,sc} \rangle \simeq 0.036, \quad \langle \Delta a_0^{U,sc} \rangle \simeq 0.037 \quad (\text{mJy arcsec}^{-2}), \quad (47)$$

with combined residual amplitude $a_0^{\text{res}} = \sqrt{(a_0^{Q,sc})^2 + (a_0^{U,sc})^2}$ of median $6.7 \times 10^{-2} \text{ mJy arcsec}^{-2}$ and interquartile range $(4.4\text{--}13) \times 10^{-2} \text{ mJy arcsec}^{-2}$. The amplitudes are comparable to the per-fit statistical uncertainties (Eq. 47), demonstrating that the residuals are noise-dominated rather than systematic. As a representative case, Ophiuchus A yields $a_0^{Q,sc} = 0.009 \pm 0.013$ and $a_0^{U,sc} = 0.013 \pm 0.008 \text{ mJy arcsec}^{-2}$ —fully consistent with zero at 1σ and an order of magnitude below the raw intercepts of Eq. (38). The final 450 μm Stokes parameters used for scientific analysis are therefore those given by Eq. (44), without further intercept subtraction.

Application to Single-Field 450 μm Magnetic Field Maps—Figure 18 demonstrates the effect of the offset correction on magnetic-field maps for six representative single-field targets spanning Ophiuchus A ($d = 139 \text{ pc}$) to Sgr B2 ($d = 8340 \text{ pc}$). Magnetic-field pseudo-vectors are obtained by rotating the polarization angle by 90° under the standard assumption of optically thin thermal dust polarization (Hildebrand 1988; Andersson et al. 2015):

$$\chi = \frac{1}{2} \arctan\left(\frac{U_{450}^{\text{corr}}}{Q_{450}^{\text{corr}}}\right), \quad \theta_B = \chi + \frac{\pi}{2}, \quad (48)$$

with $(Q_{450}^{\text{corr}}, U_{450}^{\text{corr}})$ defined in Eq. (44) and vector segments shown only at $S/N \equiv I/\sigma_I > 30$ (Pattle et al. 2019). The default `pca.zero_snr = 3` map (brown vectors) is compared with the field-optimized + offset-corrected map (blue vectors); contours of Stokes I at $10 \times 2^n \sigma_I$ ($n = 0\text{--}10$) provide the underlying intensity reference. The correction primarily affects low-intensity regions ($\lesssim 100\sigma_I$) where the systematic zero-level contributes polarization comparable to or exceeding the intrinsic emission, removing spurious patterns and azimuthal distortions while leaving high- S/N cores ($\gtrsim 160\sigma_I$) essentially unchanged. This behavior is consistent across all six fields, including the Galactic-Center 20 km s^{-1} cloud and Sgr B2 at kiloparsec distances; the Ophiuchus A result is independently validated against SOFIA/HAWC+ $154 \mu\text{m}$ polarimetry in §3.4.8.

Independent Validation Using SOFIA/HAWC+ Polarimetry—The preceding subsection demonstrated that the $450\text{ }\mu\text{m}$ offset correction removes spurious polarization patterns in low-intensity regions while preserving robust magnetic field structures in high- S/N cores. To provide an independent validation of the corrected $450\text{ }\mu\text{m}$ data, we compare the magnetic field orientations with those obtained from SOFIA/HAWC+ observations at $154\text{ }\mu\text{m}$ (Band D) and $214\text{ }\mu\text{m}$ (Band E). The $450\text{ }\mu\text{m}$ wavelength lies between the POL-2 $850\text{ }\mu\text{m}$ and SOFIA bands, providing a critical test of whether the offset correction recovers magnetic field orientations consistent with neighboring wavelengths.

SOFIA/HAWC+ Data and Target Selection—We use archival SOFIA/HAWC+ polarimetric observations of eight BISTRO-overlap targets at Band D ($\theta_{\text{FWHM}} \approx 13''.6$ at $154\text{ }\mu\text{m}$) and Band E ($\theta_{\text{FWHM}} \approx 18''.2$ at $214\text{ }\mu\text{m}$; Harper et al. 2018): L1688 Ophiuchus A (Paper S2 Figure 27) and NGC 6334 South (Paper S2 Figure 77) in Band D, and OMC 1 (Paper S2 Figure 40), the Galactic Center 20 km s^{-1} cloud (Paper S2 Figure 87), the Brick (Paper S2 Figure 88), Sgr B2 (Paper S2 Figure 92), DR 21 (Paper S2 Figure 73), and Monoceros R2 (Paper S2 Figure 64) in Band E (PIs: D. Dowell, D. Chuss, T. Pillai, R. Pokhrel; SOFIA Science Archive). The SOFIA Stokes maps are reduced with SOFIA Redux v1.3.3 (Clarke & Vander Vliet 2023), converted into POL-2-compatible cubes via Starlink POLPACK, and reprojected onto the $12''$ POL-2 pixel grid using `reproject` (no beam convolution applied, preserving the SOFIA polarization information for direct angle comparison). For the multi-wavelength comparison we adopt $I/\sigma_I > 30$ for the POL-2 data and the standard HAWC+ quality cuts $I/\sigma_I > 200$, $PI/\sigma_{PI} > 3$, and $p < 50\%$ (e.g., Harper et al. 2018; Butterfield et al. 2024), restricting the analysis to pixels within $150''$ of the field center to avoid the ground-based POL-2 spatial-filtering edge (Paper S1 §2). DR 21 and Monoceros R2 are excluded from quantitative comparison because their median SOFIA-angle uncertainties ($\sigma_{\text{SOFIA}} = 90.4$ and 102.5 , respectively; Table 5) exceed a half-cycle of the polarization angle; the remaining six targets carry combined uncertainties $\sigma_{\text{diff}} = \sqrt{\sigma_{\text{POL2}}^2 + \sigma_{\text{SOFIA}}^2}$ in the range $13.3\text{--}20.9$, suitable for robust statistical comparison.

Qualitative Comparison: Ophiuchus A as a Representative Case—We use Ophiuchus A as the representative case because it carries the largest $450\text{ }\mu\text{m}$ offset correction among the SOFIA-validated targets (Category 1) and the highest-quality SOFIA Band D data. Figure 19 compares the inferred plane-of-sky $\vec{B}$ -field orientations from POL-2 $850\text{ }\mu\text{m}$ (green), uncorrected $450\text{ }\mu\text{m}$ (brown), corrected $450\text{ }\mu\text{m}$ (blue), and SOFIA/HAWC+ Band D $154\text{ }\mu\text{m}$ (magenta). The $850\text{ }\mu\text{m}$ –SOFIA pair agrees to within $\lesssim 10^\circ\text{--}15^\circ$ along the bright ridge (panel a); the uncorrected $450\text{ }\mu\text{m}$ vectors deviate by $\sim 20^\circ\text{--}30^\circ$ (panel b); and the offset correction reduces the $450\text{ }\mu\text{m}$ –SOFIA discrepancy to the same $\lesssim 10^\circ\text{--}15^\circ$ level achieved by the $850\text{ }\mu\text{m}$ –SOFIA reference (panel c). An additional S/N -stratified CDF check on Ophiuchus A (low- vs. high-intensity subsets) confirms that this qualitative behavior is representative across the full surface-brightness range, with no statistically significant intensity dependence given the limited per-bin sample size.

Quantitative Comparison Using Cumulative Distribution Functions—To extend the validation to the full set of validated targets, we compute cumulative distribution functions (CDFs) of the absolute polarization-angle differences $|\Delta\theta|$ between SOFIA/HAWC+ and the three POL-2 datasets ($850\text{ }\mu\text{m}$, uncorrected $450\text{ }\mu\text{m}$, corrected $450\text{ }\mu\text{m}$). Each CDF uses all matched pixels for which the angle difference is defined, so the sample size N may differ between wavelength pairs. Figure 20 presents the CDFs for the six validated targets and Table 5 summarizes the median angle differences and sample sizes.

The CDF analysis reveals three behavioral regimes among the validated targets. First, Ophiuchus A and the Brick—both Category 1 fields with significant raw offsets—show a clear improvement after correction: the median $450\text{ }\mu\text{m}$ –SOFIA difference decreases from 34.2 to 24.4 for Ophiuchus A (against the 22.1 $850\text{ }\mu\text{m}$ baseline), and from 29.2 to 20.6 for the Brick (against 16.2); two-sample Kolmogorov–Smirnov tests confirm that the corrected $450\text{ }\mu\text{m}$ CDFs become statistically consistent with the $850\text{ }\mu\text{m}$ reference in both cases. Second, NGC 6334 South—a Category 2 field where the measured offsets are consistent with noise (§3.4.7; Figure 31)—shows identical raw and corrected $|\Delta\theta|$ medians of 13.4 , validating the conservative threshold logic that applies corrections only when statistically significant offsets are detected. Third, OMC 1, the Galactic Center 20 km s^{-1} cloud, and Sgr B2 show more complex behaviors: OMC 1 has a 10.7 baseline against $15.3/16.1$ raw/corrected $450\text{ }\mu\text{m}$ medians; the Galactic Center 20 km s^{-1} cloud has 21.9 baseline against $28.3/29.0$; Sgr B2 inverts the pattern with a 27.5 baseline against $20.9/20.2$ at $450\text{ }\mu\text{m}$. These diverse behaviors plausibly reflect genuine wavelength-dependent dust or grain-alignment effects in the different environments probed. Overall, the offset correction improves agreement with independent FIR measurements where systematic biases were largest while leaving angles essentially unchanged in fields where $450\text{ }\mu\text{m}$ already agreed with $850\text{ }\mu\text{m}$.

Table 5. Summary of POL-2 and SOFIA/HAWC+ Polarization Angle Comparison

Target	λ_{SOFIA} (μm)	Category	N	σ_{POL2} (deg)	σ_{SOFIA} (deg)	σ_{diff} (deg)	850–SOFIA (deg)	450raw–SOFIA (deg)	450corr–SOFIA (deg)
<i>Valid Comparisons</i>									
Ophiuchus A	154	1	53	18.4	9.0	20.5	22.1	34.2	24.4
NGC 6334 South	154	2	88	13.8	4.4	14.4	16.4	13.4	13.4
OMC 1	214	1	47	13.1	2.0	13.3	10.7	15.3	16.1
GC 20 km s ^{−1}	214	1	79	13.9	13.4	19.3	21.9	28.3	29.0
The Brick	214	1	64	15.5	14.1	20.9	16.2	29.2	20.6
Sgr B2	214	1	101	13.2	10.0	16.6	27.5	20.9	20.2
<i>Excluded from Quantitative Analysis</i>									
DR 21	214	1	...	15.7	90.4	...	...	...	...
Monoceros R2	214	1	...	17.0	102.5	...	...	...	...

NOTE— Column descriptions: λ_{SOFIA} is the SOFIA/HAWC+ filter wavelength (154 μm corresponds to Band D; 214 μm corresponds to Band E); Category refers to the offset classification defined in §3.4.7 (1: significant offset requiring correction; 2: offset consistent with noise; see Figures 29–32); N is the number of matched pixels satisfying $I_{850}/\sigma_{I,850} > 30$ within 150'' of the field center; σ_{POL2} and σ_{SOFIA} are the median polarization angle uncertainties for POL-2 (850 μm) and SOFIA/HAWC+, respectively; $\sigma_{\text{diff}} = \sqrt{\sigma_{\text{POL2}}^2 + \sigma_{\text{SOFIA}}^2}$ is the combined angle uncertainty; the final three columns give the median absolute polarization angle differences $|\Delta\theta|$ between SOFIA and the three POL-2 datasets. DR 21 and Monoceros R2 are excluded because their SOFIA polarization angle uncertainties exceed 90°, rendering quantitative comparison impossible. NGC 6334 South is a Category 2 source where the measured 450 μm offsets were consistent with noise; the identical “450raw” and “450corr” values confirm that no significant correction was applied.

Debiasing of Polarized Intensity—The final polarimetric products ($P_{\text{MAS}}, p_{\text{MAS}}, \chi$) adopted throughout our analysis are debiased with the Modified Asymptotic Estimator of Plaszczyński et al. (2014) and Montier et al. (2015), applied to the offset-corrected Stokes parameters of Eq. (44). The Stokes-component-weighted noise term, the explicit MAS expression, and the corresponding debiased polarization fraction p_{MAS} and angle χ are documented in Appendix I.

3.4.9. Summary of Technical Validation 4

This section completes the construction stage of Pillar 3 of the technical credibility chain. The 450 μm Stokes parameters that enter the companion science analyses (Papers S1 and S2) have been brought onto a common spatial footing with the 850 μm data through three independent corrections that together render the dual-band map products quantitatively comparable: the wavelength-dependent additive offset has been removed at the ~ 1 mJy arcsec^{−2} level (residual $\lesssim 0.07$ mJy arcsec^{−2}), independently validated against SOFIA/HAWC+ polarimetry of Ophiuchus A; the 450 μm effective beam has been transformed to a 12''6 Gaussian matched to the 850 μm effective FWHM; and the 450 μm spatial-frequency recovery has been truncated to the common 12''6–300'' scale range matched to the standard 850 μm `pol2map` reduction. The remaining technical question is whether these constructions in fact preserve the *polarization-angle morphology* required by the dual-band science analyses; this is the quantitative capstone validation provided in the next section.

3.5. Technical Validation 5: Validation of Beam-Harmonized Polarization Angles

3.5.1. Motivation and overview

Using the corrected and beam-harmonized 450 μm maps from §3.4, we now assess the residual geometric polarization-angle error floor that survives the dual-band reduction procedure. Localized inter-band polarization-angle residuals of several to $\sim 10^\circ$ are visible in a limited number of regions when the final 450 and 850 μm polarization maps are compared, with the clearest example occurring in the L1688 Ophiuchus A field (companion Paper S1 F1 analysis). The

Table 6. Systematic Uncertainty Cross-Tabulation for the Dual-Band Ratio Diagnostics

Systematic source	Adopted magnitude	Effect on P_{450}/P_{850}	Effect on $\gamma - \alpha$
Absolute flux calibration — $850\ \mu\text{m}^a$	$\pm 7\%$	$\approx 0^d$	$\approx 0^d$
Absolute flux calibration — $450\ \mu\text{m}^a$	$\pm 17\%$	$\approx 0^d$	$\approx 0^d$
$450\ \mu\text{m}$ Q/U offset residual ^b	$\lesssim 0.07\ \text{mJy arcsec}^{-2}$	$\lesssim 0.02$	$\lesssim 0.04$
Stokes- I intensity threshold ^f	$I/\sigma_I > 100$ at $450\ \mu\text{m}$	filtered upstream ^g	filtered upstream ^g
Beam morphology mismatch ^c	Wiener deconvolution; $\sim 90\%$ -correlated $\pm 15\%$ beam solid-angle	$\approx 0^e$	$\approx 0^e$
Beam spatial-frequency truncation ^c	$300''$ common-scale cutoff	$\approx 0^e$	$\approx 0^e$
Practical systematic floor (combined)	—	$\lesssim 0.02$	$\lesssim 0.04$

^aExtended-source FCF uncertainties are taken from Mairs et al. (2021); the surface-brightness calibration framework is described in §3.1 and its contribution to the error budget is discussed in §4.1.1.

^bThe three-step $450\ \mu\text{m}$ offset-correction procedure is described in §3.4 and §3.4.4; the residual level is characterized in §3.4.8 and propagated in §4.1.2.

^cThe beam-harmonization procedure is described in §3.5 and §3.4.4; its contribution to the systematic budget is discussed in §4.1.3. The two beam-related rows decompose the harmonization sequence: native-to-Gaussian morphology replacement via Wiener deconvolution (first row) followed by truncation of the $450\ \mu\text{m}$ largest-recoverable scale to the common $12''6$ – $300''$ range matched to the standard $850\ \mu\text{m}$ reduction (second row).

^dFCF errors are multiplicative and cancel in $P = PI/I$. They also enter α and γ in common mode and therefore cancel in $\gamma - \alpha$.

^eResidual beam-response errors affect Stokes I , Q , and U through the same angular response to first order, preserving P after beam harmonization. Their common-mode contribution to α and γ cancels in $\gamma - \alpha$.

^fThe $I/\sigma_I > 100$ intensity threshold at $450\ \mu\text{m}$, justified empirically in §3.2, sets the entry condition for the systematic budget by removing pixels susceptible to running-median bias upstream of the dual-band derived quantities.

^gAbove the intensity threshold, polarized intensity is bias-corrected by the modified asymptotic estimator (MAS) of Plaszczynski et al. (2014) and Montier et al. (2015); the residual statistical scatter on the dual-band derived quantities is reported separately as δX_{stat} in the companion-paper uncertainty notation rather than as a systematic floor in this table.

NOTE— The cross-tabulation lists the seven systematic terms developed in §4.1 as rows against the two principal dual-band derived quantities of scientific interest as columns: the polarization-fraction ratio P_{450}/P_{850} and the spectral-index difference $\gamma - \alpha$. The seven rows decompose the budget more finely than the three narrative subsections of §4.1 (FCF, Q/U offset, beam) by separating the calibration row pair (FCF at each wavelength), the additive Q/U offset row, the sample-selection entry condition (Stokes- I intensity threshold), the two-row beam-harmonization decomposition (morphology and spatial-frequency components), and the integrated practical floor. The three statistical estimators of P_{450}/P_{850} used in the companion analyses are defined in Paper S2 §3.3. The spectral-index difference $\gamma - \alpha$ is defined in Equation (1), and its algebraic relation to P_{450}/P_{850} is given in Equation (2). The practical-floor row of $\sigma(\gamma - \alpha) \lesssim 0.04$ is an order of magnitude smaller than the observed median displacement $|\gamma - \alpha| \approx 0.52$ (Paper S1 F2), confirming that the spectral-index difference is not produced by the systematic effects quantified here.

technical question is whether these residuals indicate intrinsic wavelength-dependent differences in the traced magnetic-field geometry, or whether they are primarily introduced by the $450\ \mu\text{m}$ beam- and spatial-frequency harmonization procedures applied during the dual-band reduction. To isolate these two possibilities we perform an explicit internal validation that compares three independently processed $450\ \mu\text{m}$ polarization products derived from the same underlying Stokes observations: (i) the *direct native-beam* product (`pol2map` flag `smooth450=no`, default $450\ \mu\text{m}$ spatial-frequency coverage), which uses the native two-component compound $450\ \mu\text{m}$ point-spread function modelled by Equation (10) of Mairs et al. (2021): $G_{\text{total}} = \alpha G_{\text{MB}} + \beta G_{\text{S}}$ with main-beam FWHM $\theta_{\text{MB}} = 6''.2$ and amplitude $\alpha = 0.89$, secondary-beam FWHM $\theta_{\text{S}} = 18''.8$ and amplitude $\beta = 0.11$ (Mairs+2021 Table 6); (ii) an analytically *deconvolved-reconvolved*

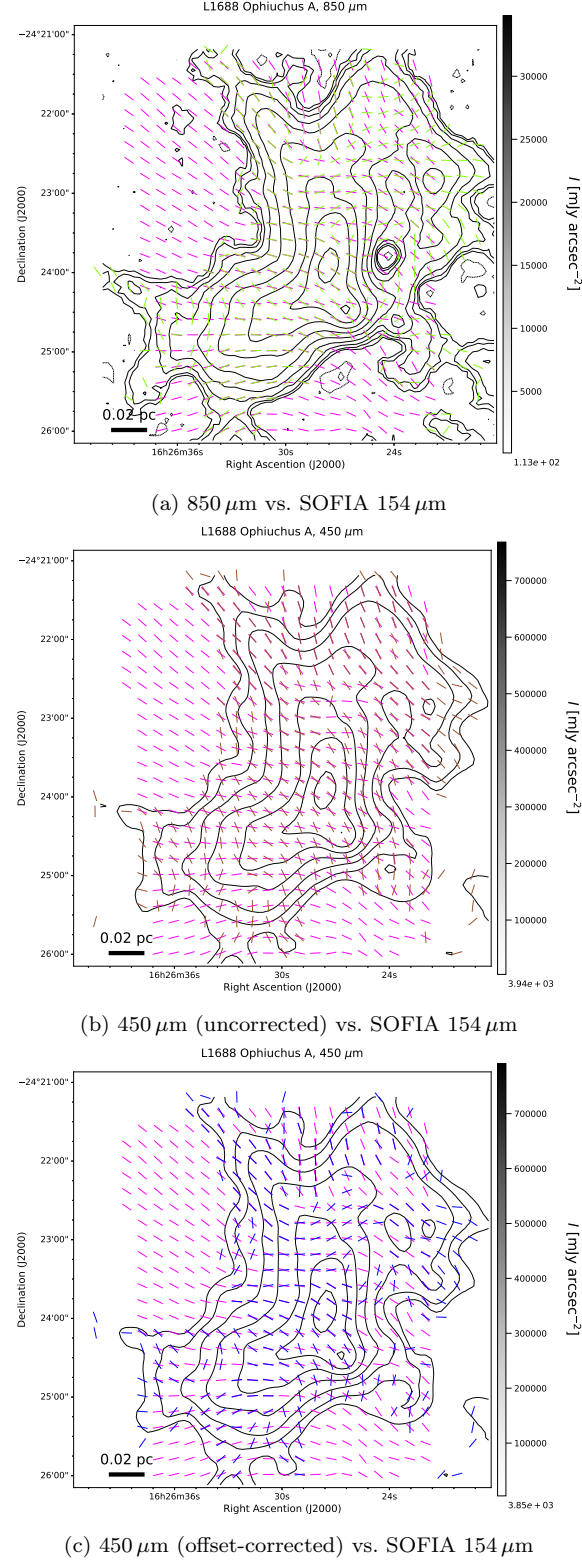


Figure 19. Comparison of magnetic field orientations in the central region of L1688 Ophiuchus A at 850 μm (green), at 450 μm before (brown) and after (blue) offset correction, and from SOFIA/HAWC+ Band D at 154 μm (magenta). The POL-2 pseudo-vectors are shown where $I/\sigma_I > 30$ and $PI/\sigma_{PI} > 0$, while the SOFIA vectors are selected using $I/\sigma_I > 200$, $PI/\sigma_{PI} > 3$, and $p < 50\%$. All vectors are displayed over the corresponding POL-2 Stokes I contours, which begin at 10σ with σ estimated from blank-sky regions. Panel (a) shows the good agreement between the 850 μm and 154 μm orientations. Panel (b) reveals a systematic angular offset between the uncorrected 450 μm and 154 μm vectors across the bright ridge. Panel (c) demonstrates that the offset-corrected 450 μm map becomes consistent with both 154 μm and 850 μm , confirming the effectiveness of the offset-removal procedure (§3.4.6). A quantitative comparison of the angular differences is presented in Figure 20 and §3.4.8.

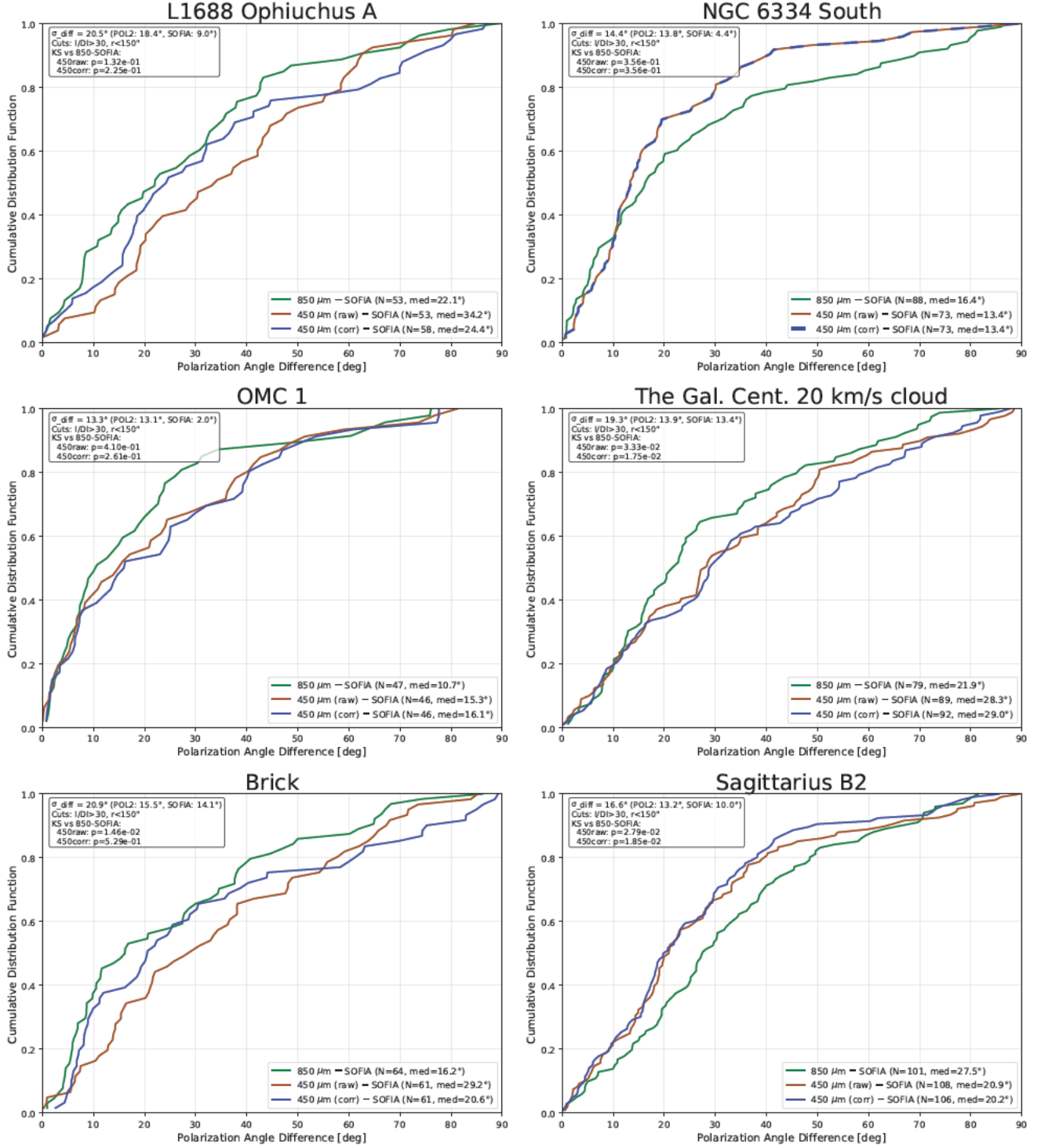


Figure 20. Cumulative distribution functions (CDFs) of the absolute polarization angle differences $|\Delta\chi|$ between SOFIA/HAWC+ and POL-2 data for the six validated targets. The green curves show the 850 μm –SOFIA comparisons, which serve as the baseline expectation for wavelength-to-wavelength variations. The brown curves show the uncorrected 450 μm –SOFIA comparisons. The blue curves show the offset-corrected 450 μm –SOFIA comparisons. For NGC 6334 South, the blue curve is shown as a dashed line to indicate that raw and corrected 450 μm data are identical, confirming that no significant offset correction was required for this Category 2 source (§3.4.7; Figure 31). Quantitative statistics for each target are provided in Table 5. The Ophiuchus A panel is reproduced here together with the other SOFIA-validated fields to enable direct comparison across §3.4.8.

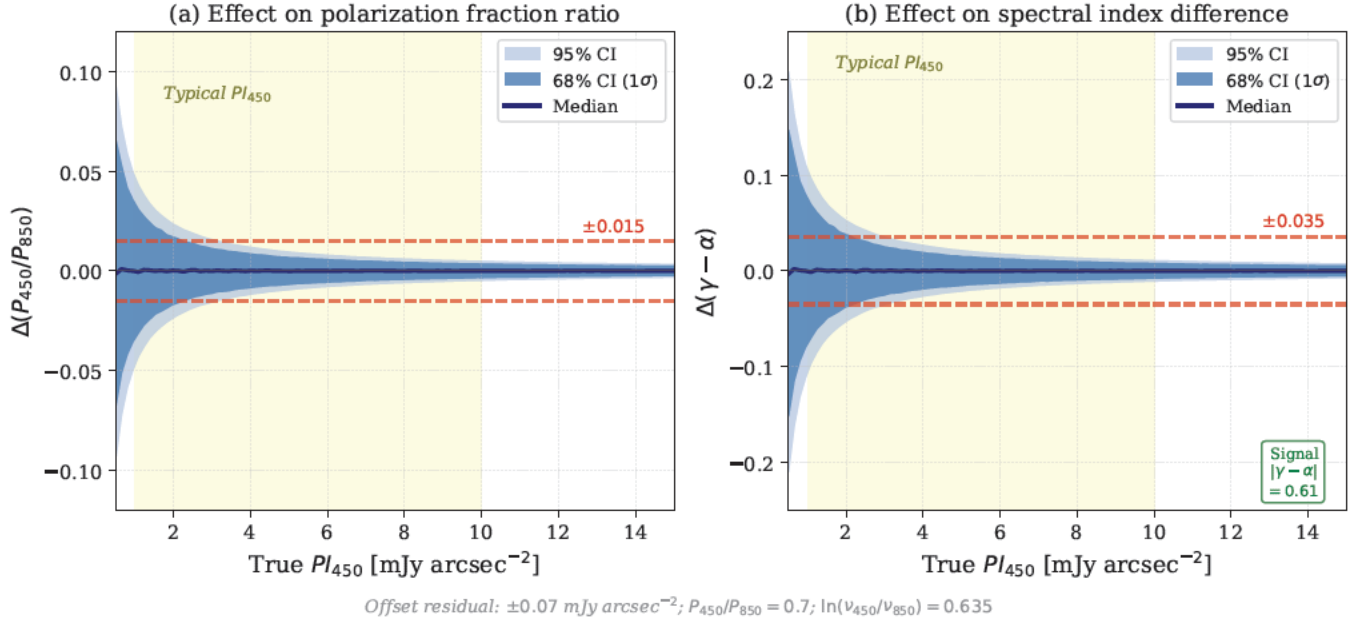


Figure 21. Monte Carlo simulation of systematic errors from the 450 μm offset correction residual ($\pm 0.07 \text{ mJy arcsec}^{-2}$; §3.4.8). (a) Effect on the polarization fraction ratio P_{450}/P_{850} (computed via Methods 1–3 of Paper S2 §3.3). (b) Effect on the spectral index difference $\gamma - \alpha$ (Equation 1). Shaded bands show the 68% (dark blue) and 95% (light blue) confidence intervals from 10^4 Monte Carlo realizations at each PI_{450} value. The yellow shaded region indicates the typical range of PI_{450} in our sample (1–10 mJy arcsec^{-2}). Red dashed lines mark the $\pm 1\sigma$ values quoted in Table 6. In panel (b), the green annotation indicates the magnitude of the detected signal ($|\gamma - \alpha| = 0.52$; Paper S1 F2 (observed $|\gamma - \alpha| \approx 0.52$)), demonstrating that systematic errors are small compared to the astrophysical signal.

product (`smooth450=yes`, default 450 μm spatial-frequency coverage), in which the native two-component compound beam has been replaced by a single 12".6 Gaussian beam matched to the 850 μm effective FWHM, while the 450 μm map-making retains its native largest-recoverable-scale of $\sim 600''$; and (iii) the *final science product* (`smooth450=yes` together with `flt.filt_edge_largescale=300`), in which the 450 μm data are both beam-matched to the same 12".6 Gaussian and spatial-frequency-truncated so that both wavelengths sample the *common* angular-scale range 12".6–300" matched to the standard 850 μm `pol2map` reduction.

Because all three products originate from identical 450 μm input visibilities, the (i)-to-(ii) and (ii)-to-(iii) comparisons cleanly factorize the harmonization pipeline into two physically distinct stages: the (i)-to-(ii) step isolates the contribution of *beam morphology*, in which the native compound point-spread function is replaced by a single 12".6 Gaussian, while the (ii)-to-(iii) step measures the additional effect of *spatial-frequency-range matching*, in which the 450 μm recovery is truncated from its native $\sim 600''$ largest scale to the 300" cutoff enforced by the standard 850 μm reduction. The canonical residual statistic adopted throughout this section is the source-level median absolute polarization-angle difference, denoted $|\widehat{\Delta\chi}|$, with the median absolute deviation (MAD) of the underlying $\Delta\chi$ distribution reported as a parenthetical scatter descriptor.

This section serves as the *quantitative capstone of Pillar 3 of the methodology* and, together with the calibration and noise pillars established in §§3.1–3.3, establishes the survey-level systematic-reliability framework that defines the polarization-angle uncertainty floor adopted throughout the companion science papers (Papers S1 and S2). The dual-band beam construction step itself is described in §3.4; the present section provides the empirical demonstration that the constructed product preserves the polarization-angle information required by the subsequent science analyses (Pillar 3 capstone entry of Figure 1 and Table 1).

We proceed in two stages: a representative high- S/N case study on L1688 Ophiuchus A (§3.5.2, with both 12" polarization-catalog and 4" Stokes-map robustness checks) followed by a survey-level statistical generalization across the dual-band sample (§3.5.3, with per-source diagnostics in Figure 25 as a full-transparency reference layer). The

L1688 case study provides the physical intuition for *how* the harmonization improves the angle field, while the survey-level test provides the *statistical guarantee* that this behaviour is generic across the BISTRO 450/850 μm sample.

3.5.2. Representative high- S/N example: L1688 Ophiuchus A

Figure 22 presents the polarization-catalog validation for L1688 Ophiuchus A, evaluated on the final 12''-sampled polarization catalog at $I/\sigma_I > 30$ in both compared products. Panel (a) shows the native 450 μm two-component beam decomposition $\alpha G_{\text{MB}} + \beta G_{\text{S}}$ adopted from Equation (10) and Table 6 of Mairs et al. (2021) (main-beam FWHM 6''.2, $\alpha = 0.89$; secondary-beam FWHM 18''.8, $\beta = 0.11$), together with the 12''.6 Gaussian deconvolution–reconvolution target. The direct comparison of the native-beam product against the final beam-harmonized science product (panel b) yields a median signed offset of only $+2''.5$ but a sample-wide $|\widetilde{\Delta\chi}| = 9''.3$ (MAD = $8''.7$); the same comparison after the deconvolve–reconvolve step (panel c) reduces the residual to $|\widetilde{\Delta\chi}| = 4''.3$ (MAD = $4''.0$). A two-sample Kolmogorov–Smirnov test on the underlying $|\Delta\chi|$ distributions yields $D = 0.31$ with $p < 10^{-10}$, formally rejecting the null hypothesis that the pre- and post-harmonization residual distributions are drawn from the same parent. This statistically robust factor-of-two contraction at fixed S/N demonstrates that *the majority of the apparent local angular mismatch in L1688 originates from the morphology of the native 450 μm compound beam rather than from intrinsic changes in the polarization-angle field.*

A supplementary pixel-domain comparison on the underlying 4'' Stokes-map grid (i.e., on the native $\{I, Q, U\}$ extended-emission maps prior to 12'' polarization-vector resampling) confirms the same picture (Figure 23). At $I/\sigma_I > 30$ the direct native-beam comparison gives $|\widetilde{\Delta\chi}| \approx 16''.9$, decreasing to $\approx 5''.5$ after deconvolution and reconvolution; at the more conservative $I/\sigma_I > 100$ threshold the corresponding values are $\approx 10''.1$ and $\approx 4''.1$. The two-sample Kolmogorov–Smirnov test on the higher-statistics 4'' map sample yields $D = 0.45$ with $p < 10^{-55}$ at $I/\sigma_I > 30$, an even more decisive distributional separation than the 12'' catalog result and confirming that the pre- and post-harmonization $|\Delta\chi|$ distributions are statistically incompatible at the per-pixel level. The agreement between the 12'' polarization-catalog and 4'' Stokes-map tests shows that the post-harmonization improvement is *not* an artefact of 12'' vector sampling, but reflects genuine stabilization of the underlying polarization-angle morphology after beam replacement.

3.5.3. Survey-level validation across the dual-band statistical sample

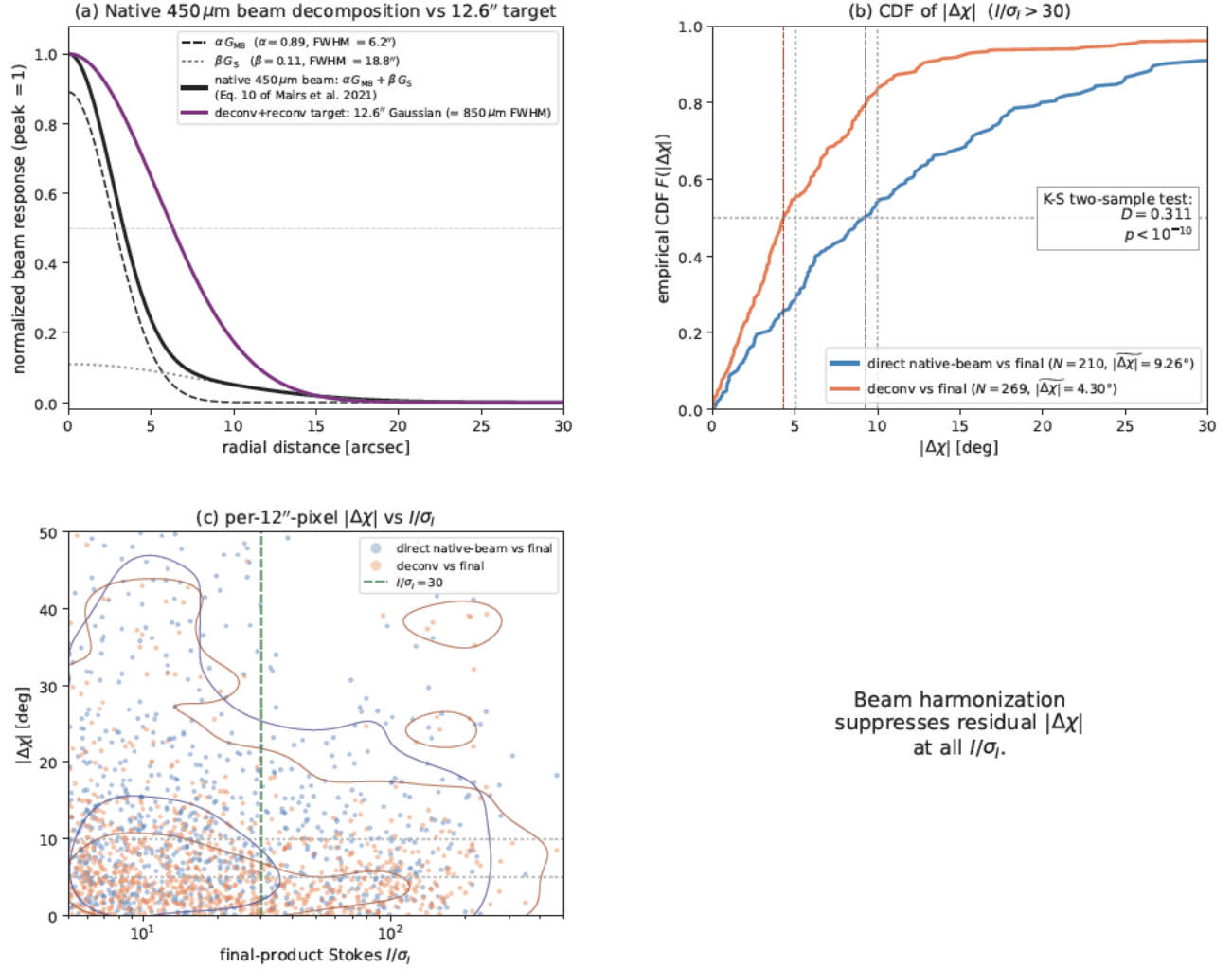
To establish that the L1688 result is representative of the broader 450 μm polarization sample rather than peculiar to a single bright field, we extend the same three-product comparison to all sources in the SPIDEX sample of Paper S1 for which the trio of products (i)–(iii) is available and which yield ≥ 20 matched pixels at $I/\sigma_I > 30$ in both compared catalogs. This filter retains 24 of the 28 sources: three sources processed only through the mosaicked-pipeline path lack the native-beam product (i), and one further source (Sgr B2) provides too few high- S/N pixels for a robust per-source statistic. Figure 24 summarizes the resulting per-source comparison.

The headline survey-level results, computed across the 24 statistical-sample sources, are:

1. the *source-median* $|\widetilde{\Delta\chi}|$ decreases from $5''.9$ before harmonization to $4''.9$ after the deconvolve–reconvolve step (a per-source median improvement factor of 1.31, 16–84 percentile range 1.00–1.77);
2. 83% of sources show a measurable improvement (improvement factor > 1);
3. all 24 sources (100%) reach a post-harmonization $|\widetilde{\Delta\chi}|$ *below* the $10''$ physical-tolerance scale adopted in companion Paper S1 F1; and
4. 58% of sources reach a post-harmonization $|\widetilde{\Delta\chi}|$ below $5''$.

The per-environment breakdown is consistent with this picture: the LMSF subset moves from a source-median $|\widetilde{\Delta\chi}|$ of $6''.6$ to $5''.0$, the CMZ subset from $7''.9$ to $4''.8$ —the largest fractional improvement, indicating that the harmonization is effective even in the spatially complex Galactic-Centre environment—and the HMSF subset from $5''.2$ to $5''.1$. The small HMSF improvement is the expected outcome rather than a pathology: the HMSF sample is dominated by bright, compact polarized peaks in which the native 450 μm compound beam already produces a relatively modest local angular mismatch, leaving correspondingly little room for post-harmonization gain.

For external corroboration we further apply the same comparison to the 147 sources in the 450 μm archival reduction pool that have all three reduction variants of §3.5 available—(i) the native compound 450 μm beam with the default



Beam harmonization
suppresses residual $|\Delta\chi|$
at all I/σ_I .

Figure 22. Representative beam-harmonization validation in the final 12''-sampled polarization catalog for L1688 Ophiuchus A. The same three 450 μm products described in the text are compared at $I/\sigma_I > 30$ (in both products). (a) Native 450 μm beam decomposition from Equation (10) and Table 6 of Mairs et al. (2021): main-beam component αG_{MB} (FWHM $\theta_{\text{MB}} = 6''.2$, $\alpha = 0.89$; black dashed), secondary-beam component βG_{S} (FWHM $\theta_{\text{S}} = 18''.8$, $\beta = 0.11$; grey dotted), and the combined two-component profile (black solid), shown together with the 12.6'' Gaussian deconvolution–reconvolution target (purple solid; equal to the 850 μm effective FWHM). (b) Empirical cumulative distribution function (CDF) of $|\Delta\chi|$ per-pixel for the *direct native-beam vs final* comparison (blue) and the *deconvolve–reconvolve vs final* comparison (orange) against the same final beam-harmonized science product; vertical dashed lines mark the per-distribution medians, light dotted lines mark the 5° and 10° guides, and the inset reports the two-sample Kolmogorov–Smirnov test statistic D and p -value testing the null hypothesis that the two $|\Delta\chi|$ distributions are drawn from the same parent distribution. (c) Per-pixel $|\Delta\chi|$ as a function of the final-product Stokes I/σ_I for both comparison pairs (per-pixel scatter plus 50/90 percentile KDE contours). The deconvolve–reconvolve step alone reduces the median absolute angle residual from $|\Delta\chi| = 9.3$ to 4.3 at $I/\sigma_I > 30$, isolating the beam-morphology contribution to the local PA mismatch from the spatial-frequency-truncation contribution that the subsequent (ii)-to-(iii) step would add.

$\sim 600''$ largest-recoverable angular-scale coverage; (ii) the analytically deconvolved-and-reconvolved 12.6'' Gaussian beam with the same native spatial-frequency range retained; and (iii) the final science product, combining the same 12.6'' Gaussian beam with spatial-frequency truncation to the common 12.6''–300'' scale range matched to the standard 850 μm reduction—and 94 of these pass the $N_{\text{samp}} \geq 20$ filter and span the 24 statistical-sample sources together with 70 additional fields (compact-source calibrators, mosaic-component fields, and other archival sources). The aggregate metrics on the 94-source pool are quantitatively consistent with the 24-source statistical sample at the level of the source-median residuals (6.4 before, 4.9 after), the per-source improvement factor (1.31), and the fraction of sources

L1688 Oph A: 4'' map-domain beam-processing sensitivity ($|\widetilde{\Delta\chi}|_{\text{native}} \rightarrow |\widetilde{\Delta\chi}|_{\text{deconv}} = 16.8^\circ \rightarrow 5.5^\circ$ at $I/\sigma_I > 30$, $= 10.1^\circ \rightarrow 4.1^\circ$ at $I/\sigma_I > 100$)

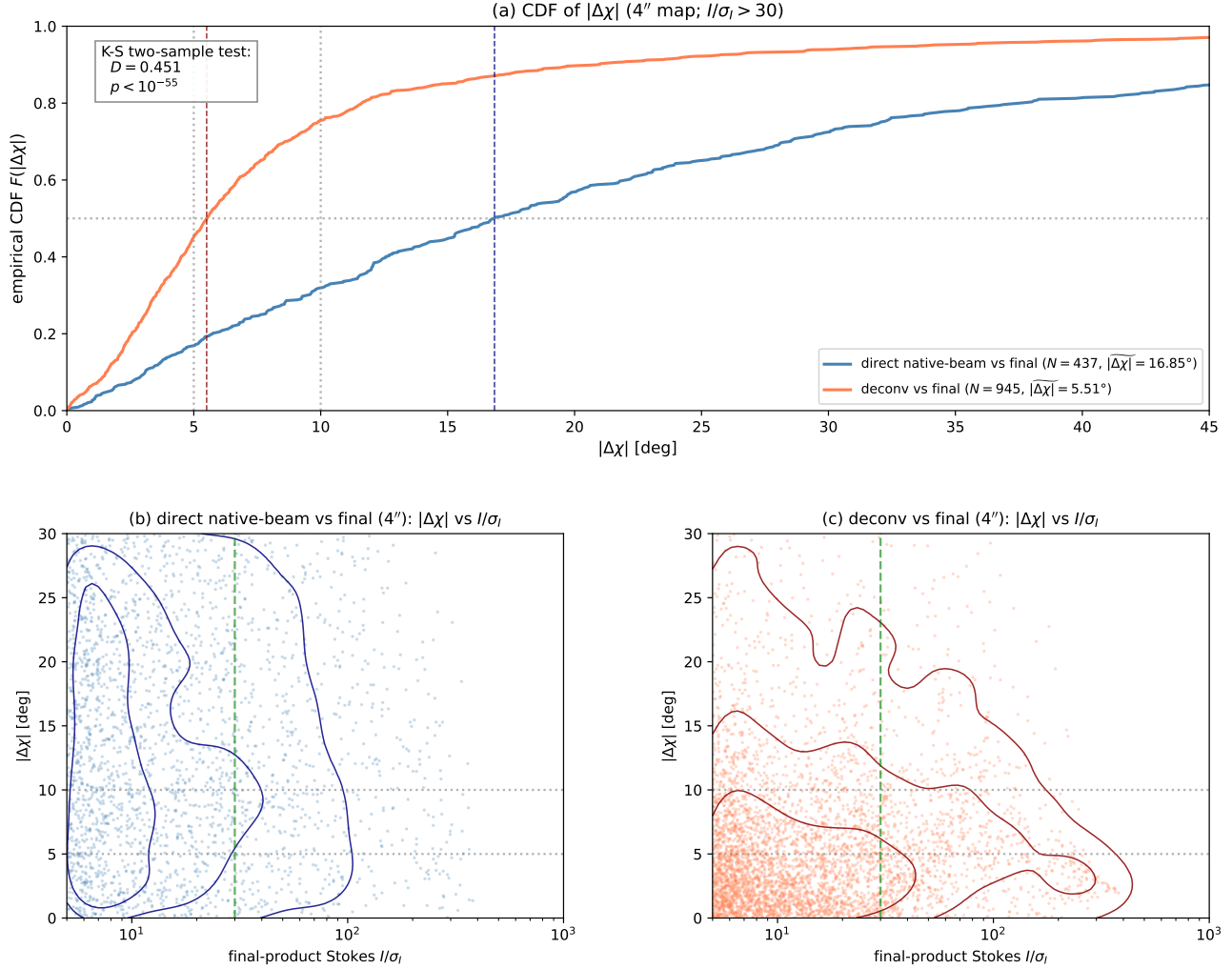


Figure 23. Supplementary beam-harmonization validation on the native 4'' Stokes-map pixel grid for L1688 Ophiuchus A. The same three $450\ \mu\text{m}$ products as in Figure 22 are compared directly in the original map domain prior to $12''$ polarization-vector sampling. **(a)** Empirical CDF of $|\Delta\chi|$ for the direct native-beam vs final (blue) and deconvolve–reconvolve vs final (orange) comparisons at $I/\sigma_I > 30$, with the two-sample Kolmogorov–Smirnov test statistic and p -value reported in the inset and the per-distribution medians and $5^\circ/10^\circ$ guides marked as in Figure 22. **(b, c)** Per-pixel $|\Delta\chi|$ versus final-product Stokes I/σ_I for the direct native-beam vs final and deconvolve–reconvolve vs final comparisons respectively (per-pixel scatter plus 50/75/90 percentile KDE contours). At $I/\sigma_I > 30$ the median absolute angle residual decreases from $|\widetilde{\Delta\chi}| \approx 16.9$ in the direct native-beam comparison to ≈ 5.5 after deconvolution and reconvolution; at $I/\sigma_I > 100$, the corresponding values are ≈ 10.1 and ≈ 4.1 . The qualitative and quantitative consistency with Figure 22 confirms that most of the apparent local angle mismatch is introduced by the native $450\ \mu\text{m}$ compound beam, rather than by intrinsic astrophysical wavelength dependence or by the $12''$ vector-sampling step.

reaching $|\widetilde{\Delta\chi}|_{\text{deconv}} < 10^\circ$ (94%), demonstrating that the harmonization performance reported above is not specific to the science-grade subset. Per-source histograms for the 24 statistical-sample sources are shown in Figure 25.

3.5.4. Implications for inter-band polarization-angle science

Combining the L1688 representative single-source demonstration, the dual-domain robustness check, and the survey-level ensemble validation, we demonstrate that the beam-harmonization procedure produces a *statistically significant*

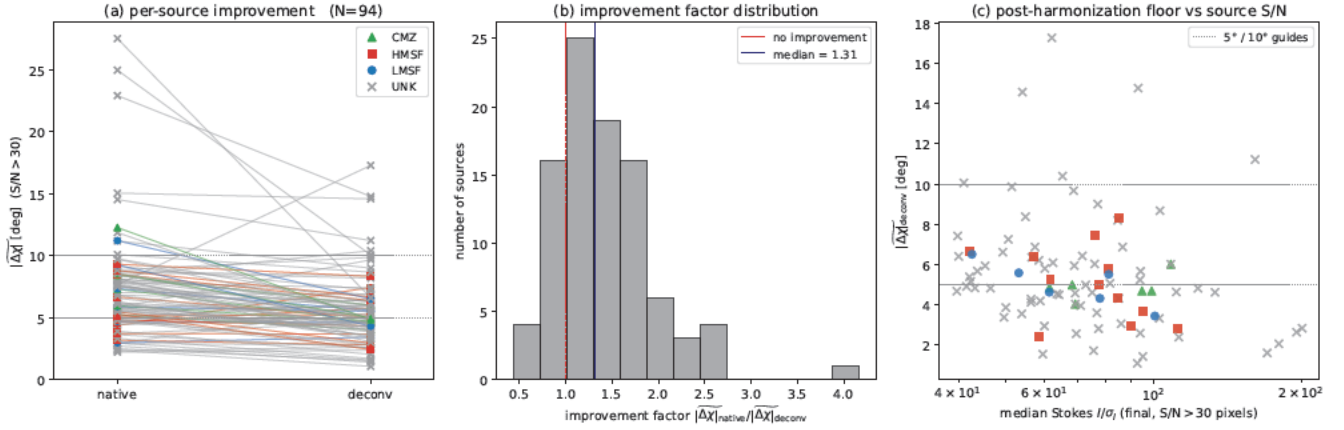


Figure 24. Survey-level beam-harmonization validation across the 24-source subset of the dual-band statistical sample of Paper S1 for which the three-product comparison is feasible at $I/\sigma_I > 30$ and $N_{\text{samp}} \geq 20$. (a) Per-source paired tracks of the median absolute polarization-angle residual $|\Delta\chi|$ from the direct native-beam comparison (left) to the deconvolve–reconvolve comparison (right) against the final science product, color-coded by environment (LMSF, HMSF, CMZ); the broader 94-source pool used as a corroboration set (calibrators and additional fields, plotted in grey, see text) is overlaid. (b) Distribution of the per-source improvement factor $|\Delta\chi|_{\text{native}}/|\Delta\chi|_{\text{deconv}}$. (c) Post-harmonization $|\Delta\chi|_{\text{deconv}}$ as a function of the median final-product Stokes I/σ_I in the eligible pixel set, with the 5° and 10° guide lines for reference. The figure makes explicit that the L1688 example of Figure 22 is a relatively beam-sensitive case ($|\Delta\chi|_{\text{native}} = 9^\circ.3$, well above the survey median of $5^\circ.9$) which converges to the survey-typical post-harmonization floor.

and survey-wide contraction of the inter-product polarization-angle residual distribution. The factored design (i)–(ii)–(iii) attributes this contraction to its two physical causes: the (i)–to–(ii) step—replacement of the native two-component compound 450 μm beam by a single $12''.6$ Gaussian via the analytical deconvolve–reconvolve operation—accounts for the dominant fraction of the observed reduction, while the (ii)–to–(iii) step—truncation of the 450 μm recovery from its native $\sim 600''$ largest scale to the common $12''.6$ – $300''$ range matched to the standard 850 μm pol2map reduction—absorbs the remaining residual into the final science product. The two-sample Kolmogorov–Smirnov tests on the L1688 single-source comparison reject the null hypothesis of equal pre- and post-harmonization $|\Delta\chi|$ distributions at $D = 0.31$, $p < 10^{-10}$ in the $12''$ catalog domain (Figure 22b), and at $D = 0.45$, $p < 10^{-55}$ in the independent $4''$ map domain (Figure 23a); the two-domain agreement excludes catalog-construction artefacts as the source of the contraction. At the survey scale, all 24 sources of the dual-band statistical sample reach a post-harmonization $|\Delta\chi|$ below the 10° physical-tolerance scale of Paper S1 F1 (Figure 24c), with a sample-wide source-median floor of $|\Delta\chi| \sim 4^\circ$ – 6° across high-quality regions and 58% of sources reaching $|\Delta\chi|_{\text{deconv}} < 5^\circ$. Application of the same comparison to the broader 94-source archival pool yields quantitatively consistent floors, demonstrating that the result is not driven by source selection.

The localized 5° – 10° inter-band angle residuals visible in individual fields prior to harmonization are therefore accounted for by the native 450 μm compound-beam morphology and by the wider native 450 μm spatial-frequency coverage, rather than requiring intrinsically different magnetic-field geometries between the 450 and 850 μm bands. The measured $|\Delta\chi| \sim 4^\circ$ – 6° instrumental floor is the practical technical basis against which the survey-wide inter-band angle agreement reported by Paper S1 should be interpreted: localized residuals within this empirical post-harmonization budget are not direct evidence of wavelength-dependent magnetic-field geometry, while excursions substantially above the floor (notably the Mon R2 outlier discussed in Paper S1) remain candidates for genuine source-physics interpretation.

3.5.5. Summary of Technical Validation 5

This survey-wide quantification closes Pillar 3 of the technical credibility chain. Every dual-band polarization-angle observable in the companion science papers—the local angle agreement, the angle-difference distributions, the magnetic-field morphology comparisons, and the angle-dispersion statistics derived from them—is interpreted against a quantified instrumental floor of $|\Delta\chi| \sim 4^\circ$ – 6° and an empirically demonstrated $< 10^\circ$ all-source guarantee at

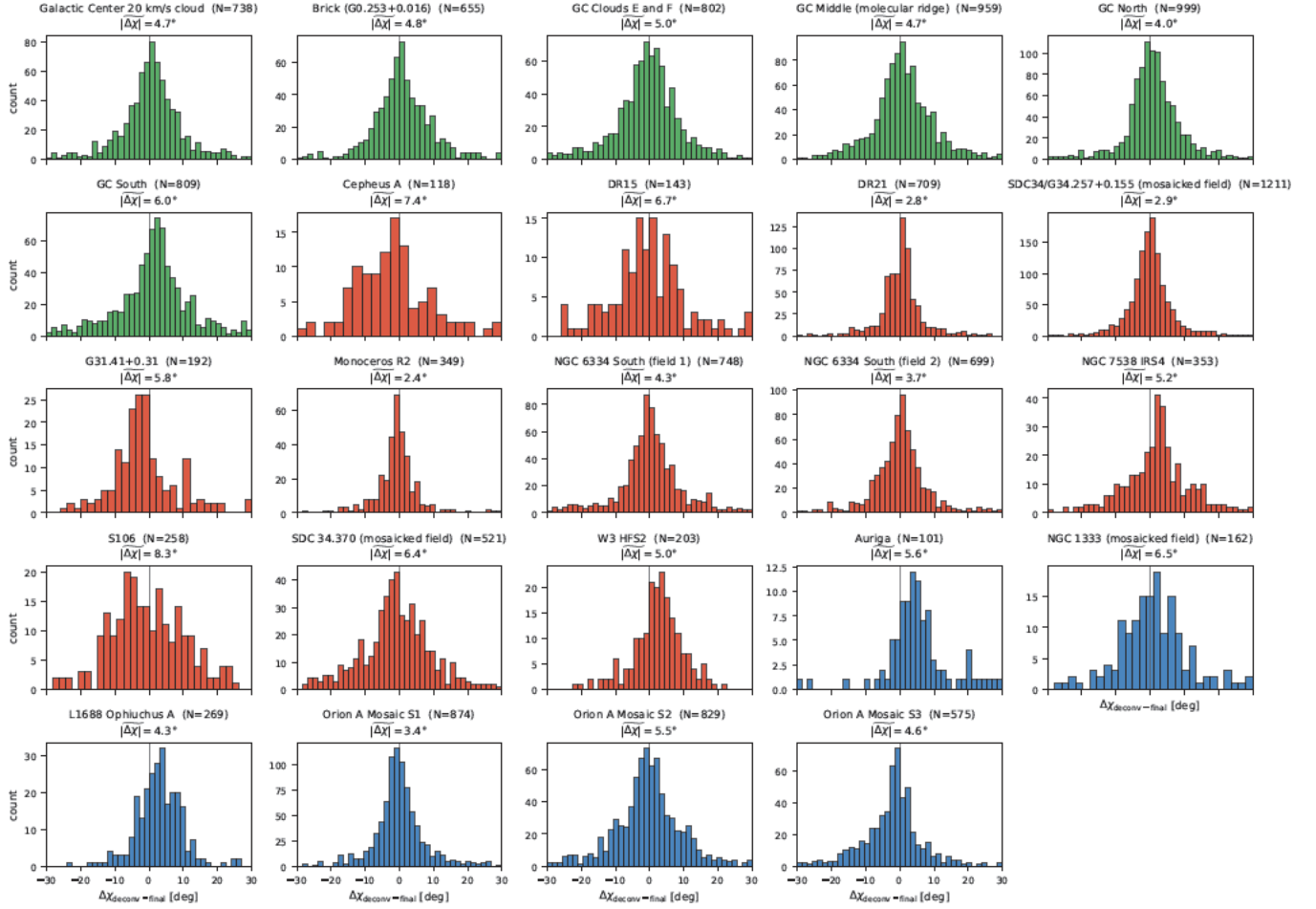
Per-source $\Delta\chi$ histograms (deconv-final, $I/\sigma_I > 30$)

Figure 25. Per-source histograms of the deconvolve-reconvolve vs final $\Delta\chi$ residuals at $I/\sigma_I > 30$ for the 24 sources used in the survey-level validation of Figure 24. Panels are grouped by environment (CMZ in green, HMSF in red, LMSF in blue) and labelled with the source-level $|\Delta\chi|$ printed at the top of each panel; histogram bin widths use the Freedman–Diaconis rule.

$I/\sigma_I > 30$. Combined with Pillar 1 (§3.1: photometric legitimacy) and Pillar 2 (§§3.2–3.3: statistical legitimacy), *all* principal dual-band science observables in Papers S1 and S2—the polarization-fraction ratio P_{450}/P_{850} , the spectral-index difference $\gamma - \alpha$, and the dual-band magnetic-field morphology—therefore carry quantified systematic floors set by the present three-pillar framework rather than by ad hoc per-source assessment, and the consolidated propagation through to the principal derived quantities is presented in the Discussion that follows (§4).

4. DISCUSSION

4.1. Integrated Systematic Uncertainty Budget for the Dual-Band Derived Quantities

The preceding subsections §§3.1–3.5 establish five independent technical ingredients required for a physically interpretable dual-wavelength polarization survey: (1) consistent surface-brightness calibration (§3.1), (2) empirically justified Stokes- I intensity thresholds (§3.2), (3) analytic propagation of Stokes noise into the polarized observables P and polarization angle (§3.3), (4) additive Stokes- Q/U zero-level offset correction of the $450\mu\text{m}$ Stokes maps (§3.4), and (5) beam-harmonized validation of residual inter-band angle offsets (§3.5). We now consolidate these ingredients into a survey-wide systematic-uncertainty budget for the dual-band derived quantities P_{450}/P_{850} , whose three statistical estimators are defined in Paper S2 §3.3, and $\gamma - \alpha$ (Equation 1). These two quantities are related through Equation (2). This consolidation corresponds to the cross-pillar entry of Figure 1 and Table 1, and defines the technical precision floor on which the survey-level conclusions of companion Paper S1 and Paper S2 rest. The budget below

focuses on *systematic* contributions; statistical uncertainties from pixel-to-pixel scatter are reported separately in the companion papers.

4.1.1. Absolute Flux Calibration

The extended-source flux conversion factors carry uncertainties of $\pm 7\%$ at 850 μm and $\pm 17\%$ at 450 μm (Mairs et al. 2021), as applied in the data preparation (Paper S1 §2). For an individual total-intensity spectral index $\alpha = \log(I_{450}/I_{850})/\log(\nu_{450}/\nu_{850})$ (Equation 1), the combined FCF uncertainty contributes

$$\sigma_{\alpha}^{\text{cal}} \simeq \frac{\sqrt{0.17^2 + 0.07^2}}{\ln(\nu_{450}/\nu_{850})} \simeq 0.29. \quad (49)$$

The polarized-intensity spectral index γ carries the same contribution. However, because wavelength-dependent FCF errors multiply both Stokes I and polarized intensity PI by the same calibration factors, their effect is common-mode in α and γ . Thus the calibration contribution cancels in the difference:

$$\delta(\gamma - \alpha)^{\text{FCF}} = \delta\gamma^{\text{FCF}} - \delta\alpha^{\text{FCF}} = 0. \quad (50)$$

For the same reason, the polarization fraction $P = PI/I$ is unaffected by FCF errors, since both numerator and denominator scale with the same flux conversion factor at a given wavelength. Consequently, P_{450}/P_{850} is immune to the absolute FCF uncertainty.

4.1.2. 450 μm Offset-Correction Residuals

The three-step regression procedure (§3.4.6) leaves residual offsets at the level of $\lesssim 0.07 \text{ mJy arcsec}^{-2}$ (§3.4.8; Equations 45–47). Unlike multiplicative FCF errors, these residuals are additive and affect the corrected 450 μm Stokes maps. They therefore perturb Q_{450} , U_{450} , and hence PI_{450} directly, while leaving the total-intensity scale largely unchanged. Among the systematic terms considered here, this is the only contribution that enters the derived quantities non-trivially.

Figure 21 quantifies this effect through Monte Carlo simulations. For each assumed true value of PI_{450} , we draw 10^4 realizations of the residual offset from a uniform distribution over $\pm 0.07 \text{ mJy arcsec}^{-2}$ and propagate the perturbation to P_{450}/P_{850} and $\gamma - \alpha$. The resulting systematic uncertainty decreases with increasing PI_{450} because the additive residual becomes a smaller fractional perturbation. For the typical range in the survey, $PI_{450} \sim 1\text{--}10 \text{ mJy arcsec}^{-2}$, the simulations give

$$\sigma(P_{450}/P_{850}) \lesssim 0.02, \quad \sigma(\gamma - \alpha) \lesssim 0.04. \quad (51)$$

Sources with very low polarized intensity, for which this systematic becomes non-negligible, are flagged as “noise-dominated” in the [per-source polarization-fraction table of Paper S2](#) and excluded from the statistical analyses.

The Monte Carlo calculation is defined at the pixel level and therefore most closely corresponds to the equal-weighted, pixel-based estimator of [Method 3 in Paper S2 §3.3](#). Because the same pixel-level quantities enter [Methods 1 and 2 of Paper S2 §3.3](#) through intensity-weighted sums and weighted correlation fits, respectively, this calculation provides a conservative upper limit for the offset-residual contribution applicable to all three estimators. The empirical consistency among the three estimators demonstrated in [Paper S2 §3.3](#) further supports applying this systematic floor across the survey-level P_{450}/P_{850} measurements.

This offset-correction budget applies specifically to single-field observations classified as Category 1 (Figures 29 and 30). Mosaicked data and mosaic-component reductions cannot be corrected with the same procedure because the field-dependent PCA thresholds are not propagated reliably through the current `pol2map` mosaicking step (§3.4.7). Consequently, in the [per-source \$P_{450}/P_{850}\$ and spectral-index tables of Paper S2](#), the offset-residual systematic documented here applies only to entries corresponding to single-field Category 1 sources; mosaicked sources are processed without the three-step correction and are not assigned this particular systematic term.

4.1.3. Beam Mismatch

Beam convolution acts as a linear operator on the intrinsic sky brightness, so comparing extended emission observed with different effective beam profiles requires explicit beam transformation rather than simple smoothing (e.g., Furuya et al. 2006, Appendix A). In this work, the 450 μm maps are transformed to the 850 μm angular response using a Wiener deconvolution kernel (§3.4.4):

$$K = \mathcal{F}^{-1} \left[\frac{\tilde{B}_{850}}{\tilde{B}_{450}} \right], \quad (52)$$

where $\tilde{B}_{450}$ and $\tilde{B}_{850}$ are the Fourier transforms of the two-component beam models from Mairs et al. (2021). This procedure transforms the effective 450 μm beam into the 850 μm beam response rather than merely broadening the map.

Residual beam-matching errors arise from uncertainties in the adopted beam parameters. Although Mairs et al. (2021) report $\pm 15\%$ uncertainties in the beam solid angles, the two wavelengths are observed simultaneously with the same telescope and calibrated through the same methodology, so the beam-profile uncertainties are expected to be substantially correlated. Monte Carlo simulations assuming $\sim 90\%$ correlation, motivated by these common instrumental and calibration systematics, yield $\sigma_{\alpha}^{\text{beam}} \approx 0.05$ for an individual total-intensity spectral index.

For the ratio quantities of interest here, the relevant contribution is smaller still. Beam convolution affects Stokes I , Q , and U through the same angular response at a given wavelength; after beam harmonization, the polarization fraction $P = \sqrt{Q^2 + U^2}/I$ is therefore preserved to first order under residual multiplicative beam-response errors. Similarly, the total-intensity and polarized-intensity spectral indices receive the same beam-related contribution, which cancels in their difference:

$$\delta(\gamma - \alpha)^{\text{beam}} = \delta\gamma^{\text{beam}} - \delta\alpha^{\text{beam}} = 0. \quad (53)$$

Thus residual beam mismatch does not contribute significantly to either P_{450}/P_{850} or $\gamma - \alpha$.

The preservation of polarization-angle morphology under the same beam-matching procedure is assessed separately through the empirical beam-matching residual $|\Delta\chi|$ established in §3.5. That residual does not propagate into P_{450}/P_{850} or $\gamma - \alpha$, but it defines the dual-band angle-comparison floor used in companion Paper S1 and Paper S2.

4.2. Survey-Wide Technical Precision Floor and Implications for the Companion Papers

The preceding error-source accounting establishes not only the magnitude of each systematic contribution, but also which terms survive in the survey-level ratio diagnostics. Table 6 summarizes the resulting systematic budget. The central result is that the dominant multiplicative calibration uncertainties cancel in the derived quantities that carry the primary science interpretation: FCF errors cancel in $P = PI/I$, and hence in P_{450}/P_{850} , while both FCF and beam-response uncertainties cancel in $\gamma - \alpha$ because they affect γ and α in common mode. The only systematic term that contributes appreciably is the additive 450 μm offset-correction residual, and even this contribution remains small over the polarized-intensity range that defines the statistical sample.

The resulting practical systematic floors are

$$\sigma(P_{450}/P_{850}) \lesssim 0.02, \quad \sigma(\gamma - \alpha) \lesssim 0.04, \quad (54)$$

for typical survey pixels with $PI_{450} \sim 1\text{--}10 \text{ mJy arcsec}^{-2}$. These values are substantially smaller than the survey-level signals reported in the companion papers. In particular, the $\gamma - \alpha$ floor is an order of magnitude below the observed median spectral-index displacement, $|\gamma - \alpha| \approx 0.52$ (Paper S1 F2), demonstrating that the detection of $\gamma < \alpha$ is not an artifact of flux calibration, beam mismatch, or residual 450 μm offset correction. Likewise, the systematic floor on P_{450}/P_{850} is far below the observed displacement of the polarization-fraction ratio from unity (Paper S1 F3; Paper S2 §3.3).

This cancellation property is central to the design of the companion science analyses. Although individual quantities such as P_{450} , PI_{450} , α , or γ can carry larger calibration- or beam-related uncertainties when interpreted in isolation, the paired dual-band diagnostics P_{450}/P_{850} and $\gamma - \alpha$ are substantially more robust because the leading multiplicative systematics divide out. Thus the principal observables used in Paper S1 and Paper S2 are not merely measured quantities, but deliberately constructed ratio diagnostics with controlled technical floors.

Together with the polarization-angle floor established in §3.5, this completes the technical closure of the three-pillar framework. The present paper sets the quantitative instrumental limits for all three principal dual-band observables used in the companion papers: P_{450}/P_{850} , $\gamma - \alpha$, and the inter-band polarization-angle morphology. The science interpretation in Paper S1 and Paper S2 therefore rests on survey-level observables whose dominant calibration, noise, beam, and offset-related systematics have been explicitly propagated and shown to be smaller than the astrophysical signals they are used to interpret.

Table 7 summarises the technical-framework deliverables that the present paper exports as primitives to the companion science papers S1 and S2.

Table 7. Paper T technical-framework deliverables exported to companion Papers S1 and S2

Deliverable	Established here	Used in companion papers
Pillar 1: extended-source FCF + Planck-corrected $\beta = \alpha - C(T_d)$	§3.1	Paper S1 § 3.3 (α, γ for F2); Paper S2 § 3.4 (F4 spectral-index inventory)
Pillar 2: Stokes- $I + Q, U$ analytic noise model	§§3.2–3.3	Paper S1 §§ 3.1–3.2 (F1, F3 detection masks); Paper S2 § 3.6 (F6 per-pixel masks)
Pillar 3: 450 μm Q/U offset correction + beam harmonization	§§3.4–3.5	matched-resolution dual-band products throughout companion Papers S1 and S2
Systematic-uncertainty floor $\sigma(P_{450}/P_{850}) \lesssim 0.02$, $\sigma(\gamma - \alpha) \lesssim 0.04$	§4.1 (Table 6)	δX_{sys} entries throughout companion Paper S1 § 3 and Paper S2 § 3
Sample-inventory architecture (174/37/24/9 source-based; 417 Method-3 pixel-pair)	§2.2 (Table 2)	Paper S1 28-source SPIDX + 21-source dual-band-angle + 9-region HGBS subsets; Paper S2 69-source POLSRC + 7-source CMZ + 417-pixel HGBS

NOTE— The five rows summarise the technical-framework outputs of Paper T that are imported as primitives by the companion science papers S1 and S2. Paper T does not import science-level findings from the companion papers; the technical-to-science chain is unidirectional. The reverse-direction view (S1 / S2 imports from Paper T) is given in companion Paper S1 Table 3 and Paper S2 Table 3.

5. SUMMARY

This paper establishes the technical analysis framework that enables the first quantitatively controlled dual-wavelength 450/850 μm polarimetric census of the BISTRO survey. We have presented a complete methodology for the reduction, noise characterization, and systematic calibration of JCMT SCUBA-2/POL-2 polarimetric observations at 450 μm , validated against the BISTRO 450/850 μm dual-wavelength sample. The framework is organised around three interlocking methodological pillars (summarised in Figure 1 and cross-indexed in Table 1) that together establish the end-to-end technical credibility of the JCMT 450 μm dual-band polarization products on which all survey-level science conclusions of companion Paper S1 and Paper S2 rest:

1. *Pillar 1 — Photometric legitimacy.* The Planck-corrected spectral-index framework $\beta = \alpha - C(T_{\text{d}})$, verified empirically on the 37-source FCF-validation sample, resolves a deterministic $\Delta\beta = -1.06$ bias that naive peak-brightness-FCF application would introduce (§3.1). This pillar establishes that the inferred 450 μm fluxes and dust-emissivity spectral slopes are physically meaningful for the extended-emission regime relevant to dual-band polarimetry.
2. *Pillar 2 — Statistical legitimacy.* A hybrid additive-noise estimator for Stokes I (§3.2), combined with a two-component noise model for Stokes Q and U (§3.3), characterizes the 450 μm polarization noise environment across 174 POL-2 fields in the JCMT archive. The 450 μm surface-brightness noise is median-38 $\times$ larger than its 850 μm counterpart (interquartile range 21.6–60.2), with a superlinear power-law scaling $\sigma_{450} \propto \sigma_{850}^{2.59 \pm 0.26}$ that reflects the non-proportional impact of atmospheric transmission variability on the shorter-wavelength band. The Stokes Q, U component of the model separates cleanly into an atmosphere-dominated regime at $p \lesssim 5\%$ and a residual-systematic-dominated regime at $p \gtrsim 10\%$; empirical verification of isotropy ($\sigma_Q \simeq \sigma_U$, $\text{Cov}(Q, U) \approx 0$) and jackknife cross-validation across the 174-field sample confirm the model’s self-consistency.
3. *Pillar 3 — Map-domain comparability legitimacy and the global polarization-angle uncertainty architecture.* Pillar 3 has two stages. In the *construction* stage (§3.4), a wavelength-dependent additive Stokes- Q/U zero-level offset—invisible in single-wavelength analyses but inconsistent with the physical requirement that regions with $Q_{850} = U_{850} = 0$ must also yield $Q_{450} = U_{450} = 0$ under optically thin conditions—is identified at the $\sim 1 \text{ mJy arcsec}^{-2}$ level in the 450 μm Stokes- Q and Stokes- U maps, where required. A three-step PCA-based correction procedure reduces the offset to a residual of $\lesssim 0.07 \text{ mJy arcsec}^{-2}$, and the corrected 450 μm maps are independently validated against SOFIA/HAWC+ 154/214 μm polarimetry of Ophiuchus A. In the *capstone validation* stage (§3.5), a three-product internal comparison applied at three successively broader scales (single-source 12'' polarization-catalog, independent 4'' Stokes-map, and 24-source dual-band statistical-sample ensemble) demonstrates that the beam-harmonization procedure produces a statistically significant and survey-wide contraction of the inter-product polarization-angle residual distribution: the source-median residual after harmonization is $|\widetilde{\Delta\chi}| \sim 4^\circ\text{--}6^\circ$, with *all* sources reaching $|\widetilde{\Delta\chi}| < 10^\circ$ at $I/\sigma_I > 30$, well below the 10° angle-tolerance scale used in the companion Paper S1 dual-band angle-consistency analysis (two-sample Kolmogorov–Smirnov $D = 0.31$, $p < 10^{-10}$ in the 12'' polarization-catalog domain and $D = 0.45$, $p < 10^{-55}$ in the independent 4'' Stokes-map domain for the L1688 representative example). This survey-level $|\widetilde{\Delta\chi}|$ floor and the $< 10^\circ$ all-source guarantee constitute the *global polarization-angle uncertainty architecture* adopted throughout the companion science papers: every dual-band angle observable in Papers S1 and S2—the local angle agreement, the angle-difference distributions, and the magnetic-field morphology comparisons—is interpreted against the present empirical floor rather than against ad hoc per-source assessment.

The consolidated three-pillar systematic-uncertainty budget (§4.2) shows that the leading multiplicative systematics largely cancel in the paired dual-band diagnostics. FCF errors cancel in $P = PI/I$ and hence in P_{450}/P_{850} , while both FCF and beam-response uncertainties cancel to first order in $\gamma - \alpha$ because they affect γ and α in common mode. The remaining non-trivial contribution is the additive 450 μm offset-correction residual, which gives practical systematic floors of $\sigma(P_{450}/P_{850}) \lesssim 0.02$ and $\sigma(\gamma - \alpha) \lesssim 0.04$ over the typical survey range. Together with the polarization-angle floor established by Pillar 3, this completes the end-to-end credibility chain: all principal dual-band observables in Papers S1 and S2— P_{450}/P_{850} , $\gamma - \alpha$, and the inter-band polarization-angle morphology—carry quantified systematic floors set by the present framework.

This framework enables robust 450 μm polarization-fraction determinations with $p \gtrsim 2\sigma$ for pixels meeting the validation thresholds, and is applicable to current and future JCMT 450 μm polarimetric analyses as well as archival

re-analyses of existing POL-2 data. The scientific findings built on this methodology are reported in the companion papers: three empirical constraints on grain-alignment physics in [Paper S1 \(ApJ, in preparation\)](#), and the complete 69-source dual-wavelength polarimetric census including the environment-stratified analysis across low-mass, high-mass, and Galactic Center sub-environments—visualised as a $(\gamma - \alpha, R_3)$ phase diagram ([Paper S2 Figure 11](#))—in [Paper S2 \(ApJ, in preparation\)](#).

Together, the present technical framework and the companion science papers constitute the BISTRO 450/850 μm dual-band polarimetric legacy. All dual-band scientific claims in [Paper S1](#) and [Paper S2](#) rest on the quantitatively controlled instrumental foundation established here, providing a reference standard against which future multi-band polarimetric surveys can be calibrated.

All technical-validation data products described in this paper, including the beam-harmonized 450 μm Stokes Q/U maps and the offset-correction parameters of Table 9, will be made available electronically as supplementary material to this paper.

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APPENDIX

A. COMPARISON OF RICEAN EXPECTATION FORMULATIONS

A.1. Motivation

The Ricean analyses invoked elsewhere in this paper series employ the exact Ricean expectation value using modified Bessel functions I_0 and I_1 . Under the isotropic polarization-noise condition validated in §3.3 ($\sigma_Q \simeq \sigma_U \equiv \sigma_{QU}$,

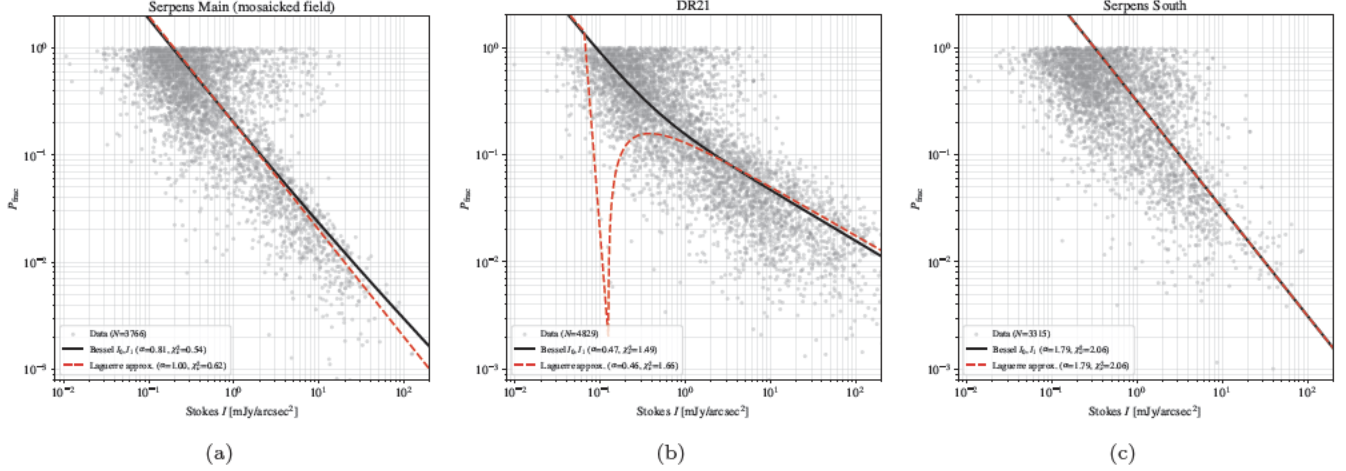


Figure 26. Comparison of the exact Ricean expectation using modified Bessel functions I_0 , I_1 (black solid curves) and the Laguerre polynomial approximation (red dashed curves) for three representative targets. The vertical blue dashed line indicates σ_{QU} , below which the high signal-to-noise asymptotic expansion underlying the Laguerre approximation becomes unreliable. (a) Serpens Main: The Laguerre approximation yields $\alpha_{\text{dep}} = 1.00$, systematically biased toward the noise-dominated regime, while the Bessel formulation recovers the physical depolarization index $\alpha_{\text{dep}} = 0.81$. (b) DR21: Both methods yield similar α_{dep} values ($\alpha_{\text{dep}} \approx 0.47$), but the Laguerre approximation exhibits a non-physical “hump” near $I \sim \sigma_{QU}$ due to breakdown of the asymptotic expansion at intermediate S/N ratios. (c) Serpens South: In the noise-dominated regime ($\alpha_{\text{dep}} \approx 1.8$), both formulations yield identical results, as expected when the true polarization signal is weak relative to noise throughout the fitted intensity range.

$\text{Cov}(Q, U) \approx 0$, this expectation is

$$\langle PI_{\text{obs}} \rangle = \sigma_{QU} \sqrt{\frac{\pi}{2}} \exp\left(-\frac{\xi^2}{4}\right) \left[\left(1 + \frac{\xi^2}{2}\right) I_0\left(\frac{\xi^2}{4}\right) + \frac{\xi^2}{2} I_1\left(\frac{\xi^2}{4}\right) \right], \quad (\text{A1})$$

where $\xi \equiv PI_{\text{true}}/\sigma_{QU}$ is the signal-to-noise ratio of the true polarized intensity. Equation A1 is exact and valid for all ξ , from the noise-dominated regime ($\xi \ll 1$) through intermediate signal-to-noise ($\xi \sim 1$) to the high signal-to-noise limit ($\xi \gg 1$). This Bessel-form comparison is essential because the choice of Ricean expectation formulation directly affects the inferred depolarization index α_{dep} , particularly in the intermediate signal-to-noise regime ($\xi \sim 1$ – 10) that dominates our $450 \mu\text{m}$ data. Here we compare the exact Bessel formulation (Equation A1) with the Laguerre polynomial approximation used in previous polarimetric studies (e.g., Wardle & Kronberg 1974; Montier et al. 2015; Vidal et al. 2016; Pattle et al. 2019), demonstrating where the approximation remains valid and where it breaks down.

A.2. Laguerre Polynomial Approximation

In the high signal-to-noise limit ($\xi \gg 1$), the exact Ricean expectation (Equation A1) can be approximated using Laguerre polynomials (e.g., Wardle & Kronberg 1974):

$$\langle PI_{\text{obs}} \rangle \approx PI_{\text{true}} \left(1 - \frac{1}{2\xi^2}\right). \quad (\text{A2})$$

This approximation provides computational simplicity and has been widely adopted in submillimeter polarimetry (e.g., Pattle et al. 2019). However, it is important to emphasize that Equation (A2) is not a general approximation to the Ricean expectation, but a strictly asymptotic expansion valid only in the limit $\xi \gg 1$. When applied outside this regime, the approximation can produce mathematically well-defined but physically meaningless behavior.

A.3. Coupling with the Power-Law Model

The pathological behavior of the Laguerre approximation does not require taking formal limits ($\xi \rightarrow 0$ or $\xi \rightarrow \infty$). In practical applications, the breakdown occurs already at intermediate S/N ratios ($\xi \sim$ a few to tens), where the multiplicative correction term couples nonlinearly with the intrinsic power-law model. To see this explicitly, we

substitute the power-law model $P_{\text{true}} = P_0(I/\sigma_{QU})^{1-\alpha_{\text{dep}}}$ into the S/N ratio $\xi = PI_{\text{true}}/\sigma_{QU}$, yielding

$$\xi = \frac{P_{\text{true}} \cdot I}{\sigma_{QU}} \propto I^{2-\alpha_{\text{dep}}}. \quad (\text{A3})$$

Consequently, the Laguerre correction factor becomes intensity-dependent:

$$1 - \frac{1}{2\xi^2} = 1 - C I^{-2(2-\alpha_{\text{dep}})}, \quad (\text{A4})$$

where C is a constant that depends on P_0 and σ_{QU} . The observed polarization fraction is then

$$\langle P_{\text{frac}} \rangle \propto I^{-\alpha_{\text{dep}}} \left\{ 1 - C I^{-2(2-\alpha_{\text{dep}})} \right\}. \quad (\text{A5})$$

In log-log space, this becomes

$$\log \langle P_{\text{frac}} \rangle = -\alpha_{\text{dep}} \log I + \log \left\{ 1 - C I^{-2(2-\alpha_{\text{dep}})} \right\}. \quad (\text{A6})$$

The second term introduces curvature that distorts the intrinsic linear relation $\log P \propto -\alpha_{\text{dep}} \log I$. This curvature can produce an artificial flattening or even a spurious upturn at high intensities, even when ξ remains well within the nominal fitting range. The exact Bessel formulation of Eq. (A1) does not suffer from this artifact because the modified Bessel functions I_0 and I_1 correctly capture the full nonlinear behavior of the Rice distribution without introducing spurious intensity-dependent corrections.

A.4. Behavior at Limiting Cases

For completeness, we also examine the formal limiting behavior. In the noise-dominated limit ($\xi \rightarrow 0$), the exact Bessel formulation (Equation A1) correctly approaches

$$\lim_{\xi \rightarrow 0} \langle PI_{\text{obs}} \rangle = \sigma_{QU} \sqrt{\frac{\pi}{2}} \approx 1.25 \sigma_{QU}, \quad (\text{A7})$$

representing the expected value of polarized intensity from pure Gaussian noise in Q and U . In contrast, the Laguerre approximation (Equation A2) gives

$$\lim_{\xi \rightarrow 0} PI_{\text{true}} \left(1 - \frac{1}{2\xi^2} \right) \rightarrow -\infty, \quad (\text{A8})$$

which is manifestly non-physical. The correction factor $(1 - 1/2\xi^2)$ passes through zero at $\xi = 1/\sqrt{2} \approx 0.71$ and becomes increasingly negative for smaller ξ . This produces the characteristic “hump” artifact visible in Figure 26(b) near $I \sim \sigma_{QU}$. In the high signal-to-noise limit ($\xi \rightarrow \infty$), both formulations correctly recover $\langle PI_{\text{obs}} \rangle \rightarrow PI_{\text{true}}$. However, the curvature effect described by Equations (A5)–(A6) can still produce visible artifacts at finite ξ values typically encountered in real data.

A.5. Comparison Across Representative Targets

Figure 26 compares the two formulations for three representative targets spanning different regimes. In each panel, we fit the observed $P_{\text{frac}}-I$ relation using both the exact Bessel formulation (Equation A1; black solid curves) and the Laguerre approximation (Equation A2; red dashed curves). The vertical blue dashed line indicates σ_{QU} , marking the boundary below which the asymptotic approximation becomes unreliable.

We identify three distinct behaviors depending on the intrinsic polarization properties of the target. First, for targets with moderate depolarization indices ($\alpha_{\text{dep}} \sim 0.5-0.9$), the Laguerre approximation tends to yield α_{dep} values systematically biased toward unity. Panel (a) shows Serpens Main, where the Bessel formulation recovers $\alpha_{\text{dep}} = 0.81$ while the Laguerre approximation yields $\alpha_{\text{dep}} = 1.00$. In practice, this occurs because the Laguerre approximation underestimates $\langle PI_{\text{obs}} \rangle$ at intermediate ξ , forcing the fitted power-law slope toward the noise-dominated limit $P \propto I^{-1}$ in order to compensate for the suppressed model amplitude.

Second, even when both methods yield similar α_{dep} values, the Laguerre approximation can produce non-physical model curves at low intensities. Panel (b) shows DR21, where both formulations give $\alpha_{\text{dep}} \approx 0.47$, but the Laguerre

model exhibits a characteristic “hump” feature near $I \sim \sigma_{QU}$. This artifact results directly from the sign change in the correction factor $(1 - 1/2\xi^2)$ at $\xi \approx 0.71$, as described above.

Third, for noise-dominated targets ($\alpha_{\text{dep}} \gtrsim 1.5$), both formulations produce identical results. Panel (c) shows Serpens South, where $\alpha_{\text{dep}} \approx 1.79$ for both methods. In this regime, the intrinsic polarization signal is weak relative to noise ($\xi \lesssim 1$ throughout the fitted intensity range), and the distinction between formulations becomes negligible because both are fitting to noise-dominated data where the precise form of the Ricean expectation has minimal influence on the inferred slope.

A.6. Recommendation

Based on these comparisons, we adopt the exact Bessel formulation (Equation A1) throughout this work. The computational cost of evaluating modified Bessel functions is negligible with modern numerical libraries, and this approach ensures physically meaningful model curves across the full intensity range without introducing the curvature artifacts inherent to the Laguerre approximation. It is worth noting that the Laguerre approximation remains appropriate when the fitted data are restricted to the high signal-to-noise regime (e.g., $\xi \gtrsim 5$ throughout), as demonstrated by the consistency between formulations in such cases. The Bessel formulation simply extends the applicable range to include intermediate signal-to-noise data, which are essential for constraining α_{dep} in the $450 \mu\text{m}$ observations presented here.

B. STOKES Q, U RELIABILITY DIAGNOSTICS FOR BRIGHT $850 \mu\text{m}$ FIELDS (W51 AND DR21)

This appendix supplements the Stokes- I noise characterization of §3.2 by examining the corresponding uncertainties in Stokes Q and U for two exceptionally bright $850 \mu\text{m}$ fields, W51 and DR21. The diagnostic identifies algorithmic systematics that the Stokes- I analysis (§3.2) cannot fully expose on its own, and refines the pixel-selection criteria adopted in the polarization noise model of §3.3.

To understand why the polarization uncertainties provide diagnostic information about the total intensity reliability, it is necessary to consider the POL-2 measurement principle.

In POL-2 observations, the signal recorded by each bolometer can be idealized as

$$d(t) = I(t) + Q(t) \cos\{4\gamma(t)\} + U(t) \sin\{4\gamma(t)\} + n(t), \quad (\text{B9})$$

where $\gamma(t)$ is the half-wave plate rotation angle and $n(t)$ represents noise and systematics. Crucially, the Stokes parameters I , Q , and U are *all* extracted from sinusoidal fitting to this same time-series data: Q and U correspond to the amplitude and phase of the 4γ modulation, while I corresponds to the mean signal level. This is the opposite of the intuitive assumption that total intensity is measured first and polarization derived subsequently; in fact, the quality of the Q and U extraction directly reflects the fidelity of the underlying time-series data from which I is also derived.

When the bolometer operates in its linear regime, the sinusoidal demodulation cleanly separates the polarized signal from the total intensity. However, detector saturation or strong non-linearity can severely compromise this process. Signal clipping truncates the sinusoidal waveform, causing amplitude and phase errors in the fitted Q and U . Gain variations that depend on signal level introduce waveform distortions that generate spurious harmonics. These distortions can leak intensity fluctuations into the 4γ component, creating artificial polarization signals through leakage of Stokes I into Stokes Q and U . Since Q and U represent small modulation amplitudes relative to I , even minor distortions in the strong I signal can produce proportionally large errors in the derived polarization parameters. In other words, anomalous behavior in ΔQ and ΔU serves as a sensitive indicator of time-stream corruption, which necessarily implies that the corresponding Stokes I values are also unreliable, even if this is not immediately obvious from I alone.

We note that the present analysis is based on the final map products rather than the raw time-stream data. Consequently, we do not directly verify the sinusoidal fitting process itself; instead, we infer its quality from the statistical properties of the reconstructed Stokes parameters. This approach is nonetheless diagnostic: if the time-series fitting is compromised by saturation or map-making artifacts, the degradation should manifest as correlated anomalies in ΔQ and ΔU —and by the measurement principle above, such anomalies imply concurrent unreliability in Stokes I .

Figure 27 presents Case 2 (Stokes I -referenced) noise diagnostics for the $850 \mu\text{m}$ observations of W51, following the dual-regime analysis framework established in §3.3. The horizontal axis extends to *negative* Stokes I values, reaching $I \simeq -60 \text{ mJy arcsec}^{-2}$ in the central region. Since thermal dust emission is intrinsically non-negative ($I \geq 0$), the presence of negative Stokes I values constitutes direct, unambiguous evidence of “negative bowling” artifacts introduced

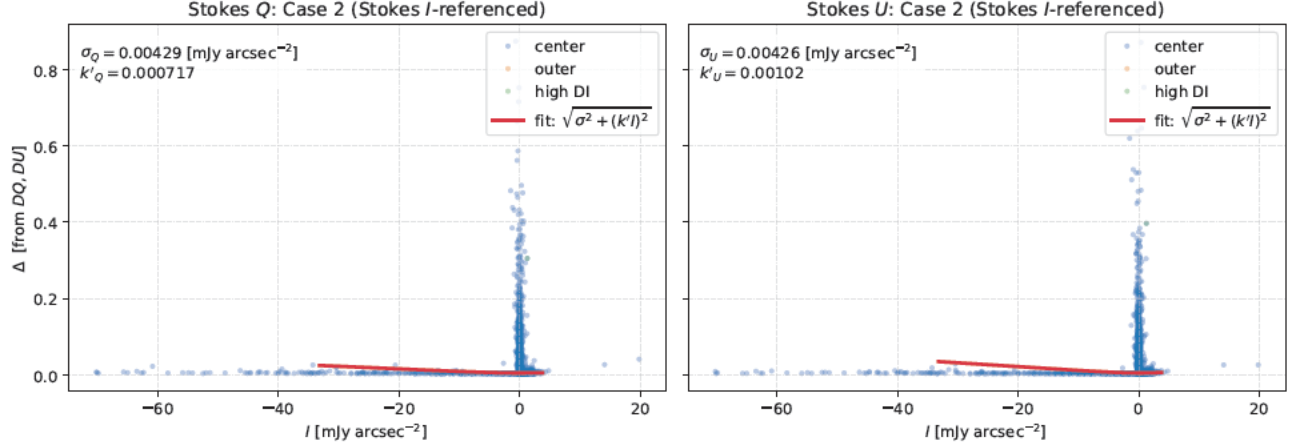


Figure 27. Stokes Q and U uncertainty diagnostics for the $850\mu\text{m}$ observations of W51 (Case 2; Stokes I -referenced). The red curves show the best-fit empirical noise model, $\Delta_{Q,U}(I) = \sqrt{\sigma_{Q,U}^2 + (k'_{Q,U}I)^2}$. The presence of data points with negative Stokes I (extending to $I \simeq -60 \text{ mJy arcsec}^{-2}$) provides direct evidence for “negative bowling” artifacts. Since thermal dust emission is intrinsically non-negative, negative I values unambiguously indicate algorithmic overcorrection in the iterative map-making process. Because all Stokes parameters are derived from the same time-series data via sinusoidal fitting (Eq. B9), the coincidence of negative I values with elevated ΔQ and ΔU demonstrates that map-making artifacts affect the full Stokes vector simultaneously. Data were reduced using Starlink version of 2021 July 23.

by the iterative map-making algorithm (Chapin et al. 2013). These artifacts arise when spatial filtering and common-mode subtraction overcorrect the time-stream data in the vicinity of bright, compact sources, producing localized negative residuals in the reconstructed map.

The concurrent elevation of ΔQ and ΔU in these negative- I pixels (reaching $\sim 0.2\text{--}0.8 \text{ mJy arcsec}^{-2}$) demonstrates that the map-making systematics propagate into all three Stokes parameters simultaneously. This is precisely the expected behavior given Eq. (B9): any systematic affecting the time-stream quality—whether detector non-linearity, clipping, or algorithmic artifacts—will degrade I , Q , and U concurrently, since all three are extracted from the same corrupted data. The $850\mu\text{m}$ observations of DR21 exhibit a similar but less pronounced signature, with pixels from the central region clustering at anomalously low reconstructed intensities with elevated uncertainties.

Importantly, the fitted additive noise floors for both fields remain comparable to those of other survey fields when evaluated using only the well-behaved outer regions: $\sigma_Q \simeq \sigma_U \simeq 0.003\text{--}0.004 \text{ mJy arcsec}^{-2}$. This demonstrates that the instrumental performance is nominal outside the affected central pixels. These findings provide independent confirmation that the central pixels of the $850\mu\text{m}$ W51 and DR21 maps are affected by algorithmic systematics, validating the exclusion of these regions from quantitative polarization analysis. The $I/\sigma_I > 10$ selection criterion effectively isolates the problematic regions: pixels with low or negative reconstructed I combined with inflated σ_I are automatically rejected, while valid data in the surrounding regions are preserved.

Crucially, this exclusion is not universal for all high-intensity sources. OMC-1 and Sgr B2, which are among the brightest sources in our sample, do not exhibit the anomalous negative- I pixels or elevated $\Delta Q/\Delta U$ values observed in W51 and DR21. The central regions of these two fields maintain $I/\sigma_I \gg 10$ throughout, with Stokes I values remaining strictly positive and noise characteristics consistent with the two-component model established in §3.3. This contrast indicates that the artifacts in W51 and DR21 are source-specific—likely reflecting differences in observing conditions, scan patterns, or source morphology—rather than a general limitation of POL-2 observations at high intensity.

Subsequent analysis of the $450\mu\text{m}$ observations reveals that W51 and DR21 are similarly affected at the shorter wavelength. Stokes I signal-to-noise ratio maps show that 41–46% of pixels exhibit negative intensity values in both the $450\mu\text{m}$ and $850\mu\text{m}$ data for these two fields, with negative-to-positive pixel ratios of 0.68–0.85. This contrasts sharply with the behavior expected for thermal dust emission, where $I \geq 0$ by definition. The pervasive nature of the artifacts—affecting comparable fractions of pixels at both wavelengths—indicates that the underlying cause is not wavelength-specific but rather reflects the extreme brightness of these sources exceeding the dynamic range of the observing and data reduction system. Consequently, both W51 and DR21 are excluded entirely from the quantitative dual-wavelength polarization analysis presented in this work.

Table 8. External comparison between the empirical BISTRO noise model and the official JCMT SCUBA-2/POL-2 sensitivity calculator (Calculator v3.2.0, Band 1 weather, Daisy 3-arcmin map, 12'' pixel sampling, median across four declinations $\delta = \{+45^\circ, +30^\circ, -5^\circ, -25^\circ\}$), evaluated at a representative exposure time of $t = 13.5$ hr. All quantities are expressed in mJy arcsec $^{-2}$.

Band / convention	Empirical $\Delta I_\lambda^{\text{emp}}$	ITC $\Delta I_\lambda^{\text{ITC}}$	$\mathcal{R}_{\text{ITC}}$
850 μm	0.0088	0.00124	7.1
450 μm , matched-resolution ($\Omega_{450} = 240$ arcsec 2)	0.339	0.0079	42.9
450 μm , native beam ($\Omega_{450} = 122$ arcsec 2)	0.339	0.0156	21.7

NOTE—The two 450 μm entries reflect the effective beam-area convention used to express the proposal-level RMS on the mJy arcsec $^{-2}$ scale: the matched-resolution convention adopts the 12-arcsec Gaussian beam of the Pillar 3 capstone (§3.5); the native-beam convention uses the original 8-arcsec 450 μm sampling.

C. COMPARISON WITH THE JCMT POL-2 SENSITIVITY CALCULATOR

The empirical scaling in Figure 8 shows that the final BISTRO map noise decreases more slowly than the ideal $t^{-0.5}$ expectation for uncorrelated white noise. As an external sanity check, we compare the same Stokes- I noise floors with the official JCMT SCUBA-2/POL-2 sensitivity calculator (Calculator v3.2.0). All quantities are placed on the same surface-brightness scale, mJy arcsec $^{-2}$, consistent with the flux-density convention adopted throughout this paper series. This comparison is not intended as a recalibration of the integration-time calculator: the calculator provides proposal-level sensitivity estimates, whereas the BISTRO products are reduced science maps of bright and spatially structured star-forming regions, whose noise includes map-making residuals, source-structure coupling, masking choices, and downstream systematics not represented in a proposal-level estimate.

We define the empirical-to-ITC ratio as

$$\mathcal{R}_{\text{ITC}} = \frac{\Delta I_\lambda^{\text{emp}}}{\Delta I_\lambda^{\text{ITC}}}, \quad (\text{C10})$$

where $\Delta I_\lambda^{\text{emp}}$ is the BISTRO empirical power-law fit (§3.2.3) evaluated at a representative integration time, and $\Delta I_\lambda^{\text{ITC}}$ is the calculator-predicted RMS placed on the matching surface-brightness scale.

For a representative total exposure time of $t = 13.5$ hr, the empirical BISTRO fit gives $\Delta I_{850}^{\text{emp}} = 0.0088$ mJy arcsec $^{-2}$ and $\Delta I_{450}^{\text{emp}} = 0.339$ mJy arcsec $^{-2}$. The ITC prediction for Band 1 conditions (Daisy 3-arcmin map, 12'' pixel sampling, median across four representative declinations $\delta = \{+45^\circ, +30^\circ, -5^\circ, -25^\circ\}$) gives $\Delta I_{850}^{\text{ITC}} = 0.00124$ mJy arcsec $^{-2}$ at 850 μm . At 450 μm the predicted value is $\Delta I_{450}^{\text{ITC}} = 0.0079$ mJy arcsec $^{-2}$ when expressed using the matched-resolution effective beam area ($\Omega_{450} = 240$ arcsec 2), or 0.0156 mJy arcsec $^{-2}$ on the native-beam convention ($\Omega_{450} = 122$ arcsec 2). The resulting ratios are $\mathcal{R}_{\text{ITC}} = 7.1$ at 850 μm and $\mathcal{R}_{\text{ITC}} = 21.7\text{--}42.9$ at 450 μm , summarized in Table 8.

These ratios should not be interpreted as a failure of the ITC prediction. The ITC provides an ideal proposal-level sensitivity estimate based on the radiometer equation, whereas the empirical BISTRO values are measured from final science maps of bright, structured regions after map-making, filtering, smoothing, masking, and downstream polarization analysis. The magnitude of $\mathcal{R}_{\text{ITC}}$ is of the same broad scale as the independently inferred smoothing/map-making efficiency factors, with median $f_{\text{smooth}} \simeq 5.3$ at 850 μm and $\simeq 10.6$ at 450 μm (Figure 9). The ITC comparison therefore provides an external consistency check on the empirical noise model rather than a replacement for it, and the empirical noise model of §3.3 is retained for all downstream polarization analyses.

D. NULL HARDWARE TESTS FOR THE 450 μm POLARIZATION OFFSET

This appendix presents the three null hardware tests summarized in §3.4.3: telescope pointing/attitude, source-brightness dependence, and the instrumental-polarization (IP) correction model. These tests required observations of exceptionally bright fields to achieve the S/N ratio necessary for offset characterization at the individual Minimum Science Block (MSB) level (typically 40–42 minutes per observation).

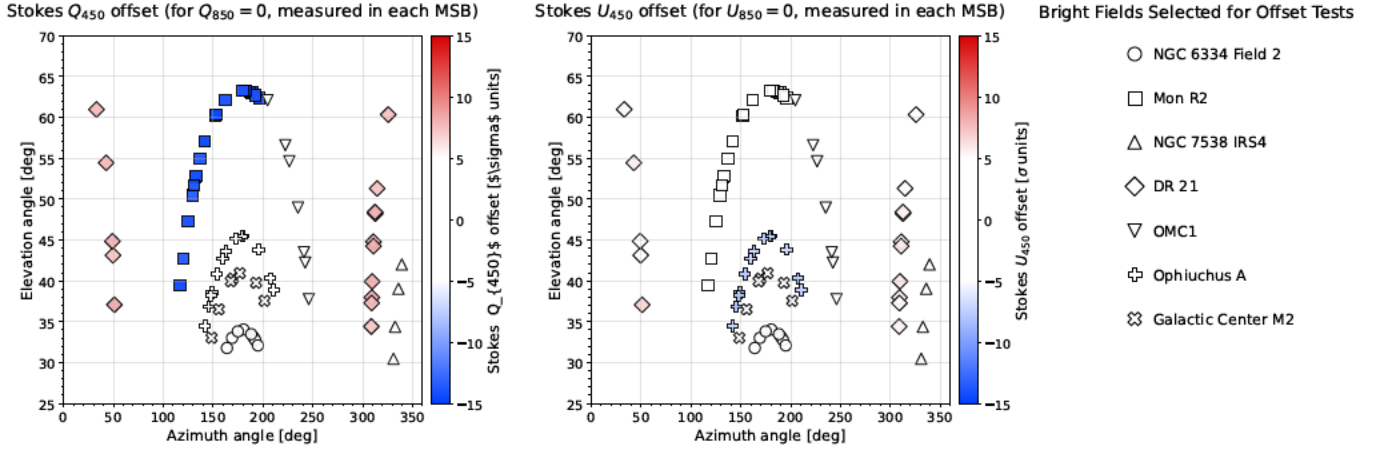


Figure 28. Azimuth–elevation distribution of the residual 450 μm offsets in Stokes Q (left panel) and U (right panel), measured when the corresponding 850 μm component equals zero. Each point represents a single MSB (typically 40–42 minutes of observation) from one of seven bright fields selected for their exceptional S/N ratios. The color scale represents the offset amplitude in units of field-level uncertainty (σ), with a white plateau for values within $\pm 5\sigma$ and seismic coloring (blue for negative, red for positive) beyond this range. The absence of systematic structure across the azimuth–elevation plane demonstrates that the offsets are uncorrelated with telescope attitude, pointing geometry, or tracking parameters. Both Stokes components exhibit comparable amplitude distributions and spatial symmetry, confirming that hardware-related effects cannot account for the observed 450 μm polarization offsets. The limited sample of bright fields shown here was sufficient to exclude telescope-dependent origins.

D.1. Telescope Pointing and Attitude Parameters

The first hypothesis tested was whether the 450 μm offsets correlate with the geometry of the telescope pointing or attitude parameters. Figure 28 presents the azimuth–elevation distribution of the residual Stokes Q and U offsets, evaluated at the MSB level for seven bright star-forming regions selected from the larger survey sample because their exceptional brightness enabled offset measurements with sufficient precision in individual 40–42 minute observations. The offsets are expressed in units of their field-level uncertainties (σ), with values within $\pm 5\sigma$ rendered in white and larger deviations transitioning to blue (negative) or red (positive). The distribution of points across the azimuth–elevation plane shows no systematic structure, gradient, or clustering that would indicate telescope-dependent effects; both Stokes components exhibit similar amplitude distributions and spatial symmetry. Additional tests confirmed the absence of correlations with declination, parallactic angle, or hour angle in the analyzed fields, definitively excluding telescope pointing and tracking geometry as sources of the offset.

D.2. Source Brightness Dependence

The MSB-level analysis necessarily focused on bright fields because of signal-to-noise requirements, so we investigated whether the 450 μm offsets might scale with source brightness, which could indicate detector non-linearity or signal-dependent bolometer response. Across the dynamic range accessible in our sample—spanning peak intensities from approximately 100 mJy beam $^{-1}$ to several Jy beam $^{-1}$ —the offsets remained constant within measurement uncertainties when expressed in surface-brightness units (mJy arcsec $^{-2}$). This insensitivity to source flux demonstrates that saturation effects, non-linear detector response, and flux-dependent calibration errors do not contribute to the observed phenomenon. The most luminous targets in our sample (including OMC-1 and NGC 6334) exhibited offset magnitudes comparable to fainter sources when normalized by their respective noise levels.

D.3. Instrumental Polarization Correction

The JCMT POL-2 data-reduction pipeline applies an elevation-dependent IP correction following the model of Friberg et al. (2018). To assess whether uncertainties in this correction could produce the observed offsets, we reprocessed the data with perturbed IP parameters in the range -0.6% to $+0.6\%$ relative to the nominal values—perturbations that exceed the expected IP-model uncertainty by approximately a factor of three. The resulting changes in Stokes Q , U , and polarized intensity $PI = \sqrt{Q^2 + U^2}$ were negligible compared to observational noise levels. For most fields, the 450 and 850 μm channels responded identically to variations in the IP parameters, preserving

the ratio PI_{450}/PI_{850} to within 1%. A small subset of fields exhibited differential responses between wavelengths, but the magnitude of these effects ($< 0.1 \text{ mJy arcsec}^{-2}$) was insufficient to explain the observed offsets, which typically exceed $1 \text{ mJy arcsec}^{-2}$ in bright regions.

D.4. Summary of Null Hardware Tests

These three independent null tests collectively rule out hardware-related origins of the $450 \mu\text{m}$ polarization offset and motivate the software-origin diagnosis pursued in §3.4.4. In particular, the identification of the `pca.zero_snr` parameter in the SMURF/pol2map pipeline as the source of the effect (§3.4.4) is a direct consequence of having ruled out telescope-, source-brightness-, and IP-related origins through the diagnostics presented above.

E. PHYSICAL ASSUMPTIONS BEHIND THE ZERO-INTERCEPT CONDITION

The body-text statement at §3.4.5 rests on a single physical conclusion—“under optically thin dust emission with wavelength-common alignment geometry, $Q_{850} = U_{850} = 0$ implies $Q_{450} = U_{450} = 0$.” This appendix collects the six standard radiative-transfer and grain-alignment assumptions required to derive that conclusion, together with the supporting citations. The assumption labels are referenced from the body text of §3.4.

1. **Astrophysical origin of the $850 \mu\text{m}$ reference polarization.** All $850 \mu\text{m}$ polarizations used as reference are of astrophysical origin, confirmed by instrumental polarization characterization (Friberg et al. 2018). The magnetic field directions remain coherent along the line of sight within each resolution element, preventing vector cancellation effects that could artificially reduce polarization (beam depolarization; Jones et al. 2015).
2. **Potential instrumental contamination at $450 \mu\text{m}$.** The fully calibrated $450 \mu\text{m}$ data may contain residual atmospheric contributions that affect the measured Stokes parameters; these instrumental effects must be identified and corrected before scientific analysis.
3. **Optically thin dust emission.** The dust emission is optically thin at both wavelengths: $\tau_{450} < 0.01$ even at column densities $N_{\text{H}_2} \sim 10^{24} \text{ cm}^{-2}$, and $\tau_{850} \ll 0.01$ (Hildebrand 1983; Ossenkopf & Henning 1994). This ensures that the observed polarization traces the entire column without self-absorption effects.
4. **Line-of-sight integration of volume emissivity.** Observed intensities represent line-of-sight integrations of volume emissivity, following standard radiative transfer (Draine 2003).
5. **Stokes parameters as projections of polarization pattern.** Stokes Q and U equal the product of emissivity and polarization efficiency, fundamental to emission polarimetry (Hildebrand 1988).
6. **Wavelength-independent grain alignment directions.** Grain alignment directions are wavelength-independent between $450\text{--}850 \mu\text{m}$, though alignment efficiency may vary with wavelength (consistent with RAT theory; Andersson et al. 2015; Planck Collaboration et al. 2020).

Collectively, assumptions (1), (3), (4), and (5) imply that the observed Stokes Q and U parameters at both wavelengths are projections of a single intrinsic polarization pattern that is common to 450 and $850 \mu\text{m}$. Assumption (6) requires the polarization position angle to be wavelength-independent, while the proportionality between the two wavelengths—equivalently, the ratio of polarized intensities, quantified by the slopes a_1^Q and a_1^U in Eq. 37—may differ owing to wavelength-dependent alignment efficiency. Under these conditions, any sky position satisfying $Q_{850} = U_{850} = 0$ must also satisfy $Q_{450} = U_{450} = 0$, because both wavelengths trace the same column density and polarization pattern. Therefore, the linear regressions of Q_{450} versus Q_{850} and of U_{450} versus U_{850} must pass through the origin, implying $a_0^Q = a_0^U = 0$.

The observed non-zero intercepts (Eq. 36) thus indicate systematic effects that violate at least one of the assumptions above. Because assumption (1) is supported by the IP characterization of Friberg et al. (2018) and assumption (3) holds for all sources in our sample, the most plausible explanation is assumption (2): residual instrumental or processing-related contamination in the $450 \mu\text{m}$ Stokes parameters. The hardware investigation (§3.4.3; Appendix D) rules out telescope-related effects, and the software-origin analysis in §3.4.6 identifies the specific pipeline parameter responsible. Accordingly, the three-step correction procedure described in §3.4.6 is designed to remove this contamination and to restore the physically required condition $a_0^Q = a_0^U = 0$.

F. PCA-THRESHOLD GRID ATLAS FOR THE 450 μm OFFSET OPTIMIZATION

This appendix presents the field-by-field response of the 450 μm offset to the PCA-threshold sweep that underpins the three-step correction of §3.4.6. We describe the empirical origin of the offsets in the time-stream interpolation and Fourier filtering of `makemap` (§F.1), the grid-search formalism that converts the per-field offsets into a quantitative significance test (§F.2), the limitations of the mosaicking step at 450 μm that restrict the optimization framework to single-field reductions (§F.3), and the optimal-threshold ranges as a function of field morphology (§F.4). Table 9 lists the explored grid of (`pca.zero_snr`, `pca.zero_snrlo`) combinations, and Figures 29–32 show the resulting Category 1–3 atlas of a_0^{tot} versus `pca.zero_snr`, with the shaded band indicating the noise-consistency condition $|a_0| < \sigma_{\text{eff}}$ (Eq. F12).

F.1. Origin of the 450 μm Offset

Through systematic testing of the `makemap` algorithm (Chapin et al. 2013), we trace the 450 μm offsets to the construction of low-frequency background models in the `pol2map` pipeline. The FLT, PCA, and COM stages of `makemap` replace masked time-stream samples with linearly interpolated values prior to Fourier-domain filtering. Because the 450 μm time-stream noise exceeds the 850 μm value by approximately an order of magnitude (Drabek et al. 2012; Holland et al. 2013; Mairs et al. 2021), these interpolated segments differ systematically between bands and inject wavelength-dependent zero-level biases into the FFT, where sharp data-edges produce ringing that propagates as negative bowls or spatial ripples in the final map (Mairs et al. 2021; Friberg et al. 2023).

The biases do not depend monotonically on PCA-mask area: fields with comparable mask sizes can exhibit offsets differing by more than an order of magnitude. This indicates that the offset depends on the *interaction* between mask geometry, interpolation noise, and FFT edge response during the early filtering iterations rather than on mask size alone. Reducing the spatial extent of the FLT, PCA, and COM masks via the (`pca.zero_snr`, `pca.zero_snrlo`) pair limits the artificial interpolated content and extrapolation in poorly constrained regions; bright targets retain non-zero masks to suppress ringing, while faint or smoothly varying fields reach minimal offsets with minimal masking. Table 9 below summarizes the explored parameter grid; the standard configuration (`pca.zero_snr`, `pca.zero_snrlo`) = (3, 2) used in previous POL-2 studies (e.g., Ward-Thompson et al. 2017; Pattie et al. 2017; Kwon et al. 2018; Soam et al. 2018) is one point on this grid, and the present work shows that the optimal threshold is field-dependent.

F.2. Optimization Procedure

For each field in the survey—both individual pointings and the component tiles of mosaicked targets—we reprocess the data across the grid of `pca.zero_snr`/`pca.zero_snrlo` values listed in Table 9 and measure the baseline terms (a_0^Q, a_0^U) via weighted regression. We define the combined offset and the effective noise as

$$a_0^{\text{tot}} = \sqrt{(a_0^Q)^2 + (a_0^U)^2}, \quad (\text{F11})$$

$$\sigma_{\text{eff}} = \sqrt{\sigma_Q \sigma_U}, \quad (\text{F12})$$

where $\sigma_Q \simeq \sigma_U$ and $\text{Cov}(Q, U) \approx 0$ are empirically validated in Eq. (33) of §3.3.6. This formulation lets us assess any residual baseline directly in units of the polarization-noise amplitude, which feeds the offset-significance test that produces the field categorization summarized in body §3.4.7.

F.3. Limitations of Mosaicking for 450 μm Offset Analysis

The optimization procedure is reliable only for single-field reductions. For 450 μm mosaicked and mosaic-component products, (a_0^Q, a_0^U) becomes nearly independent of `pca.zero_snr`: the curves flatten, minima disappear, and the resulting `polvecat` catalogs at `pca.zero_snr` = 3 are bitwise identical to those at optimized thresholds, indicating that the public `pol2map` Step 3 implementation does not propagate field-dependent PCA thresholds through the mosaicking step at 450 μm . No analogous failure is seen at 850 μm , so the issue reflects a wavelength-dependent interaction between time-stream noise, PCA masking, and the mosaicking algorithm rather than a general POL-2 limitation. As a consequence we exclude mosaicked and mosaic-component reductions from the offset diagnostics and from the three-step correction (§3.4.6), and apply the corrected three-step framework only to single-field 450 μm data. Diagnostic case studies of mosaicking failure modes for OMC 1, NGC 1333, and NGC 2264 are presented in Appendix G.

Table 9. Tested combinations of `pca.zero_snr` and `pca.zero_snrlo` parameters in the masking grid search. The nominal SCUBA-2 default is `pca.zero_snr` = 3 and `pca.zero_snrlo` = 2 (Chapin et al. 2013).

<code>pca.zero_snr</code>	<code>pca.zero_snrlo</code>
2	1.2
3	2
5	3
10	6
17	10
25	15
50	30
100	60
150	90
200	120
300	180
400	240
800	480
1600	960

F.4. Optimal Threshold Ranges as a Function of Field Morphology

The grid search yields optimal single-field thresholds spanning `pca.zero_snr` $\simeq$ 150–450, depending on field morphology: bright, high-contrast regions (e.g., OMC 1, Sgr B2) prefer more aggressive masking ($\sim$ 300–400), while smoother fields (e.g., Ophiuchus A, OMC 2–3, NGC 6334 South) reach minimal a_0^{tot} at $\sim$ 150–250. The minima are generally broad (Δ `pca.zero_snr` $\sim$ 50–100 of near-optimal performance), and adopting the default `pca.zero_snr` = 3 leaves residual offsets of $\sim$ 0.2–0.4 mJy arcsec $^{-2}$ that the optimized thresholds reduce to a few $\times 10^{-2}$ mJy arcsec $^{-2}$. The two key practical conclusions are: (i) the significance of (Q, U) baselines must be evaluated against σ_{eff} rather than against an absolute amplitude criterion, and (ii) field-dependent PCA-threshold optimization is essential for bias-free dual-band 450 μ m polarimetry but is currently feasible only for single-field reductions.

G. MOSAICKING FAILURE MODES FOR THE 450 μ m OFFSET DIAGNOSTICS

The body-text statement at §3.4.7 that 450 μ m offset diagnostics fail under mosaicking is supported by direct case-study comparisons among (i) single-field reductions, (ii) mosaic-component reductions, and (iii) mosaicked products for several POL-2 targets observed with multiple overlapping tiles. This appendix collects the diagnostic case studies and documents the failure-mode signatures.

For 450 μ m single-field reductions, the offset amplitude a_0^{tot} shows the expected dependence on `pca.zero_snr`: bright fields (e.g., OMC 1, Mon R2, OMC 2, NGC 6334 North, Ophiuchus A) exhibit clear minima at finite thresholds, enabling reliable optimization and application of the three-step correction (§3.4.6). When identical data are processed as mosaic-component fields, however, the offset behavior at 450 μ m changes qualitatively: (a_0^Q, a_0^U) becomes nearly independent of `pca.zero_snr`—the curves flatten, minima disappear, and the amplitudes can increase relative to the single-field reductions. For example, the OMC 1 component maintains offsets of $(a_0^Q, a_0^U) \approx (+0.7, +0.2)$ mJy arcsec $^{-2}$ for `pca.zero_snr` = 2–25 and then transitions abruptly to $\approx (-0.8, -0.7)$ mJy arcsec $^{-2}$ at larger thresholds with little variation thereafter. NGC 1333 and NGC 2264 show analogous behavior: their single-field tiles exhibit modest threshold dependence, whereas the mosaic-component reductions hold nearly constant offsets across the full parameter grid.

A second failure mode is the propagation and amplification of residual offsets in the final mosaicked products. For NGC 1333 the southern component shows individual offsets of only $\sim$ 0.05–0.10 mJy arcsec $^{-2}$, but yields $(a_0^Q, a_0^U) \approx (-0.20, +0.05)$ mJy arcsec $^{-2}$ under mosaic-component processing, and the final mosaic inherits the larger values, placing it in Category 1. NGC 2264 shows an even stronger effect: the northern component maintains $a_0^U \approx +0.34$ mJy arcsec $^{-2}$ across nearly all tested thresholds, a factor of two larger than any offset seen in its single-field reduction.

Finally, for several mosaicked fields the polarization catalogs (`polvecat`) generated with `pca.zero_snr` = 3 are bitwise identical to those generated with optimized thresholds. This null catalog-level response indicates that the

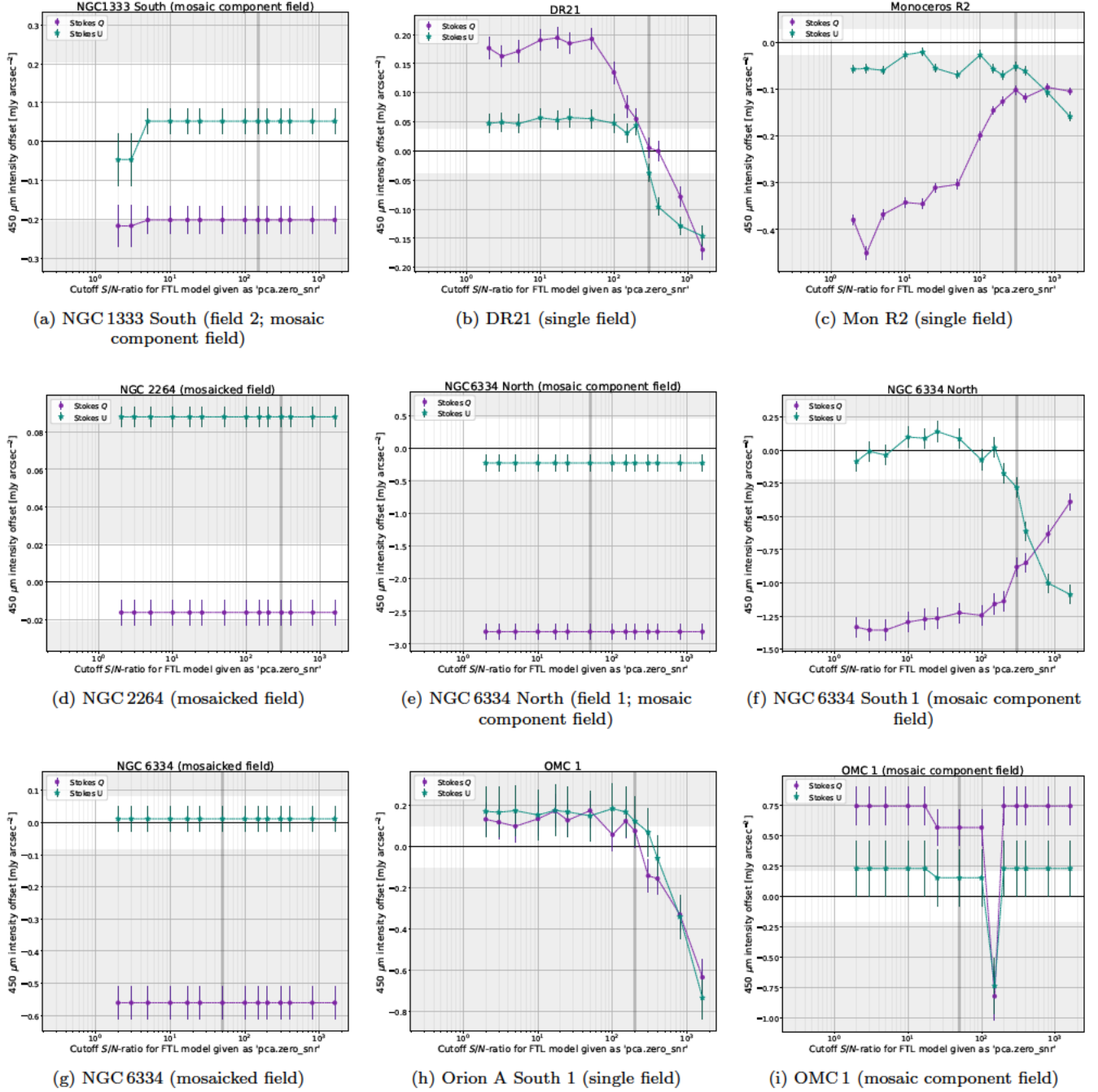


Figure 29. Category 1. Fields where 450 μm Stokes Q and U offsets exceed the effective noise σ_{eff} for the default `pca.zero_snr`=3 and remain substantial across all tested thresholds, requiring optimized masking (§3.4.7). Labels indicate the field type: “single field” denotes an individual POL-2 pointing, “mosaic component field” refers to a single tile within a multi-tile mosaic, and “mosaicked field” designates the combined product of multiple component fields.

public `pol2map` implementation does not propagate field-dependent PCA-mask configurations through the mosaicking step at 450 μm . No analogous failure is observed at 850 μm , so the issue reflects a wavelength-dependent interaction between time-stream noise, PCA masking, and the mosaicking algorithm rather than a general POL-2 limitation. Together, these signatures justify the body-text exclusion of mosaicked and mosaic-component products from the 450 μm offset analysis and from the corrected dual-band Stokes maps used in the companion science papers.

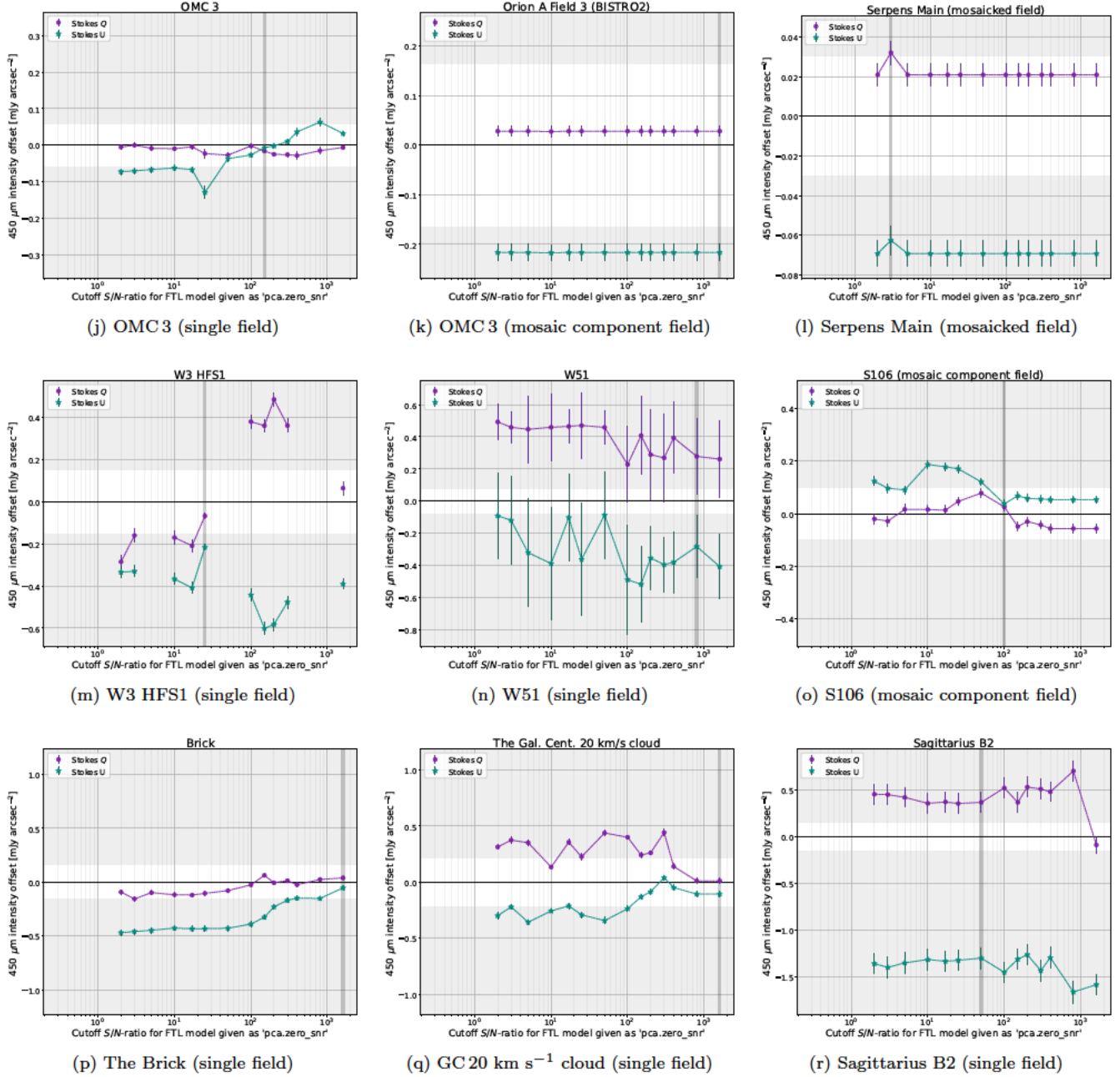


Figure 30. Category 1 (*Continued*). Additional fields showing statistically significant $450\,\mu\text{m}$ baseline offsets that exceed σ_{eff} across the full threshold range. These targets demonstrate that bright, structured regions such as W51, the Central Molecular Zone (CMZ), and Orion A consistently require optimized PCA masking to suppress interpolation-induced zero-level errors.

H. REPRODUCIBLE POL2MAP REDUCTION CONFIGURATION FOR THE $450\,\mu\text{m}$ SINGLE-FIELD PRODUCTS

This appendix documents the reproducible `pol2map` command set and the field-dependent modifications adopted for the $450\,\mu\text{m}$ single-field reductions used throughout this work (body-text summary in §3.4.7). All $450\,\mu\text{m}$ reductions follow the standard three-step `pol2map` structure with two modifications: (i) `flt.filt_edge_largescale=300` enforcing the $12''$ – $300''$ common spatial-scale range, and (ii) field-specific (`pca.zero_snr`, `pca.zero_snrlo`) thresholds drawn from the optimization grid in Table 9.

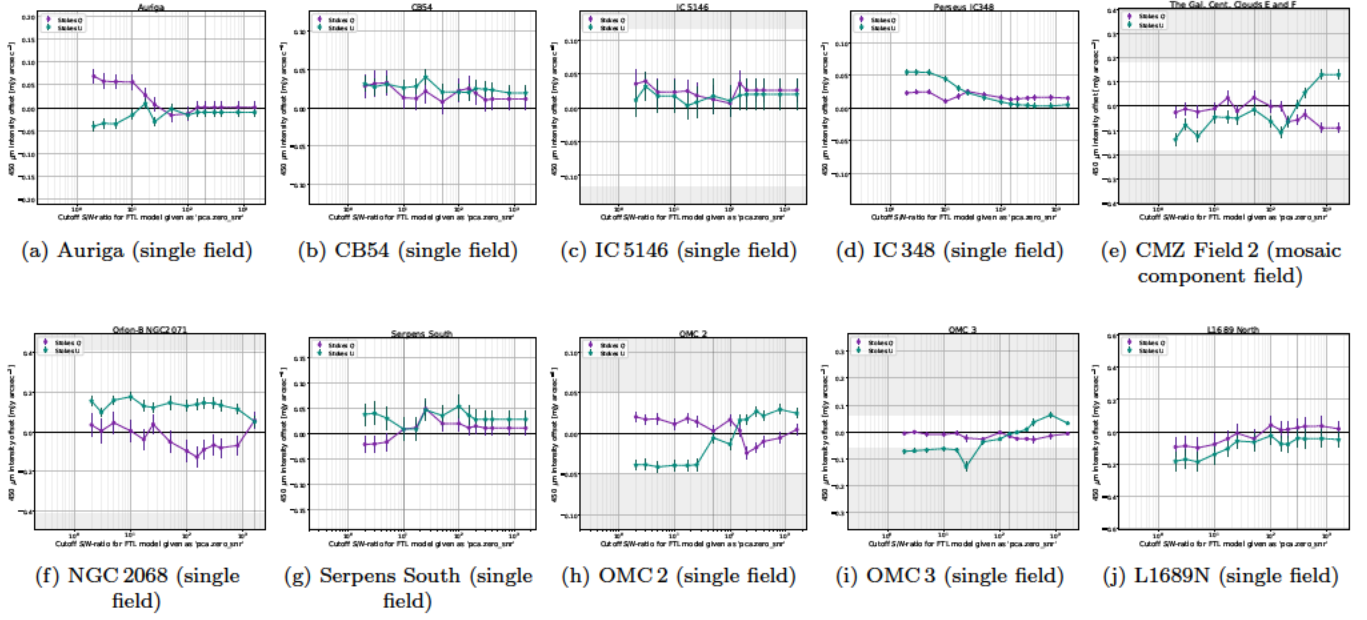


Figure 31. Category 2. Fields whose optimal `pca.zero_snr` differs from the default value of three, but for which the combined offset amplitude a_0^{tot} remains fully consistent with the effective noise level σ_{eff} (see §3.4.7). For these targets, the default `pca.zero_snr` = 3 remains appropriate for downstream analysis. In several panels the shaded band indicating $|a_0| < \sigma_{\text{eff}}$ is not visible; in these cases the offsets are sufficiently small that the auto-scaled y axis compresses the band below the plotting range.

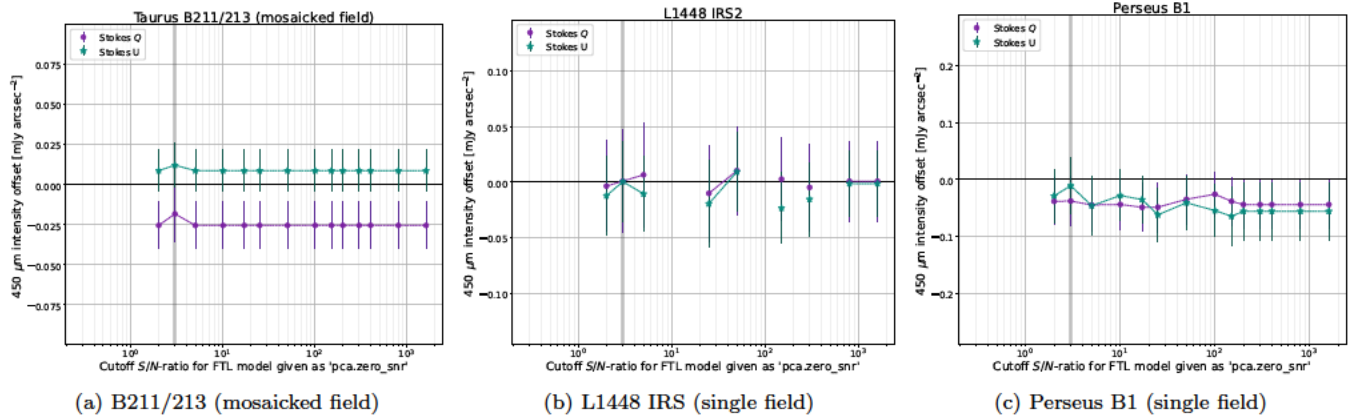


Figure 32. Category 3. Fields where the 450 μm offsets remain consistent with noise for all cutoff values (see §3.4.7). For these targets, the total offset amplitude a_0^{tot} of Eq. (F11) stays within the noise band σ_{eff} across the entire threshold range. No offset correction is required, and the standard mask threshold is adequate. In these panels the $|a_0| < \sigma_{\text{eff}}$ shaded band does not appear, because the offsets are sufficiently small that the auto-scaled axes compress the band below the plotting range, consistent with the noise-dominated character of Category 3.

H.1. *pol2map* Step 1: auto-masked total intensity map

We first add `flt.filt_edge_largescale=300` to the configuration file, limiting sensitivity to structures larger than $\sim 300''$. Combined with `smooth450=yes`, which deconvolves the 450 μm double-Gaussian beam and reconvolves to $12''.6$, this ensures that both wavelengths sample identical spatial scales ($12''.6$ – $300''$). The auto-masked reference map is produced with:

```
pol2map config=~conf in=~infiles iout=iauto qout=! uout=! \
ref=iext850.sdf mapdir=maps qudir=qudata pixsize=4 smooth450=yes
```

Here `ref=iext850.sdf` guarantees that the 450 and 850 μm maps are pixel-matched, allowing direct pixel-by-pixel regression.

H.2. *pol2map Step 2: externally-masked intensity map*

The masks are refined and inter-observation normalization applied:

```
pol2map config=~conf in=qdata/* iout=iext qout=! uout=! mapdir=maps mask=iauto \
    maskout1=astmask maskout2=pcamask BINSIZE=12 mapvar=yes skyloop=yes \
    obsweight=no normalise=yes smooth450=yes retain=yes
```

The `normalise=yes` setting stabilizes the large-scale background across multiple MSBs.

H.3. *pol2map Step 3: final Q and U maps with optimized PCA thresholds*

The configuration file is modified to include the field-specific thresholds (`pca.zero.snr`, `pca.zero.snrlo`) drawn from Table 9, where these per-field thresholds minimize the offset for each field (Figures 29–32). The final Stokes Q and U maps are generated with:

```
pol2map config=~conf2 in=qdata/* iout=! qout=qext uout=uext mapdir=maps mask=iauto \
    maskout1=astmask maskout2=pcamask ipref=iext mapvar=yes skyloop=yes \
    obsweight=no normalise=yes smooth450=yes cat=polvecat.FIT BINSIZE=12 PIXSIZE=4 \
    debias=yes JY=no retain=yes
```

H.4. *Note on debiasing*

The `debias=yes` option applies the Modified Asymptotic Estimator (Plaszczynski et al. 2014), the explicit form of which is given in Appendix I. Users who require alternative statistical treatments of the polarization amplitude should run Step 3 with `debias=no` and apply their preferred correction during post-processing.

H.5. *Note on flux calibration*

We use `JY=no` to retain native units (pW) appropriate for extended emission. The alternative `JY=yes` applies point-source flux conversion that is unsuitable for molecular clouds. Extended-emission FCFs from Mairs et al. (2021) and POL-2 attenuation factors (Friberg et al. 2023) are applied during post-processing.

I. MODIFIED ASYMPTOTIC DEBIASING OF THE 450 μm POLARIMETRIC PRODUCTS

This appendix documents the explicit form of the Modified Asymptotic Estimator (MAS) of Plaszczynski et al. (2014) and Montier et al. (2015) adopted for the final 450 μm polarimetric products ($P_{\text{MAS}}, p_{\text{MAS}}, \chi$) summarized in §3.4.8.

Following the offset removal of §3.4.4, the corrected Stokes parameters are given by Eq. (44). The polarized intensity is computed as $P^{\text{corr}} = \sqrt{Q_{450}^{\text{corr},2} + U_{450}^{\text{corr},2}}$, but this raw amplitude is positively biased in the presence of Gaussian noise. The MAS estimator removes this bias in a stable, noise-robust manner well suited to POL-2 data.

Let σ_Q and σ_U denote the uncertainties in the corrected Stokes Q and U parameters. Defining $S^2 = Q_{450}^{\text{corr},2} + U_{450}^{\text{corr},2}$, the noise term entering the Rice distribution is

$$\sigma_P^2 = \frac{Q_{450}^{\text{corr},2} \sigma_Q^2 + U_{450}^{\text{corr},2} \sigma_U^2}{Q_{450}^{\text{corr},2} + U_{450}^{\text{corr},2}}. \quad (\text{I13})$$

The MAS-debiased polarized intensity is then

$$P_{\text{MAS}} = P^{\text{corr}} - \frac{\sigma_P^2}{2P^{\text{corr}}} \left[1 - \exp\left(-\frac{S^2}{\sigma_P^2}\right) \right], \quad (\text{I14})$$

where $P^{\text{corr}} = \sqrt{S^2}$ is the raw amplitude. By construction, P_{MAS} is set to zero when the bracketed expression becomes negative, ensuring physically meaningful behavior in the low- S/N regime. The corresponding debiased polarization fraction is

$$p_{\text{MAS}} = \frac{P_{\text{MAS}}}{I}, \quad (\text{I15})$$

where I is the corresponding Stokes I intensity. The procedure is applied after the offset-correction steps of Eqs. (38)–(40), so that the statistics of the corrected Stokes parameters satisfy the assumptions of the MAS estimator. Together with the polarization angle χ defined in Eq. (48), the debiased quantities ($P_{\text{MAS}}, p_{\text{MAS}}, \chi$) constitute the final polarimetric products used in our scientific analysis (and in companion Papers S1 and S2).

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